

README and notes about the template for programmers

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This document is a work in progress.

This document describes the thesis template that I have developed for use at KTH Royal Institute of Technology (KTH) and provides some background on why it is structured in this way. This document provides information for programmers about the template, particularly on how it can be utilized in conjunction with specific programs to facilitate the thesis process for students, faculty, and administrators. The document is primarily intended for readers who are comfortable with programming or who need to maintain or modify the template or the programs I have written.

⌘ This is a very deep dive that is probably only of interest to a *very* small number of readers.

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1 Introduction

The focus in the first part of this document is on how to use the information in the thesis and captured by the $\text{\LaTeX}$ template and stored in the `fordiva.json` file to do other tasks:

1. The announcements and calendar events of thesis presentations or defenses. Moreover, thesis presentations and defenses are supposed to be public, so it would be useful and foster their visibility if it were easy to make such announcements (in both English and Swedish).
2. Both the metadata and the thesis itself need to be entered into Digitala Vetenskapliga Arkivet (DiVA)*.
3. Both English and Swedish titles will be reported along with the decision of the grading committee in Lokalt adb-baserat dokumentationssystem (LADOK).

A high-level overview was given in the `README_supervisor_notes.tex` file. This document covered “Accuracy and Economics”, “Make it simple from the start” by collecting the metadata, ‘Automating later steps in the degree project’, ‘Making an announcement’, ‘Making a cover and applying it’, ‘Can someone else use these programs?’, ‘ $\text{\LaTeX}$ template in Overleaf’, and ‘Put information into $\text{\LaTeX}$ template to generate a draft or final thesis”.

Section 17 discusses some enhancements. Some readers may be interested in accessibility, see Section 9. Section 10 discusses the structure of the template and the report. Section 11 shows how one can generate $\text{\LaTeX}$ commands for use in the student’s thesis source file, specifically for the `custom_configuration` file. Section 3 presents the big picture of how the different data and programs interact.

Section 12 presents how the copyright or Creative Commons License can be realized on the `bookinfo` page. Section 4.3 describes how one can add the `acronyms.tex` file to the Portable Document Format (PDF) both as text and attach it to the PDF file. Section 16.1 presents the details of how to add additional metadata to the PDF file, such that human readers and other tools can access this data. This is a rather expansive area, as there are those who would like to be able to mechanically extract data about the thesis and extract data from it (*e.g.*, citation

*There is a separate issue regarding whether the full-text is publicly available or not, but this is outside the scope of this document.

data). Finally, Section 18 suggests some future work, such as how to better exploit unicode, better math support in DiVA, and Cora-based DiVA.

I hope that this document contains sufficient information to enable someone else to address any remaining problems (that will inevitably occur) and extend it in new ways to better support the whole thesis process and the various readers.

2 Overall flow with an extension

The approach taken in the overall work was to make data readily available via the template and collect & output it in a “For DiVA” set of information at the end of the PDF document and a `fordiva.json` file. Additionally, we try to minimize the need to manually enter information by taking information from a Canvas course room (if one is relevant).

The overall flow is shown in Figure 1. Notice that this figure has been extended to include a `JSON_to_Cora_DiVA` process that can use the Cora Application Programming Interface (API) to directly create an entry in the new Cora-based DiVA:

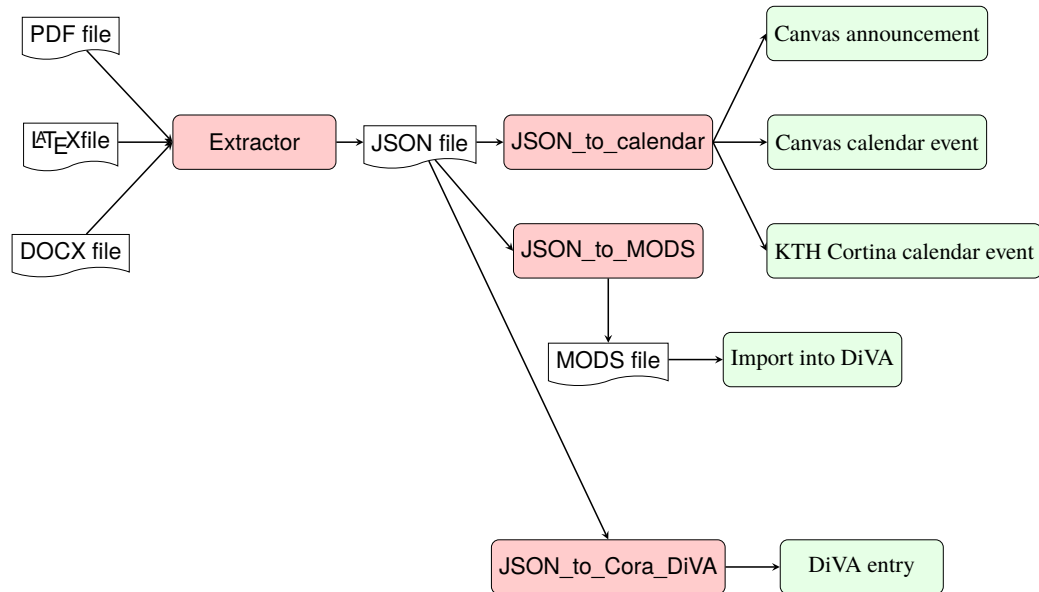


Figure 1: From a PDF file, a LaTeX project, or Microsoft Word Open XML Document (DOCX) file – generate an announcement, calendar entries, and a Metadata Object Description Schema (MODS) file for import into DiVA

If you are still reading at this point, it is going to get deeper!

3 The big picture

Table 1 shows the relationship between the information and the scripts that use it. The sections highlighted in yellow are optional. For example, only 1st cycle theses will have a second author. While all theses will have two supervisors, some may have more (the scripts support up to 10).

Individuals can be associated with KTH, in this case, they have a Local User Id and an organization element (with the implicit top level (L0) being KTH). The organization levels of L1 == School and L2 == Department. It is possible to extend this further, but the programs currently only support L1 and L2. The Title and Alternative title fields are used to support the English and Swedish title (or visa versa) — similarly, the abstracts and keywords must include English and Swedish versions; with additional support for abstracts and keywords in many languages. Where a school should be specified, one can use `\schoolAcronym{xxxx}`.

The Canvas course room's `course_id` is often an argument to my scripts. One reason for this is to be able to use the information in the course room to translate a local user ID (*i.e.*, a `kthid`) to an integration ID (*i.e.*, a LADOK user ID) or to be able to look up details of an author or supervisor.

Note that for third-cycle degrees, the degree information is a little different as there are no program codes, but rather education codes. Also there is a `\degreeModifier{Philosophy}` macro to enable the user to specify Philosophy as a modifier to the degree name.

Table 1: Information and how it is used

Top level	elements	subelements	LaTeX Macro	DOCX DocProperty or fields	
Author1	Last name	L1	\authorsLastname{} \authorsFirstname{} \kthid{} \email{}	Author1_Last_name	x
	First name			Author1_First_name	x
	Local User Id			Author1_Local User Id	x
Author2	E-mail	L1	\secondAuthorsLastname{} \secondAuthorsFirstname{} \secondkthid{} \secondemail{}	Author1_E-mail	x
	organisation			Author1_organization_L1	x
	Other organisation			Author1_organization_L2	x
Author2	Last name	L1	\secondAuthorsLastname{} \secondAuthorsFirstname{} \secondkthid{} \secondemail{}	Author2_Last_name	x
	First name			Author2_First_name	x
	Local User Id			Author2_Local User Id	x
Cycle	E-mail	L1	\secondAuthorsLastname{} \secondAuthorsFirstname{} \secondkthid{} \secondemail{}	Author2_E-mail	x
	organisation			Author2_organization_L1	x
	Other organisation			Author2_organization_L2	x
Course code			\courseCycle{} \courseCode{}	Cycle	x
				Course_code	i

Table 1 – Continued from previous page

Top level	elements	subelements	LaTeX Macro	DOCX DocProperty or fields	calendar cover	LADOK MODS
Credits				Credits		
Degree1	Educational program			Educational program	x	x
	programcode		Derived from programcode	programcode	i	i
	Degree		\degreeName{ }	Degree	x	x
	subjectArea		\subjectArea	subjectArea	x	x
Degree2	Educational program			second_Educational program	x	x
	programcode		Derived from secondProgramcode	second_Programcode	i	i
	Degree		\secondProgramcode{ }	second_Degree	x	x
	subjectArea		\secondDegreeName{ } \secondSubjectArea{ }	second_subjectArea	x	x
Title	Main title		\title{ }	(existing document property) Subtitle	x	x
	Subtitle		\subtitle{ }	Subtitle	x	x
	Language		Derived from option to document class		x	x
Alternative title	Main title		\alttitle{ }	Alternative_main_title	x	x
	Subtitle		\altsubtitle{ }	Alternative_subtitle	x	x
	Language		Derived from option to documentclass		x	x
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Table 1 – Continued from previous page					LADOs	
Top level	elements	subelements	LaTeX Macro	DOCX DocProperty or fields	calendar cover	big picture
Supervisor1	Last name		\supervisorAsLastname{ }	Supervisor1_Last_name	x	x
	First name		\supervisorAsFirstname{ }	Supervisor1_First_name	x	x
	Local User Id		\supervisorAsKTHID{ }	Supervisor1_Local User Id		x
	E-mail organisation		\supervisorAsEmail{ }	Supervisor1_E-mail		x
	Other organisation	L1 L2	\supervisorAsSchool{ } \supervisorAsDepartment{ } \supervisorAsOrganization{ }	Supervisor1_organisation_L1 Supervisor1_organisation_L2 Supervisor1_Other_organisation		x o
Supervisor2	Last name		\supervisorBsLastname	Supervisor2_Last_name	x	x
	First name		\supervisorBsFirstname{ }	Supervisor2_First_name	x	x
	Local User Id		\supervisorBsKTHID{ }	Supervisor2_Local User Id		x
	E-mail organisation		\supervisorBsEmail{ }	Supervisor2_E-mail		x
		L1	\supervisorBsSchool{ }	Supervisor2_organisation_L1		x

Table 1 – Continued from previous page

Top level	elements	subelements	LaTeX Macro	DOCX DocProperty or fields	calendar cover LADOK MODS
Supervisor3	Other organisation	L2	$\backslash$ supervisorBsDepartment{ } $\backslash$ supervisorBsOrganisation{ }	Supervisor2_organisation_L2 Supervisor2_Other_organisation	0
	Last name		$\backslash$ supervisorCsLastname{ }	Supervisor3_Last_name	x
	First name		$\backslash$ supervisorCsFirstname{ }	Supervisor3_First_name	x
	Local User Id E-mail organisation		$\backslash$ supervisorCsKTHID{ } $\backslash$ supervisorCsEmail{ }	Supervisor3_E-mail	0
Cooperation	Other organisation	L1	$\backslash$ supervisorCsSchool{ }	Supervisor3_organisation_L1	x
		L2	$\backslash$ supervisorCsDepartment{ } $\backslash$ supervisorCsOrganisation{ }	Supervisor3_organisation_L2 Supervisor3_Other_organisation	0 x
					x
Cooperation	Partner_name		$\backslash$ hostcompany{ } or $\backslash$ hostorganization{ }	Cooperation_Partner_name	0
National Subject Categories			$\backslash$ nationalsubjectcategories{ }	National Subject Categories	the big picture 7
Other information					

Table 1 – Continued from previous page					LADOK MODS	
Top level	elements	subelements	LaTeX Macro	DOCX DocProperty or fields	calendar cover	big picture
Series	Year		\the\year	(derived from document property Date)		
	Number of pages		(derived from labels: pg:lastPageofPreface and pg:lastPageofMainmatter)	(derived from bookmarks: lastPageofPreface and lastPageofMainmatter)		
	Title of series					
Opponents Presentation	No. in series		\tritaTRITA-EECS- EX2021:00	Series_name	x	x
			\opponentsNames{ }	Number_in_series	x	x
				Opponents_Name	x	
	Date		\presentationDateAnd- TimeISO{ }	Presentation_Date	x	x
	Language		\presentationLanguage	Presentation_Language	x	x
	Room		\presentationRoom{ }	Presentation_Room	x	x
	Address		\presentationAddress	Presentation_Address	x	x
	City		\presentationCity	Presentation_City	x	x
Number of lang instances abstracts				[Manually entered as 2, 3, 4, ...]		
	eng		(saved in Lua table)	(derived from bookmark: EnglishAbstract and then text inserted)	x	x

4 Use of Lua and Babel

One of the major changes introduced in this template is the use of Lua and Babel. Lua is a simple programming language. This language is available to the user of Lua_{TEX}. Lua's tables offer a very powerful way to store structured data. We use a Lua package (`dkjson.lua`) to write a Lua table containing all of the metadata to a `fordiva.json` file. The format of this file is JavaScript Object Notation (JSON). Not only is this file generated but it is both attached to the PDF file and included in a listing on the "For DiVA" page.

The primary use of the `fordiva.json` file is to facilitate reporting the thesis in DiVA. However, a secondary use of this information is to potentially automate many of the steps in announcing the oral presentation (as was described in the section "Making an announcement" in the file `README_supervisor_notes.tex`). For example, the student's organization or main supervisor's organization is used to determine which local part of the KTH calendar a calendar announcement should appear in, as the Cortina calendar is divided by the school and then by department.

Note that the department name must be in Swedish for Cortina. Ideally, this is something that a program using the JSON data should handle for the user.

In addition to the English and Swedish abstracts and keywords information, there are a number of skeleton files in the folder `Additional_Abstracts`. While not an exhaustive list of possibilities, they represent the languages for which I found several abstracts in DiVA*. Since 2000, the language code 'scc' (for Serbo-Croatian) is considered deprecated and has been replaced by 'srp' for Serbian and 'hrv' for Croatian (for which I have not implemented a skeleton). Otherwise, there is a skeleton abstract for all of the language codes used in one or more abstracts in this language in DiVA.

The `\foreignlanguage{xxxx}{xxxxxxx}` command can be used to indicate that the string `xxxxxxx` is in language `xxxx` and the use of the `\selectlanguage{xxxx}` command to switch to the language `xxxx`. The first command makes it easy to include text in a language that is not the main language of the thesis at any point in the thesis. The second command can be used to switch to a specific language at a certain point in the text. This second command is used to facilitate the inclusion of abstracts and keywords in various languages.

The `babel` package is used to handle different languages and automatically switch to the appropriate font for each language. If one does **not** use `babel`

*Except for Estonian, which had an entry but no longer has an entry.

configured for the languages that you use, you will find messages in the `output.log` file, such as those shown in Listing 1. While it is possible to work around this on a character-by-character basis by using suitable fonts and adding mappings, such as shown in Listing 2, a better approach is described in Section 4.1.

Listing 1: Examples of missing characters from the `output.log` file

```
Missing character: There is no (U+4E2D) in font TeX
↳ Gyre
↳ Cursor/OT:script=latn;language=dflt;-liga;+lnum;+tnum;
↳ mapping=tex-text;!
Missing character: There is no (U+6587) in font TeX
↳ Gyre
↳ Cursor/OT:script=latn;language=dflt;-liga;+lnum;+tnum;
↳ mapping=tex-text;!
Missing character: There is no (U+6458) in font TeX
↳ Gyre
↳ Cursor/OT:script=latn;language=dflt;-liga;+lnum;+tnum;
↳ mapping=tex-text;!
Missing character: There is no (U+8981) in font TeX
↳ Gyre
↳ Cursor/OT:script=latn;language=dflt;-liga;+lnum;+tnum;
↳ mapping=tex-text;!
Missing character: There is no (U+4E2A) in font TeX
↳ Gyre
↳ Heros/OT:script=latn;language=dflt;mapping=tex-text;!
Missing character: There is no (U+5173) in font TeX
↳ Gyre
↳ Heros/OT:script=latn;language=dflt;mapping=tex-text;!
Missing character: There is no (U+952E) in font TeX
↳ Gyre
↳ Heros/OT:script=latn;language=dflt;mapping=tex-text;!
Missing character: There is no (U+8BCD) in font TeX
↳ Gyre
↳ Heros/OT:script=latn;language=dflt;mapping=tex-text;!
```

Listing 2: Examples of new unicode character definitions

```

\wlog{[+1.059s] Shipping out page 11}

\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+4E2A - CJK Unified Ideograph-4E2A
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+4E2D - CJK Unified Ideograph-4E2D
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+5173 - CJK Unified Ideograph-5173
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+6458 - CJK Unified Ideograph-6458
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+6587 - CJK Unified Ideograph-6587
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+8981 - CJK Unified Ideograph-8981
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+8BCD - CJK Unified Ideograph-8BCD
\newunicodechar{ }\iffontchar\font` \else{\CJKfonttt
↪ }\fi} % U+952E - CJK Unified Ideograph-952E

```

If the user uses characters that require fonts other than TeX Gyre Termes and has not configured babel with a font for this language, the contents of the abstracts in these languages may not appear on the “For DiVA” page but will be in the `fordiva.json` file. This will occur because the file simply contains the UTF-8 encoded unicode for the abstracts.

4.1 Better handling of fonts

Manually or programmatically adding all of the missing characters to the relevant `unicode_blocks` file was used to work with polyglossia and Xe_{La}TeX. However, it took a lot of work.

Therefore, a decision was made to switch to Lua_{La}TeX and babel. This combination has well-integrated `fontspec` support for automatically switching language and fonts. Listing 3 shows how easy it is to set up a font that can be used when setting content written in Hindi, *i.e.*, babel language hindi. This font will automatically be invoked when you issue a `\selectlanguage{hindi}` **and** when you use one of the unicode characters with code points in the Devanagari range, *i.e.*, U+0900 to U+097F.

Listing 3: Establishing fonts to be used when setting content in Hindi

```
\babelprovide[import, onchar=ids fonts]{hindi}
\babelfont[hindi]{rm}{Noto Serif Devanagari}
\babelfont[hindi]{sf}{Noto Sans Devanagari}
```

Similar functionality is utilized by the template with the babel languages: arabic, centralkurdish, chinese-simplified, chinese-traditional greek, hebrew, japanese, russian, and vietnamese.

One of the advantages of switching to babel is that it allows lazy loading of languages and lazy loading of fonts – thus unless the language is used or the font is used, they do not have to be loaded.

Unfortunately, luaotfload program must still resolve the font name into a file to load the font from, then it transforms this font into Lua code and compiles it into bytecode (that later can be loaded – if the font is used). The time required to do all of this can be considerable and may lead to timeouts. For this reason, I have introduced `warmup.tex` that when compiled will cause the luaotfload program to load the fonts used by `examplethesis.tex` into the working font cache. Addition, it will store a copy of these font files into a working directory. By following the directions in `Saving_and_restoring_font_cache.tex` you can download the saved fonts and then upload them for use by the document `restore_font_cache.tex` that when compiled will copy the uploaded saved fonts to the working font cache.

4.1.1 Babel ranges

The file `babel-data-bidi.lua` * defines the Babel table. This is shown in Listing 4. The format of the entries is the starting codepoint of a range, the ending codepoint of a range (inclusive), and the direction. The direction is defined according to “Table 4. Bidirectional Character Types” in *Unicode® Standard Annex #9: The Bidirectional Algorithm* [1]. Accordingly, ‘r’ is right-to-left, ‘l’ is left-to-right (the default), ‘al’ is right-to-left Arabic, and ‘on’ is other neutrals.

Listing 4: Babel table and Babel-ranges supplemented with additional comments about Unicode blocks

```
Babel = Babel or {}

Babel.ranges={
```

*Perhaps under `/usr/share/texmf/tex/generic/babel/`

14 | Use of Lua and Babel

```
{0x0590, 0x05FF, 'r'}, -- Hebrew, U+0590 - U+05FF
{0x0600, 0x07BF, 'al'}, -- Arabic, U+0600 - U+06FF;
                        -- Syriac, U+0700 - U+074F;
                        -- Arabic Supplement, U+0750 - U+077F;
                        -- Thaana, U+0780 - U+07BF
{0x07C0, 0x085F, 'r'}, -- Nko, U+07C0 - U+07FF;
                        -- Samaritan, U+0800 - U+083F;
                        -- Mandaic, U+0840 - U+085F
{0x0860, 0x086F, 'al'}, -- Syriac Supplement, U+0860 - U
    ↪ +086F
{0x08A0, 0x08FF, 'al'}, -- Arabic Extended-A, U+08A0 - U
    ↪ +08FF
{0xE000, 0xF8FF, 'on'}, -- PUA -- Private Use Area, U+E000
    ↪ - U+F8FF
{0xFB1D, 0xFB4F, 'r'}, -- Alphabetic Presentation Forms, U
    ↪ +FB00 - U+FB4F
{0xFB50, 0xFDFF, 'al'}, -- Arabic Presentation Forms-A, U+
    ↪ FB50 - U+FDFF
{0xFE70, 0xFEFF, 'al'}, -- Arabic Presentation Forms-B, U+
    ↪ FE70 - U+FEFF
{0x10800, 0x10C4F, 'r'}, -- Cypriot Syllabary, U+10800 - U
    ↪ +1083F;
                        -- Imperial Aramaic, U+10840 - U+1085F;
                        -- Palmyrene, U+10860 - U+1087F;
                        -- Nabataean, U+10880 - U+108AF;
-- hole U+108B0 - 108DF;
                        -- Hatran, U+108E0 - U+108FF;
                        -- Phoenician, U+10900 - U+1091F;
                        -- Lydian, U+10920 - U+1093F;
                        -- Meroitic Hieroglyphs, U+10980 - U
    ↪ +1099F;
                        -- Meroitic Cursive, U+109A0 - U+109FF;
                        -- Kharoshthi, U+10A00 - U+10A5F;
                        -- Old South Arabian, U+10A60 - U+10A7F;
                        -- Old North Arabian, U+10A80 - U+10A9F;
                        -- Manichaean, U+10AC0 - U+10AFF;
                        -- Avestan, U+10B00 - U+10B3F;
                        -- Inscriptional Parthian, U+10B40 - U
    ↪ +10B5F;
```

```

-- Inscriptional Pahlavi, U+10B60 - U+10
↪ B7F;
-- Psalter Pahlavi, U+10B80 - U+10BAF;
-- Old Turkic, U+10C00 - U+10C4F
{0x1E800, 0x1E8DF, 'r'}, -- Mende Kikakui, U+1E800 - U+1
↪ E8DF
{0x1E900, 0x1E95F, 'r'}, -- Adlam, U+1E900 - U+1E95F
{0x1F300, 0x1F9FF, 'on'}, -- Miscellaneous Symbols and
↪ Pictographs, U+1F300 - U+1F5FF;
-- Emoticons, U+1F600 - U+1F64F;
-- Ornamental Dingbats, U+1F650 - U+1
↪ F67F;
-- Transport and Map Symbols, U+1F680 -
↪ U+1F6FF;
-- Alchemical Symbols, U+1F700 - U+1
↪ F77F;
-- Alchemical Symbols, U+1F700 - U+1
↪ F77F;
-- Supplemental Arrows-C, U+1F800 - U+1
↪ F8FF;
-- Supplemental Symbols and Pictographs
↪ , U+1F900 - U+1F9FF:
{0xF0000, 0xFFFFD, 'on'}, -- PUA -- Supplementary Private
↪ Use Area-A, U+F0000 - U+FFFFF; does not include U+
↪ FFFFE and U+FFFFF
{0x100000, 0x10FFFFD, 'on'} -- PUA -- Supplementary Private
↪ Use Area-B, U+100000 - U+10FFFFF; does not include U
↪ +10FFFE or U+10FFFFF
}

```

Listing 5 shows part of the `Babel.characters` table. The direction table (with `d` in it) has additional character types: ‘an’ for Arabic number, ‘b’ for paragraph separator, ‘bn’ for boundary neutral, ‘cs’ for common number separator, ‘en’ for European number, ‘es’ for European number separator, ‘et’ for European number terminator, ‘fsi’ for first strong isolate, ‘lre’ for left-to-right embedding, ‘lri’ for left-to-right isolate, ‘lro’ for left-to-right override, ‘nsm’ for nonspace mark, ‘pdf’ for pop directional format, ‘pdi’ for pop directional isolate, ‘rle’ for right-left embedding, ‘rli’ for right-left isolate, ‘rlo’ for right-left override, ‘s’ for segment separator, and ‘ws’ for white space.

Listing 5: (part of) Babel table of characters

```

Babel.characters={
  [0x0]={d='bn'},
  :
  [0x8]={d='bn'},
  [0x9]={d='s'},
  [0xA]={d='b'},
  [0xB]={d='s'},
  [0xC]={d='ws'},
  [0xD]={d='b'},
  [0xE]={d='bn'},
  :
  [0x1B]={d='bn'},
  [0x1C]={d='b'},
  [0x1D]={d='b'},
  [0x1E]={d='b'},
  [0x1F]={d='s'},
  [0x20]={d='ws'},
  [0x21]={d='on'},
  [0x22]={d='on'},
  [0x23]={d='et'},
  [0x24]={d='et'},
  [0x25]={d='et'},
  [0x26]={d='on'},
  [0x27]={d='on'},
  [0x28]={d='on', m=0x29},
  [0x29]={d='on', m=0x28},
  [0x2A]={d='on'},
  [0x2B]={d='es'},
  [0x2C]={d='cs'},
  [0x2D]={d='es'},
  [0x2E]={d='cs'},
  [0x2F]={d='cs'},
  [0x30]={d='en'},
  :
  [0x39]={d='en'},
  [0x3A]={d='cs'},
  [0x3B]={d='on'},
  [0x3C]={d='on', m=0x3E},
  [0x3D]={d='on'},

```

```

[0x3E]={d='on', m=0x3C},
[0x3F]={d='on'},
[0x40]={d='on'},
[0x5B]={d='on', m=0x5D},
[0x5C]={d='on'},
[0x5D]={d='on', m=0x5B},
[0x5E]={d='on'},
[0x5F]={d='on'},
[0x60]={d='on'},
[0x7B]={d='on', m=0x7D},
[0x7C]={d='on'},
[0x7D]={d='on', m=0x7B},
[0x7E]={d='on'},
[0x7F]={d='bn'},
:
[0x84]={d='bn'},
[0x85]={d='b'},
[0x86]={d='bn'},
:
[0x9F]={d='bn'},
[0xA0]={d='cs'},
[0xA1]={d='on'},
[0xA2]={d='et'},
[0xA3]={d='et'},
[0xA4]={d='et'},
[0xA5]={d='et'},
[0xA6]={d='on'},
[0xA7]={d='on'},
[0xA8]={d='on'},
[0xA9]={d='on'},
[0xAB]={d='on', m=0xBB},
[0xAC]={d='on'},
[0xAD]={d='bn'},
[0xAE]={d='on'},
[0xAF]={d='on'},
[0xB0]={d='et'},
[0xB1]={d='et'},
[0xB2]={d='en'},
[0xB3]={d='en'},
[0xB4]={d='on'},

```

```
[0xB6]={d='on'},
[0xB7]={d='on'},
[0xB8]={d='on'},
[0xB9]={d='en'},
[0xBB]={d='on', m=0xAB},
[0xBC]={d='on'},
[0xBD]={d='on'},
[0xBE]={d='on'},
[0xBF]={d='on'},
[0xD7]={d='on'},
[0xF7]={d='on'},
[0x2B9]={d='on'},
[0x2BA]={d='on'},
[0x2C2]={d='on'},
:

```

Entries of the form shown in Listing 6 define **mirrored** characters.

Listing 6: Some of the matching delimiters in Babel table of characters (with added comments)

```
[0x28]={d='on', m=0x29}, -- Left Parenthesis
[0x29]={d='on', m=0x28}, -- Right Parenthesis
[0x3C]={d='on', m=0x3E}, -- Less-Than Sign
[0x3E]={d='on', m=0x3C}, -- Greater-Than Sign
[0x5B]={d='on', m=0x5D}, -- Left Square Bracket
[0x5D]={d='on', m=0x5B}, -- Right Square Bracket
[0x7B]={d='on', m=0x7D}, -- Left Curly Bracket
[0x7D]={d='on', m=0x7B}, -- Right Curly Bracket
[0xAB]={d='on', m=0xBB}, -- Left-Pointing Double Angle
    ↪ Quotation Mark
[0xBB]={d='on', m=0xAB}, -- Right-Pointing Double Angle
    ↪ Quotation Mark
[0xF3A]={d='on', m=0xF3B}, -- Tibetan Mark Gug Rtags Gyon
[0xF3B]={d='on', m=0xF3A}, -- Tibetan Mark Gug Rtags Gyas
[0xF3C]={d='on', m=0xF3D}, -- Tibetan Mark Ang Khang Gyon
[0xF3D]={d='on', m=0xF3C}, -- Tibetan Mark Ang Khang Gyas
[0x169B]={d='on', m=0x169C}, -- Ogham Feather Mark
[0x169C]={d='on', m=0x169B}, -- Ogham Reversed Feather
    ↪ Mark
:

```


The file `luababel.def` defines `Babel.script_blocks`. These are shown in Listing 7.

Listing 7: Babel table `script_blocks`

```
Babel.script_blocks = {
  ['dflt'] = {},
  ['Arab'] = {{0x0600, 0x06FF}, {0x08A0, 0x08FF}, {0x0750,
    ↪ 0x077F},
    {0xFE70, 0xFEFF}, {0xFB50, 0xFDFF}, {0x1EE00, 0
    ↪ x1EEFF}},
  ['Armn'] = {{0x0530, 0x058F}},
  ['Beng'] = {{0x0980, 0x09FF}},
  ['Cher'] = {{0x13A0, 0x13FF}, {0xAB70, 0xABBF}},
  ['Copt'] = {{0x03E2, 0x03EF}, {0x2C80, 0x2CFF}, {0x102E0,
    ↪ 0x102FF}},
  ['Cyr1'] = {{0x0400, 0x04FF}, {0x0500, 0x052F}, {0x1C80,
    ↪ 0x1C8F},
    {0x2DE0, 0x2DFF}, {0xA640, 0xA69F}},
  ['Deva'] = {{0x0900, 0x097F}, {0xA8E0, 0xA8FF}},
  ['Ethi'] = {{0x1200, 0x137F}, {0x1380, 0x139F}, {0x2D80,
    ↪ 0x2DDF},
    {0xAB00, 0xAB2F}},
  ['Geor'] = {{0x10A0, 0x10FF}, {0x2D00, 0x2D2F}},
  % Don't follow strictly Unicode, which places some Coptic
  ↪ letters in
  % the 'Greek and Coptic' block
  ['Grek'] = {{0x0370, 0x03E1}, {0x03F0, 0x03FF}, {0x1F00, 0
    ↪ x1FFF}},
  ['Hans'] = {{0x2E80, 0x2EFF}, {0x3000, 0x303F}, {0x31C0, 0
    ↪ x31EF},
    {0x3300, 0x33FF}, {0x3400, 0x4DBF}, {0x4E00, 0
    ↪ x9FFF},
    {0xF900, 0xFAFF}, {0xFE30, 0xFE4F}, {0xFF00, 0
    ↪ xFFEF},
    {0x20000, 0x2A6DF}, {0x2A700, 0x2B73F},
    {0x2B740, 0x2B81F}, {0x2B820, 0x2CEAF},
    {0x2CEB0, 0x2EBEF}, {0x2F800, 0x2FA1F}},
  ['Hebr'] = {{0x0590, 0x05FF}},
  ['Jpan'] = {{0x3000, 0x303F}, {0x3040, 0x309F}, {0x30A0, 0
    ↪ x30FF},
```

```

UUUUUUUUUUUUUUUU{0x4E00,U0x9FAF},U{0xFF00,U0xFFEF}},
UU['Khmr']_U={0x1780,U0x17FF},U{0x19E0,U0x19FF}},
UU['Knda']_U={0x0C80,U0x0CFF}},
UU['Kore']_U={0x1100,U0x11FF},U{0x3000,U0x303F},U{0x3130,U0
↪ x318F},
UUUUUUUUUUUUUUUU{0x4E00,U0x9FAF},U{0xA960,U0xA97F},U{0xAC00,U0
↪ xD7AF},
UUUUUUUUUUUUUUUU{0xD7B0,U0xD7FF},U{0xFF00,U0xFFEF}},
UU['Laoo']_U={0x0E80,U0x0EFF}},
UU['Latn']_U={0x0000,U0x007F},U{0x0080,U0x00FF},U{0x0100,U0
↪ x017F},
UUUUUUUUUUUUUUUU{0x0180,U0x024F},U{0x1E00,U0x1EFF},U{0x2C60,U0
↪ x2C7F},
UUUUUUUUUUUUUUUU{0xA720,U0xA7FF},U{0xAB30,U0xAB6F}},
UU['Mahj']_U={0x11150,U0x1117F}},
UU['Mlym']_U={0x0D00,U0x0D7F}},
UU['Mymr']_U={0x1000,U0x109F},U{0xAA60,U0xAA7F},U{0xA9E0,U0
↪ xA9FF}},
UU['Orya']_U={0x0B00,U0x0B7F}},
UU['Sinh']_U={0x0D80,U0x0DFF},U{0x111E0,U0x111FF}},
UU['Syrc']_U={0x0700,U0x074F},U{0x0860,U0x086F}},
UU['Taml']_U={0x0B80,U0x0BFF}},
UU['Telu']_U={0x0C00,U0x0C7F}},
UU['Tfng']_U={0x2D30,U0x2D7F}},
UU['Thai']_U={0x0E00,U0x0E7F}},
UU['Tibt']_U={0x0F00,U0x0FFF}},
UU['Vaii']_U={0xA500,U0xA63F}},
UU['Yiii']_U={0xA000,U0xA48F},U{0xA490,U0xA4CF}}
}

```

```

Babel.script_blocks.Cyrs_U=Babel.script_blocks.Cyrl
Babel.script_blocks.Hant_U=Babel.script_blocks.Hans
Babel.script_blocks.Kana_U=Babel.script_blocks.Jpan

```

4.2 Generating a true JSON for diva.json file

To understand if $\text{Lua}\text{\LaTeX}$ could be used to directly emit a properly formed JSON file, we used the programming features in the Lua programming language.

Section 4.2.1 shows a simple proof-of-concept for doing this*. Section 4.2.2 describes how the template actually does the full job.

4.2.1 Proof of concept

Listing 8 shows an example of a Lua program that generates a JSON file named `output.json`. The idea would be to utilize Lua data tables to store the metadata, then write it out as a complete JSON file.

Listing 8: Using Lua code to write a JSON file

```
\documentclass{article}
\usepackage{luacode}

% Use the starred environment to handle backslashes
  ↪ correctly.
\begin{luacode*}
function write_my_json_data()
  -- This data table can now have any number of keywords.
  local myData = {
    author = "Fake_A_Student",
    title = "This_is_the_title_in_the_language_of_the_thesis"
    ↪ ,
    year = 2025,
    keywords = { "KeywordA", "KeywordB", "KeywordC", "A_
    ↪ Fourth_Keyword" }
  }

  -- 1. Create a new table to hold the formatted keywords.
  local quotedKeywords = {}

  -- 2. Loop through the original keywords using ipairs to
    ↪ preserve order.
  for _, keyword in ipairs(myData.keywords) do
```

*Two of the keys to developing the various Lua functions were (i) to interact with Google Gemini 2.5 Pro to generate Lua code for a given idea and (ii) to create small Overleaf project to carry out small experiments with this code, rather than trying to put code directly into the template. There were three reasons for this: (i) quite often the code proposed by Gemini was incorrect, (ii) one can compile a small project quickly, thus increasing the speed of development, and (iii) quite often the process of prompt, get code, test code, tell Gemini the errors that were generated, and iterate – would lead to a complete redesign of the mechanism to be used.

```

-- Wrap each keyword string in quotes and add it to the
  ↪ new table.
  table.insert (quotedKeywords, '"' .. keyword .. '"')
end

-- 3. Join the quoted keywords with a comma and a space.
local keywordsString = table.concat(quotedKeywords, ', ')

-- 4. Build the final JSON string using the dynamically
  ↪ created keyword list.
local jsonString = string.format(
  '{\n  "author": "%s",\n  "title": "%s",\n  "year": %d,\n  "keywords": %s\n}',
  ↪ myData.author,
  myData.title,
  myData.year,
  keywordsString -- The placeholder is now filled by our
  ↪ joined string.
)

-- The file writing logic remains the same.
local outputFile = io.open("output.json", "w")
if outputFile then
  outputFile:write(jsonString)
  outputFile:close()
end
end
\end{luacode*}

\begin{document}

This is a LuaLaTeX document.
\directlua{write_my_json_data()}
The JSON file has been created using a loop for the keywords.

\end{document}

```

4.2.2 Using Lua in practice

Expanding on the proof of concept in Section 4.2, Listing 9 shows how Lua can be used to write the `fordiva.json` file with abstracts and keywords. David Kolf's *JSON module for Lua 5.1 - 5.4* from the file `dkjson.lua` handles the encoding of a Lua table into JSON, including ordering of the top-level keys* and escaping of the $\text{\LaTeX}$ commands. The JSON file is generated by invoking the `\GenerateFordivaJSON` macro after filling in the table in the macro `\divainfo`.

Listing 9: Lua code snippet to write `fordiva.json` file including abstracts and keywords

```
%
% --- Define the thesis abstract environment ---
\RequirePackage{environ}
\RequirePackage{verbatim}
\RequirePackage{luacode}

% --- Lua Backend ---
\begin{luacode*}
-- This global table will store the data
thesisData = {
  abstracts = {},
  keywords = {}
}

-- Generic function to add a value to our main data table
-- It can handle nested keys, e.g., set_data({"Author1", "
  ↪ Last name"}, "Student")
function set_data(keys, value)
  local t = thesisData
  for i = 1, #keys - 1 do
    local key = keys[i]
    if not t[key] then t[key] = {} end
    t = t[key]
  end
  t[keys[#keys]] = value
end
```

*Sorting is only done of the **top-level** keys. There is no facility for sorting the elements of the table.

```

-- This function reads content from a specified filename and
  ↳ adds it to our table
function add_data_from_file(category, lang, filename)
  local file = io.open(filename, "r")
  if not file then return end
  local content = file:read("*a")
  file:close()

  if thesisData[category] then
    thesisData[category][lang] = content
  end
end

-- This function generates the final JSON file
function write_json_file()
  -- This requires the dkjson.lua library in your project
  ↳ folder
  local dkjson = require("dkjson")

  -- Define the desired order of keys
  local keyorder = {
    "Author1", "Author2", "Course_Info", "Degree1", "Degree2",
    ↳ ", "Title", "Alternative_title",
    "Supervisor1", "Supervisor2", "Supervisor3", "
    ↳ Supervisor4", "Supervisor5",
    "Opponents",
    "National_Subject_Categories", "SDGs",
    "Other_information", "Series", "Copyrightleft", "
    ↳ Presentation",
    "abstracts", "keywords"
  }

  local jsonString = dkjson.encode(thesisData, { indent = true
    ↳ , keyorder = keyorder })
  local outputFile = io.open("fordiva.json", "w")
  if outputFile then
    outputFile:write(jsonString)
    outputFile:close()
  end
end

```

```

end
\end{luacode*}

% --- LaTeX Frontend ---
% Helper commands to pass LaTeX macro values to the Lua
  ↪ set_data function
% Helper commands that NOW evaluate the last argument
  ↪ before passing it to Lua.
\newcommand{\SetValue}[2]{\directlua{set_data({"#1"}, \
  ↪ luastring{\expanded{#2}})}}
\newcommand{\SetNestedValue}[3]{\directlua{set_data({"#1",
  ↪ "#2"}, \luastring{\expanded{#3}})}}
\newcommand{\SetDeepNestedValue}[4]{\directlua{set_data({"
  ↪ "#1", "#2", "#3"}, \luastring{\expanded{#4}})}}

\newwrite\tempfile

\NewEnviron{ThesisAbstract}[1]{%
  % 1. Write the raw content to a temporary file
  \immediate\openout\tempfile=abstract.tmp
  \immediate\write\tempfile{\unexpanded\expandafter{\BODY}}
  \immediate\closeout\tempfile
  % 2. Tell Lua to read the data from that file
  \directlua{add_data_from_file("abstracts", "#1", "
    ↪ abstract.tmp")}%
  % 3. Typeset the content in the document as normal
  \par
  \BODY
}

\NewEnviron{ThesisKeywords}[1]{%
  \immediate\openout\tempfile=keywords.tmp
  \immediate\write\tempfile{\expanded{\BODY}} %<-- This
    ↪ evaluates commands first
  \immediate\closeout\tempfile
  \directlua{add_data_from_file("keywords", "#1", "
    ↪ keywords.tmp")}%
  \par
  \BODY
}

```

```

}

% This command triggers the final JSON write
\newcommand{\GenerateFordivaJSON}{\directlua{write_json_file
  ↪ () }}

```

4.2.3 Two new environments for capturing abstract and keywords

Also shown in Listing 9 are two new environments: ThesisAbstract and ThesisKeywords. The first of these captures the raw $\text{\LaTeX}$ in the environment, while the second first expands the $\text{\LaTeX}$ and then saves it. These environments must be followed by the three-letter ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes.

This leads to a new style of writing abstract, as shown in Listing 10. The first step is to select the language and when using babel this will also potentially select the correct font to use. The second step is to update the term `\abstractname` to use for this abstract (as it could be an abstract or summary). While babel (or polyglossia) often provides definitions of this term, doing it as shown makes it simple for the author to change it if they need to. The whole abstract is in an `abstract` environment. This environment is important for several reasons. (i) entering this environment formats the heading in the Figtree font (or a language-specific font if Figtree does not have the requisite glyphs) and (ii) it enables the macros that warn if a citation or cross-reference has been used. The `\markboth` sets up the page heading. Next the `\begin{ThesisAbstract}{fre}` begins the environment for the thesis abstract. Following this the `\keywordHeading` macro defines the heading to be used for the keywords (in the proper typeface). The ThesisKeywords environment handles the keywords. Finally, a `\cleardoublepage` is done to advance to the next odd-numbered (i.e., right-hand) page. Note that the running heading will also be used on the page (if needed) to get to the next odd-numbered page.

As noted above, the ThesisAbstract and ThesisKeywords environments serve two purposes: to capture content for later use and to output the thesis abstract and keywords in place.

Listing 10: Example of the abstract_french.tex file

```

\selectlanguage{french}
\setlocalecaption{french}{abstract}{Résumé}
\begin{abstract}
  \markboth{\abstractname}{}
\begin{ThesisAbstract}{fre}
Résumé en français.
\end{ThesisAbstract}

\keywordHeading{Mots-clés}
\begin{ThesisKeywords}{fre}
5-6 mots-clés
\end{ThesisKeywords}
\end{abstract}
\cleardoublepage

```

4.2.4 Generating the For DiVA pages

The `\divainfo` command generates the “For DiVA” information pages by putting the metadata into a Lua table and then writing the whole table out as JSON. The `fordiva.json` file is subsequently read back into a listing.

Listing 11 shows how to configure the listings environment that you will put the `fordiva.json` contents into. One of the tricks here is that it was essential to reduce the hyphenation penalty to enable the abstract text to wrap lines nicely on the page. As an added benefit, the $\text{\LaTeX}$ syntax highlighting is on — so one can easily see the $\text{\LaTeX}$ commands that are used — as they might need manual editing before the event is announced or the data imported into DiVA.

Listing 11: Code in the document file to configure the listing environment to put the abstract into

```

\makeatletter
\let\verbatimsc\@undefined
\let\endverbatimsc\@undefined
\lst@AddToHook{Init}{\hyphenpenalty=50\relax}
\makeatother

\lstnewenvironment{verbatimsc}
{
  \lstset{%

```

```

basicstyle=\ttfamily\tiny,
%basicstyle=\tiny,
%columns=fullflexible,
columns=[1]fixed,
language=[LaTeX]TeX,
%numbers=left,
%numberstyle=\tiny\color{gray},
keywordstyle=\color{red},
breaklines=true,           % sets automatic line
    ↪ breaking
breakatwhitespace=true,    % sets if automatic breaks
    ↪ should only happen at whitespace
%keepspaces=false,
breakindent=0em,
%fancyvrb=true
}
}{}

```

4.2.5 Attach the `fordiva.json` file

To facilitate making use of the `fordiva.json` file, this file is attached to the PDF file along with the `acronyms.tex` file. Once the `fordiva.json` and `acronyms.tex` files are attached to the PDF file, you can later extract these attached files using various existing tools or my program: `extract_embedded_files_from_PDF.py`.

4.2.6 Issues due to generating proper JSON

While the above code generates proper JSON, this comes with a cost – the need to use Lua and hence Lua^ATeX. Moreover, to make the handling of fonts easier requires a change to using **only** babel.

A positive benefit of adopting both Lua and babel is that generating the *For DiVA* information is much easier and it should be easier to process the resulting files. An additional benefit is that it makes the code much simpler, especially with respect to dealing with different fonts. However, all is not perfect, as one will get warnings, such as `Font shape 'TU/NotoSerifDevanagari(0)/m/it' undefined`. Thus far, I have not been able to find how to do silent substitutions of the font shapes, as the languages are dynamically loaded when needed.

The above choice means elimination of support for $\text{\LaTeX}$.

There is a need to dig more deeply into the babel and fontspec code to understand when the fonts are dynamically loaded and see whether it is possible to change the font shape for the font being loaded. I suspect that there is probably a hook that one could use.

4.3 Adding `acronyms.tex` to For DIVA info

If the author has used acronyms in the abstract(s), then these acronym definitions need to be available in some way to the person (or program) that makes the MODS file. One solution is to add them at the end of the For DIVA information. This can be done using the code shown in Listing 12. This code simply checks for the file and if it is present, then appends the contents of this file to the text as a listing. An important part to note is the option `nolol` - this prevents this listing from appearing in the list of listings.

Listing 12: Code for adding `acronyms.tex` at the end of the For DIVA data

```
% If there is an acronyms.tex file,
% add it to the end of the For DIVA information
% so that it can be used with the abstracts
% Note that the option "nolol" stops it from being listed
% ↪ in the List of Listings
\IfFileExists{lib/acronyms.tex}{
\section*{acronyms.tex}
\lstinputlisting[language={\LaTeX TeX}, nolol, basicstyle=\
  ↪ ttfamily\color{black},
commentstyle=\color{black}, backgroundcolor=\color{white}]{
  ↪ lib/acronyms.tex}
}
{ }
```

This is **not** used in the current template, the `acronyms.tex` file is simply attached to the PDF file.

4.4 Cleaning up JSON should be *transforming* JSON

If you save the `fordiva.json` and `acronyms.tex` files locally*, then you can clean up and transform the `fordiva.json` file. After having done this, you can make a MODS file with the following commands:

Listing 13: `cleanup_pseudo_JSON-from_LaTeX` and `JSON_to_MODS.p`

```
./cleanup_pseudo_JSON-from_LaTeX.py --json xxx_fordiva.json
↪ --acronyms acronyms.tex
./JSON_to_MODS.py --json xxx_fordiva.json
```

Now all you have to do is rename the Extensible Markup Language (XML) file that was produced to `xxx.mods` and you are all set to upload it into DiVA!

In my earlier code and writing, I have referred to cleaning up the pseudo-JSON or the actual JSON. However, this is not really the correct approach to take. It would be better to talk about a transformation from JSON that contains $\text{\LaTeX}$ to JSON that contains Hypertext Mark-up Language (HTML). Doing this once would be better than putting this work into each of the different programs that take in JSON to produce calendar entries, MODS, or Cora-DiVA API calls. It will also be interesting because one of the data formats that the Cora-DiVA API calls can accept is JSON. Logically, this transformation could take place in the magic “Extractor” that was shown in Figure 1 on page 2.

This transformation process has to transform the equations. While my various programs attempt to handle some simple equations, a more comprehensive and general solution is needed. At present, it is unknown if the Cora-based DiVA will support MathJax or something similar. If it were to support MathJax, then the solution is rather simple – just pass the $\text{\LaTeX}$ equation on and let MathJax render it in the browser when someone views it.

The programs have **not** yet been updated to move the common transformation code into an explicit JSON transformer. Moreover, doing so would simplify the maintenance of these programs, as the transformations would only need to take place in one program or perhaps in one library.

*Either by extracting them from the PDF file or you can download the `acronyms.tex` file from your project and can download the `fordiva.json` with the “Other logs and files” button shown at the bottom of the log output window.

4.5 Generating plain text titles and subtitles

Many administrative systems, such as LADOK, getting an ISBN, *etc.*, require plain Unicode text in UTF-8 encoding *i.e.*, *without* markup. Therefore, I have developed some Lua code to enable Lua_AT_EX to help the author with this transformation.

Compiling `Generate_titles_in_plain_text.tex` will generate a file named `extra_plaintext.tex` containing plain text versions of the titles and subtitles (both main and alternate). (It is named this to avoid overwriting your existing `custom_configuration_plaintext.tex` file during the process.) Additionally, the plain text is set *without* ligatures.

There is another Overleaf project that provides more details on this and gives examples of additional testing; see <https://www.overleaf.com/read/dfsjdqmcjmtp#0043d3>.

I believe that this code can handle a large fraction of all of the titles and subtitles that have previously been used at KTH.

5 What can we do with the JSON file?

Once you are happy with the JSON file's contents, the next step is to generate something interesting with this data — for example, we can use the program `JSON_to_calendar.py` to create a calendar entry.

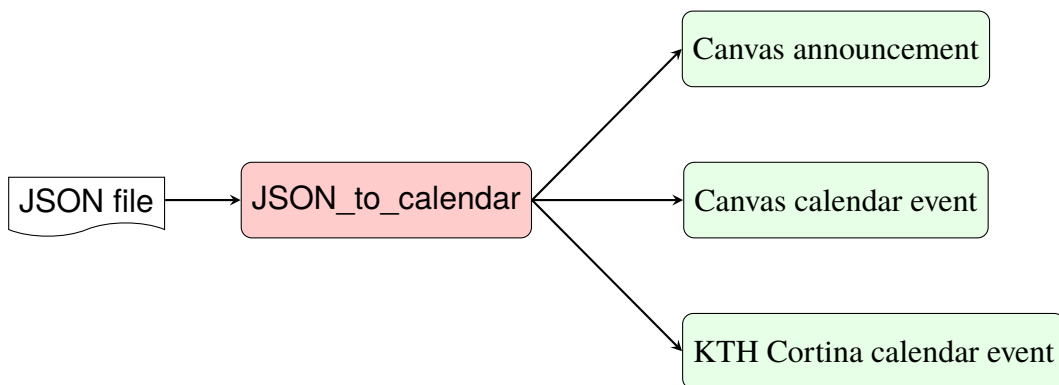


Figure 2: The several outputs of `JSON_to_calendar.py` given a JSON file

We can produce all three outputs with the command shown in Listing 14. Note that the program was run with the `event.json` file (a JSON file containing HTML) to produce a Canvas course announcement.

Listing 14: Running JSON_to_calendar.py to produce all three outputs

```
JSON_to_calendar.py -c 11 --config config-test.json
```

Note that in this case, the `config-test.json` configuration file was used to make use of the **test** Canvas course room instance as well as the **test** Cortina instance. To use this with the production instances, you would use your usual `config.json` configuration file. In either case, your configuration file needs to include a KTH API key to access Cortina.

6 Other variants of JSON_to_calendar.py

For testing purposes, I also created functionality in `JSON_to_calendar.py` to insert a fixed event (this was my first test) and to take as input a MODS file. The MODS file was created from a DiVA feed of theses presented in 2020 through to the 25th of April 2021. However, one limitation that I found is that, other than myself, few people have been entering the date and time for the oral presentation. Unfortunately, since I wanted to test making calendar announcements, I needed the date and time!

NB: I have assumed that each degree project presentation lasts one hour. This assumption was made since the KTH Cortina calendar API needs both a starting and ending time. There is currently no support for specifying the duration of third-cycle presentations. However, looking at a calendar event for a doctoral defense on 2025-06-05, only the starting time appears in the event.

These other variants probably should **not** be kept, but rather the architecture should be similar to that shown in Figure 3. Additionally, when taking data from other types of sources, one can take advantage of the data that is in a Canvas degree project course room to help when processing the source data.

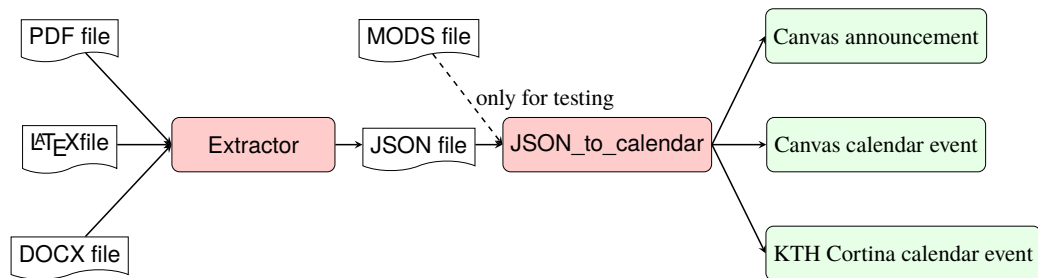


Figure 3: Several possible inputs to JSON_to_calendar.py and its outputs

7 Request an ISBN

The web page at [Create an ISBN number](#) can be used to request an International Standard Book Number (ISBN).

This form requires the TRITA series and number, as well *(i)*, as whether the request is for a Doctoral/licentiate thesis or Report, *(ii)* the title of publication, and *(iii)* first authors': (a) last name, (b) first name, (c) KTH id, and (d) e-mail address. Fortunately, all of this information should be available in the `fordiva.json` file, so it should be possible to use this to request assignment of an ISBN.

As all the required information is in the `fordiva.json` file, the program `request-ISBN-with-JSON.py` at <https://github.com/gqmagui rejr/E-learning> can be used to automate the process of making a request for an ISBN. The program generates a `assigned-ISBN.tex` file containing the assigned ISBN. This file can be uploaded to the project and will automatically be used rather than the placeholder in `custom_configuration.tex`. It can be run as shown in Listing 15.

Listing 15: Example of using `request-ISBN-with-JSON.py`

```
\texttt{./request-ISBN-with-JSON.py --json /tmp/fordiva.  
↪ json}
```

NB This program should probably be changed to use the plain text version of the title and subtitle (if they are available).

8 JSON to MODS file

In keeping with the idea of using the JSON file to drive other applications, the program `JSON_to_MODS.py` creates a MODS file using the information from the arguments and a JSON file. The program is patterned after the `JSON_to_calendar` program. The program outputs a MODS file called: `modsXML.xml`. The program is run as shown in Listing 16.

Listing 16: Example of using `JSON_to_MODS.py`

```
./JSON_to_MODS.py -c 11 --json test12.json --trita "TRITA-  
↪ EECS-EX-2021:219" --testing
```

Currently, the Canvas course information is **not** used. Note also that the “—testing” flag forces the report series to be “TRITA-ICT-EX” overriding the

actual series that is in the input version of the DiVA data. This is needed for testing with the test instance of DiVA – as the test instance only has the old series names.

If acronyms are used in the abstracts, you can also add the “`--acronyms acronyms.tex`” argument to the command line and the program will process the acronyms.

As noted above, you will get a file named `modsXML.xml` that you can manually import into DiVA. This process is shown in Section 8.2. I seem to be able to import these files into the test instance of DiVA: <https://kth.test.diva-portal.org/dream/import/importList.jsf>. Another side-effect of the testing flag is to use the test instance of DiVA. **NB:** The organization units are different between the production and test versions of DiVA. When run without the testing flag set, the MODS file can be imported into the production version of DiVA.

This program has been extended to take the TRITA number information from the MODS file.

8.1 Example of JSON to MODS

The process of importing the MODS data into DiVA begins with the extracted JSON file (note that I have changed some of the formatting of the pseudo-JSON information and added the export of the program code). An example of the JSON information is shown in Listing 17. The resulting MODS file is shown in Listing 18. [Note that there are errors in the order of unicode characters due to the use of `lstlisting`.]

Listing 17: test12.json (manually reformatted)

```
{ "Author1": { "Last_name": "Student", "First_name": "Fake_
  ↪ A.", "Local_User_Id": "u100001", "E-mail":
  ↪ "a@kth.se", "organisation": { "L1": "School_of_
  ↪ Electrical_Engineering_and_Computer_Science" } },
"Author2": { "Last_name": "Student", "First_name": "Fake_
  ↪ B.", "Local_User_Id": "u100002", "E-mail":
  ↪ "b@kth.se", "organisation": { "L1": "School_of_
  ↪ Architecture_and_the_Built_Environment" } },
"Degree": { "Educational_program": "'Bachelors_Programme_in_
  ↪ Information_and_Communication_Technology",
  ↪ "programcode": "TCOMK", "Level": "1", "Course_code":
  ↪ "IA150X", "Credits": "15.0", "Exam": "Bachelors_
```



```

    ↪ degree", "subjectArea": "Information_and_
    ↪ Communication_Technology"},
  "Title": {"Main_title": "This_is_the_title_in_the_language_
    ↪ of_the_thesis", "Subtitle": "An_subtitle_in_the_
    ↪ language_of_the_thesis", "Language": "eng"},
  "Alternative_title": {"Main_title": "Detta_är_den_svenska_
    ↪ öáversttningen_av_titeln", "Subtitle": "Detta_är_den_
    ↪ svenska_öáversttningen_av_undertiteln", "Language":
    ↪ "swe"},
  "Supervisor1": {"Last_name": "Supervisor", "First_name":
    ↪ "A._Busy", "Local_User_Id": "u100003", "E-mail":
    ↪ "sa@kth.se", "organisation": {"L1": "School_of_
    ↪ Electrical_Engineering_and_Computer_Science", "L2":
    ↪ "Computer_Science"}},
  "Supervisor2": {"Last_name": "Supervisor", "First_name":
    ↪ "Another_Busy", "Local_User_Id": "u100003", "E-mail":
    ↪ "sb@kth.se", "organisation": {"L1": "School_of_
    ↪ Architecture_and_the_Built_Environment", "L2":
    ↪ "Public_Buildings"}},
  "Supervisor3": {"Last_name": "Supervisor", "First_name":
    ↪ "Third_Busy", "E-mail": "sc@tu.va", "Other_
    ↪ organisation": "Timbuktu_University,_Department_of_
    ↪ Pseudoscience"},
  "Examiner1": {"Last_name": "Maguire_Jr.", "First_name":
    ↪ "Gerald_Q.", "Local_User_Id": "u1d13i2c", "E-mail":
    ↪ "maguire@kth.se", "organisation": {"L1": "School_of_
    ↪ Electrical_Engineering_and_Computer_Science", "L2":
    ↪ "Computer_Science"}},
  "Cooperation": {"Partner_name": "öFretaget_AB", "National_
    ↪ Subject_Categories": "10201,10206", "Other_
    ↪ information": {"Year": "2021", "Number_of_pages":
    ↪ "xxxiii,35"}, "Opponents": {"Name": "A._B._Normal_&_
    ↪ A._X._E._èNormal"}, "Presentation": {"Date":
    ↪ "2021-03-15_13:00", "Language": "eng", "Room": "via_
    ↪ Zoom_https://kth-se.zoom.us/j/ddddddddddd",
    ↪ "Address": "Isafjordsgatan_22_(åKistagngen_16)",
    ↪ "City": "Stockholm"},
  "Number_of_lang_instances": "10", "abstracts": {"eng":
    ↪ "<p>All_theses_at_KTH_are<b>required</b>_to_
    ↪ have_an_abstract_in_both_<i>English</i>_and_

```

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```

    ↪ <i>Swedish</i>.</p><p>Exchange_students_many_want_to
    ↪ include_one_or_more_abstracts_in_the_language(s)_used
    ↪ in_their_home_institutions_to_avoid_the_need_to_write
    ↪ another_thesis_when_returning_to_their_home
    ↪ institution.</p><p>Keep_in_mind_that_most_of_your
    ↪ potential_readers_are_only_going_to_read_your
    ↪ <tt>title</tt>_and_<tt>abstract</tt>._This_is_why_it
    ↪ is_important_that_the_abstract_give_them_enough
    ↪ information_that_they_can_decide_is_this_document
    ↪ relevant_to_them_or_not._Otherwise_the_likely_default
    ↪ choice_is_to_ignore_the_rest_of_your_document.</p><p>A_abstract
    ↪ should_stand_on_its_own,_i.e.,_no_citations,_cross_references
    ↪ to_the_body_of_the_document,_acronyms_must_be_spelled_out,_
    ↪ ...</p><p>Write_this_early_and_revise_as_necessary._This
    ↪ will_help_keep_you_focused_on_what_you_are_trying_to_do.</p><p>Write
    ↪ an_abstract_with_the_following_components:</p><ul><li>
    ↪ What_is_the_topic_area?(optional)_Introduces_the_subject
    ↪ area_for_the_project.</li><li>Short_problem_statement
    ↪ </li><li>Why_was_this_problem_worth_a_'Bachelors'/Masters
    ↪ thesis_project?(i.e.,_why_is_the_problem_both_significant
    ↪ and_of_a_suitable_degree_of_difficulty_for_a_'Bachelors'/Masters
    ↪ thesis_project?_Why_has_no_one_else_solved_it_yet?)</li><li>
    ↪ How_did_you_solve_the_problem?_What_was_your_method/insight?
    ↪ </li><li>Results/Conclusions/Consequences/Impact:_What
    ↪ are_your_key_results/conclusions?_What_will_others_do_based
    ↪ upon_your_results?_What_can_be_done_now_that_you_have_finished
    ↪ _that_could_not_be_done_before_your_thesis_project_was
    ↪ completed?</li></ul><p>announcement_of_the_oral_presentation
    ↪ and_for_entering_data_into_DiVA</p><p>\pi_\cdot r$ or
    ↪ \[_\int_{a}^{b}x^2\,dx</p><p></p><p>A<sup>*</sup>,_
    ↪ A<sup>&reg;</sup>,_and_A&trade;.</p><p>first_chemical_formula
    ↪ can_be_handled,_while_the_second_will_require_hand_editing</p>","
"swe" : "<p>Alla_avhandlingar_vid_KTH_<b>ämste_ha</b>
    ↪ ett_abstrakt_åp_åbde_<i>engelska</i>_och
    ↪ <i>svenska</i>.</p><p>Om_du_skriver_din_avhandling_åp
    ↪ svenska_ska_detta_ögras_öfrst_(och_placera_det_som_det
    ↪ öfrsta_abstraktet)_och_du_öbr_revidera_det_vid
    ↪ behov.</p><p>If_you_are_writing_your_thesis_in
    ↪ English,_you_can_leave_this_until_the_draft_version
    ↪ that_goes_to_your_opponent_for_the_written_opposition.</p>

```

```

    ↪ In this way you can provide the English and Swedish
    ↪ abstract/summary information that can be used in the
    ↪ announcement for your oral presentation.</p><p>If you
    ↪ are writing your thesis in English, then this section
    ↪ can be a summary targeted at a more general reader.
    ↪ However, if you are writing your thesis in Swedish, then
    ↪ the reverse is true - your abstract should be for your target
    ↪ audience, while an English summary can be written targeted
    ↪ at a more general audience.</p><p>This means that the English
    ↪ abstract and Swedish sammanfattning or Swedish abstract
    ↪ and English summary need not be literal translations of
    ↪ each other.</p><p>The abstract in the language used for
    ↪ the thesis should be the first abstract, while the Summary/Sammanfattning
    ↪ in the other language can follow.</p>",
  "fre": "<p>ééRsum en çfranais.</p>", "spa": "<p>ééRsum en
    ↪ español.</p>", "ita": "<p>Sommario in italiano.</p>",
  "nor": "<p>Sammendrag på norsk.</p>", "ger": "", "dan":
    ↪ "<p>Abstrakt på dansk.</p>",
  "dut": "<p>Zusammenfassung in Deutsch.</p><p>Samenvatting
    ↪ in het Nederlands.</p><p>Eesti keeles
    ↪ ökokkuvte.</p>", "est": "" },
  "keywords": { "eng": "Canvas Learning Management System,
    ↪ Docker containers, Performance tuning",
  "swe": "Canvas äLrplattform, åDockerbehllare,
    ↪ Prestandajustering\nNyckelord som beskriver
    ↪ åinnehllet i uppsatsrapporten\n",
  "fre": "5-6 mots-écls", "spa": "5-6 Palabras claves",
    ↪ "ita": "5-6 parole chiave",
  "nor": "5-6 ønkkelord", "ger": "5-6 üöSchlsselwrter",
    ↪ "dan": "5-6 øSgeord",
  "dut": "5-6 trefwoorden", "est": "5-6 äöMrksnad" } }

```

Listing 18: test12.mods (reformatted in EMACS using XML mode)

```

<modsCollection xmlns="http://www.loc.gov/mods/v3"
  ↪ xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  ↪ xsi:schemaLocation="http://www.loc.gov/mods/v3
  ↪ http://www.loc.gov/standards/mods/v3/mods-3-2.xsd">
  <mods xmlns="http://www.loc.gov/mods/v3"
    ↪ xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    ↪ xmlns:xlink="http://www.w3.org/1999/xlink"
    ↪ version="3.2"
    ↪ xsi:schemaLocation="http://www.loc.gov/mods/v3
    ↪ http://www.loc.gov/standards/mods/v3/mods-3-2.xsd">
    <genre authority="diva"
      ↪ type="publicationTypeCode">studentThesis
    </genre>
    <genre authority="diva" type="publicationType"
      ↪ lang="swe">Studentuppsats (Examensarbete)
    </genre>
    <genre authority="diva" type="publicationType"
      ↪ lang="eng">Studentthesis
    </genre>
    <genre authority="diva" type="publicationType"
      ↪ lang="nor">Oppgave
    </genre>
    <name type="personal" authority="kth"
      ↪ xlink:href="u100001">
      <namePart type="family">Student
    </namePart>
      <namePart type="given">Fake A.
    </namePart>
      <description>email=a@kth.se
    </description>
      <role>
    <roleTerm type="code" authority="marcrelator">aut
    </roleTerm>
      </role>
    </name>
    <name type="personal" authority="kth"
      ↪ xlink:href="u100002">
      <namePart type="family">Student
    </namePart>

```

```

<namePart type="given">Fake_B.
</namePart>
<description>email=b@kth.se
</description>
<role>
  <roleTerm type="code" authority="marcrelator">aut
</roleTerm>
</role>
</name>
<name type="personal" authority="kth"
  ↪ xlink:href="u1d13i2c">
  <namePart type="family">Maguire_Jr.
  </namePart>
  <namePart type="given">Gerald_Q.
  </namePart>
  <description>email=maguire@kth.se
  </description>
  <affiliation>KTH, School of Electrical Engineering and
  ↪ Computer Science, Computer Science
  </affiliation>
  <role>
    <roleTerm type="code" authority="marcrelator">mon
  </roleTerm>
  </role>
</name>
<name type="personal" authority="kth"
  ↪ xlink:href="u100003">
  <namePart type="family">Supervisor
  </namePart>
  <namePart type="given">A_Busy
  </namePart>
  <affiliation>School of Electrical Engineering and
  ↪ Computer Science, Computer Science
  </affiliation>
  <description>email=sa@kth.se
  </description>
  <role>
    <roleTerm type="code" authority="marcrelator">ths
  </roleTerm>
  </role>

```

```

      </name>
      <name type="personal" authority="kth"
        ↪ xlink:href="u100003">
        <namePart type="family">Supervisor
        </namePart>
        <namePart type="given">Another_Busy
        </namePart>
        <affiliation>School_of_Architecture_and_the_Built_
        ↪ Environment,_Public_Buildings
        </affiliation>
        <description>email=sb@kth.se
        </description>
        <role>
        <roleTerm type="code" authority="marcrelator">ths
        </roleTerm>
        </role>
      </name>
      <name type="personal">
        <namePart type="family">Supervisor
        </namePart>
        <namePart type="given">Third_Busy
        </namePart>
        <affiliation>Timbuktu_University,_Department_of_
        ↪ Pseudoscience
        </affiliation>
        <description>email=sc@tu.va
        </description>
        <role>
        <roleTerm type="code" authority="marcrelator">ths
        </roleTerm>
        </role>
      </name>
      <name>
        <namePart>KTH
        </namePart>
        <namePart>_School_of_Electrical_Engineering_and_
        ↪ Computer_Science
        </namePart>
        <namePart>_Computer_Science
        </namePart>

```

```

<role>
  <roleTerm type="code" authority="marcrelator">pbl
</roleTerm>
</role>
</name>
<titleInfo lang="eng">
  <title>This is the title in the language of the thesis
</title>
  <subTitle>An subtitle in the language of the thesis
</subTitle>
</titleInfo >
  <titleInfo lang="swe" type="alternative">
    <title>Detta är den svenska översättningen av titeln
    </title>
    <subTitle>Detta är den svenska översättningen av
      ↳ undertiteln
    </subTitle>
  </titleInfo >
  <subject lang="eng">
    <topic>Canvas Learning Management System
    </topic>
    <topic>Docker containers
    </topic>
    <topic>Performance tuning
    </topic>
  </subject >
  <subject lang="swe">
    <topic>Canvas Lärplattform
    </topic>
    <topic>Dockerbehållare
    </topic>
    <topic>Prestandajustering Nyckelord som beskriver
      ↳ innehållet i uppsatsrapporten
    </topic>
  </subject >
  <subject lang="fre">
    <topic>5-6 mots-clés
    </topic>
  </subject >
  <subject lang="spa">

```

```

<topic>5-6_Palabras_claves
</topic>
</subject >
<subject lang="ita">
<topic>5-6_parole_chiave
</topic>
</subject >
<subject lang="nor">
<topic>5-6_nøkkelord
</topic>
</subject >
<subject lang="ger">
<topic>5-6_Schlüsselwörter
</topic>
</subject >
<subject lang="dan">
<topic>5-6_Søgeord
</topic>
</subject >
<subject lang="dut">
<topic>5-6_trefwoorden
</topic>
</subject >
<subject lang="est">
<topic>5-6_Märksõnad
</topic>
</subject >
<abstract lang="eng">&lt;p&gt;All theses at KTH are
  &lt;b&gt;required&lt;/b&gt; to have an abstract
  in both &lt;i&gt;English&lt;/i&gt; and
  &lt;i&gt;Swedish&lt;/i&gt;. &lt;p&gt;&lt;p&gt;Exchange
  students many want to include one or more abstracts in
  the language(s) used in their home institutions to
  avoid the need to write another thesis when returning
  to their home institution. &lt;p&gt;&lt;p&gt;Keep in
  mind that most of your potential readers are only
  going to read your &lt;tt&gt;title&lt;/tt&gt; and
  &lt;tt&gt;abstract&lt;/tt&gt;. This is why it is
  important that the abstract give them enough
  information that they can decide is this document

```



```

    ↪ relevant to them or not. Otherwise the likely default
    ↪ choice is to ignore the rest of your document. </p><p>A
    ↪ abstract should stand on its own, i.e., no citations, cross
    ↪ references to the body of the document, acronyms must be
    ↪ spelled out, ... </p><p>Write this early and
    ↪ revise as necessary. This will help keep you focused on
    ↪ what you are trying to do. </p><p>Write an abstract
    ↪ with the following components: </p><ul><li>
    ↪ What is the topic area? (optional) Introduces the subject
    ↪ area for the project. </li><li> Short problem
    ↪ statement </li><li> Why was this problem worth
    ↪ a 'Bachelors' / Masters thesis project? (i.e., why is the
    ↪ problem both significant and of a suitable degree of difficulty
    ↪ for a 'Bachelors' / Masters thesis project? Why has no one
    ↪ else solved it yet?) </li><li> How did you
    ↪ solve the problem? What was your method/insight? </li><li>
    ↪ Results/Conclusions/Consequences/Impact: What are your
    ↪ key results/conclusions? What will others do based upon
    ↪ your results? What can be done now that you have finished
    ↪ - that could not be done before your thesis project was
    ↪ completed? </li></ul><p>announcement of
    ↪ the oral presentation and for entering data into DiVA </p><p>\pi
    ↪ \cdot r$ or <span class='math-tex'>\int_{a}^{b}</span>
    ↪ x^2\,dx\]</span></p><p>A<sup></sup>A<sup></sup>&reg;</sup>,
    ↪ and A&trade; .</p><p>first chemical formula
    ↪ can be handled, while the second will require hand editing</p>
</abstract >
<abstract lang="swe"><p>Alla avhandlingar vid KTH
    ↪ <b>måste ha</b> ett abstrakt på
    ↪ både <i>engelska</i> och
    ↪ <i>svenska</i> .</p><p>Om du
    ↪ skriver din avhandling på svenska ska detta göras
    ↪ först (och placera det som det första abstraktet) -
    ↪ och du bör revidera det vid
    ↪ behov. </p><p>If you are writing your
    ↪ thesis in English, you can leave this until the draft
    ↪ version that goes to your opponent for the written
    ↪ opposition. In this way you can provide the English
    ↪ and Swedish abstract/summary information that can be

```

```

    ↪ used in the announcement for your oral
    ↪ presentation. </p><p> If you are writing your
    ↪ thesis in English, then this section can be a summary targeted
    ↪ at a more general reader. However, if you are writing your
    ↪ thesis in Swedish, then the reverse is true - your abstract
    ↪ should be for your target audience, while an English summary
    ↪ can be written targeted at a more general audience. </p><p>
    ↪ means that the English abstract and Swedish sammnfattning
    ↪ or Swedish abstract and English summary need not be literal
    ↪ translations of each other. </p><p> The abstract
    ↪ in the language used for the thesis should be the first abstract,
    ↪ while the Summary/Sammanfattning in the other language
    ↪ can follow. </p>
  </abstract >
  <abstract lang="fre"><p>Résumé en
    ↪ français. </p>
  </abstract >
  <abstract lang="spa"><p>Résumé en
    ↪ español. </p>
  </abstract >
  <abstract lang="ita"><p>Sommario in
    ↪ italiano. </p>
  </abstract >
  <abstract lang="nor"><p>Sammendrag på
    ↪ norsk. </p>
  </abstract >
  <abstract lang="ger" />
  <abstract lang="dan"><p>Abstrakt på
    ↪ dansk. </p>
  </abstract >
  <abstract lang="dut"><p>Zusammenfassung in
    ↪ Deutsch. </p><p> Samenvatting in het
    ↪ Nederlands. </p><p> Eesti keeles
    ↪ kokkuvõte. </p>
  </abstract >
  <abstract lang="est" />
  <physicalDescription>
    <form authority="marcform">electronic
  </form>
  <extent>xxxiii, 35

```

```

</extent>
</physicalDescription>
<originInfo>
  <place>
    <placeTerm>Stockholm
  </placeTerm>
  </place>
  <publisher>KTH_Royal_Institute_of_Technology
</publisher>
  <dateIssued>2021
</dateIssued>
  <dateOther type="defence">2021-03-15T13:00:00
</dateOther>
</originInfo>
  <typeOfResource>text
</typeOfResource>
  <relatedItem type="series">
    <titleInfo>
      <title>TRITA-ICT-EX
    </title>
    <identifier type="local">5952
    </identifier>
    <identifier type="issue_number">2021:219
    </identifier>
  </titleInfo>
</relatedItem >
  <note lang="swe" type="level">Självständigt_arbete_på_
    ↪ grundnivå_(kandidatexamen)
  </note>
  <note lang="eng" type="degree">Bachelors_degree
  </note>
  <note lang="swe" type="universityCredits">10_poäng_/15_
    ↪ hp
  </note>
  <subject lang="swe" xlink:href="10329">
    <topic>Informations-_och_kommunikationsteknik
  </topic>
  <genre>Subject/course
  </genre>
</subject>

```

```

<subject lang="eng" xlink:href="10329">
  <topic>Information and Communication Technology
</topic>
  <genre>Subject/course
</genre>
</subject>
<subject lang="swe" xlink:href="9925">
  <topic>Teknologie kandidatexamen - Informations- och
    ↳ kommunikationsteknik
  </topic>
  <genre>Educational program
</genre>
</subject>
<subject lang="eng" xlink:href="9925">
  <topic>Bachelor of Science - Information and
    ↳ Communication Technology
  </topic>
  <genre>Educational program
</genre>
</subject>
<language objectPart="defence">
  <languageTerm type="code" authority="iso639-2b">eng
</languageTerm>
</language>
<note type="venue">via Zoom
  ↳ https://kth-se.zoom.us/j/ddddddddddd, Isafjordsgatan
  ↳ 22 (Kistagången 16), Stockholm
</note>
<note type="cooperation">Företaget AB
</note>
<subject lang="eng" authority="hsv" xlink:href="10201">
  <topic>Natural Sciences
</topic>
  <topic>Computer and Information Sciences
</topic>
  <topic>Computer Sciences
</topic>
</subject>
<subject lang="swe" authority="hsv" xlink:href="10201">
  <topic>Naturvetenskap

```

```

</topic>
<topic>Data- och informationsvetenskap
</topic>
<topic>Datavetenskap (datalogi)
</topic>
</subject>
<subject lang="eng" authority="hsv" xlink:href="10206">
  <topic>Natural Sciences
  </topic>
  <topic>Computer and Information Sciences
  </topic>
  <topic>Computer Engineering
  </topic>
</subject>
<subject lang="swe" authority="hsv" xlink:href="10206">
  <topic>Naturvetenskap
  </topic>
  <topic>Data- och informationsvetenskap
  </topic>
  <topic>Datorteknik
  </topic>
</subject>
</mods>
</modsCollection>

```

8.2 Importing MODS file to DiVA

Figure 4 to Figure 7 show the process of importing the MODS file into DiVA. Note that it skips the first user interface form (as this differs between users).

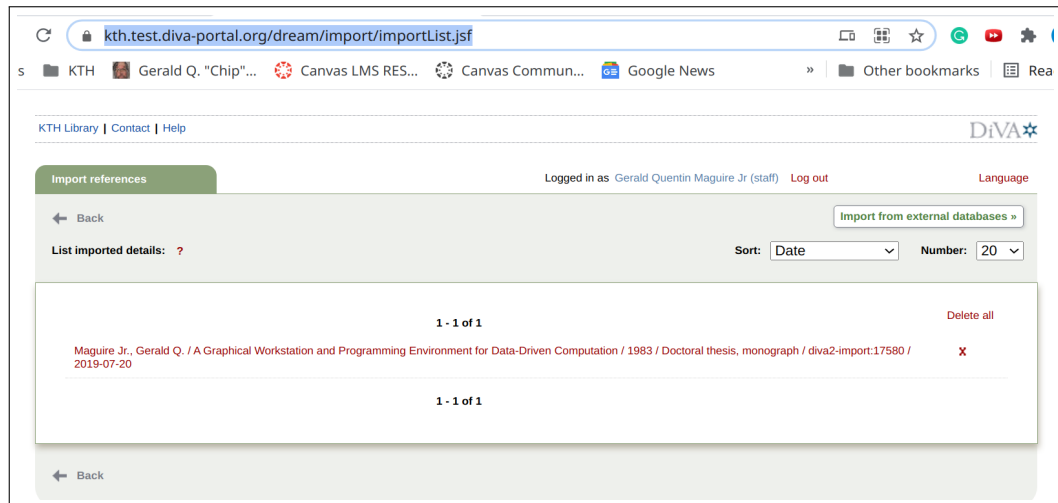


Figure 4: The import process – step 1 – click on the “Import from external database” button on the upper right corner

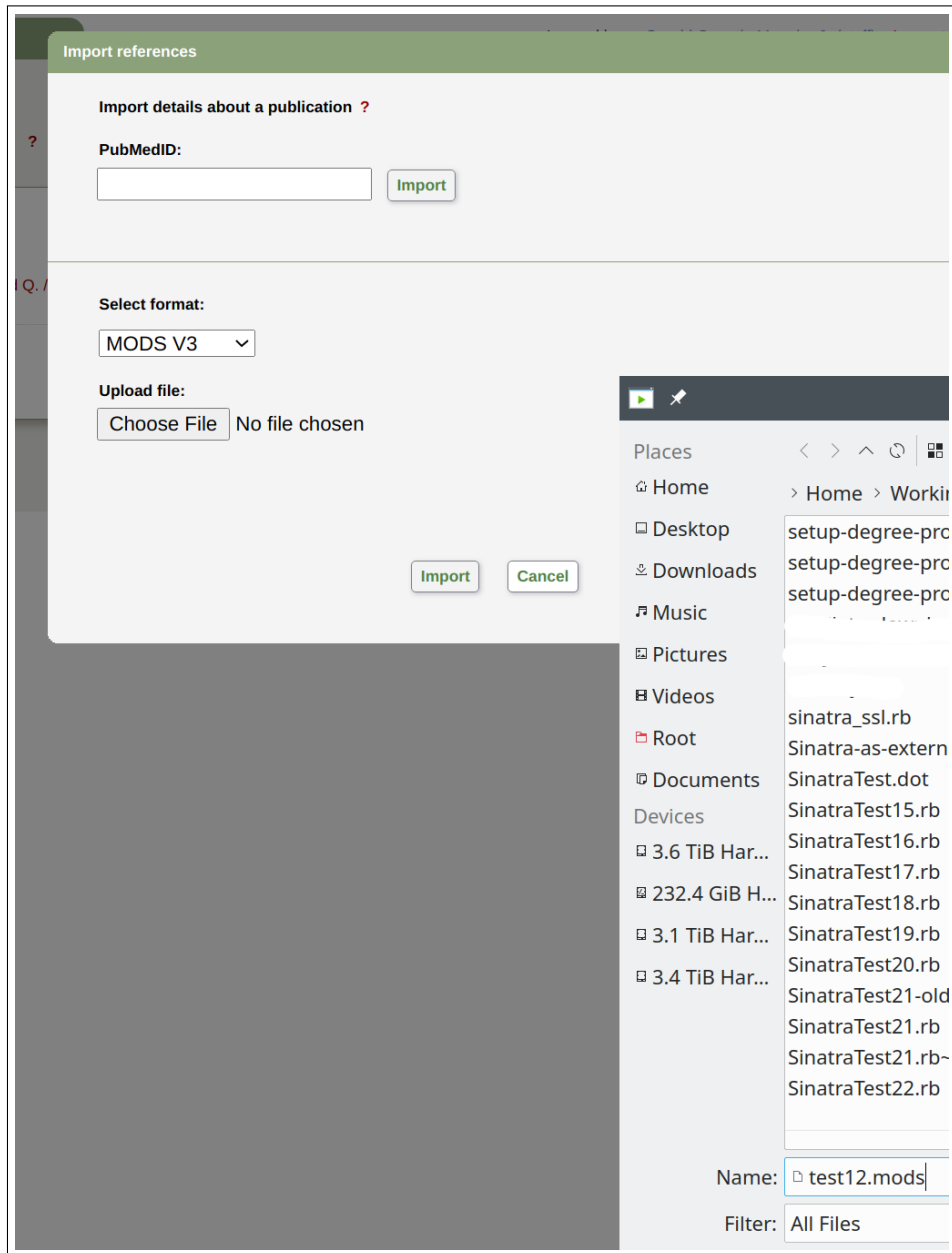


Figure 5: The import process – step 2 – choose the MODV3 format and select a file to import

Import references

Import details about a publication ?

PubMedID:

Select format:
MODS V3

Upload file:
 No file chosen

File upload successful. To complete the import click on the Import button.

Figure 6: The import process – step 3 – after clicking “Import” – it says that the file was successfully uploaded Click on “Import” again to load the file. The MODS file is loaded and shown in Figure 7.

Import references

Logged in as Gerald Quentin Maguire Jr (staff) [Log out](#) [Language](#)

[← Back](#) [Import from external databases »](#)

List imported details: ? Sort: Number:

1 - 2 of 2		Delete all
Student, Fake A.; et al. / This is the title in the language of the thesis: An subtitle in the language of the thesis / 2021 / Student thesis / diva2-import:28212 / 2021-06-26		X
Maguire Jr., Gerald Q. / A Graphical Workstation and Programming Environment for Data-Driven Computation / 1983 / Doctoral thesis, monograph / diva2-import:17580 / 2019-07-20		X

Figure 7: The import process – step 4 – after clicking “Import” once more – you can see the file has been added to the list of imported files The imported document is shown in Figure 8 to Figure 15.

Import references

Identifiers

Logged in as Gerald Quentin Maguire Jr (staff) Log out

Language

Enter information —>

Upload files —>

Review / Publish —>

Back

Cancel / Save changes

Continue →

Selected publication type: Student thesis

Change type »

Author 1 ?

Connect authority record »

Create authority record »

Get saved personal data »

Save personal data »

Last name: *

Student

Year of birth:

First name: *

Fake A.

Local User Id:

u100001

Department, unit or programme:

Choose organisation »

Research group:

E-mail:

a@kth.se

Other organisation:

↓

Author 2 ?

Connect authority record »

Create authority record »

Get saved personal data »

Save personal data »

Last name: *

Student

Year of birth:

First name: *

Fake B.

Local User Id:

u100002

Department, unit or programme:

Choose organisation »

Research group:

E-mail:

b@kth.se

Other organisation:

Another author »

↑

Cooperation ?

External cooperation

☒ Yes
 ☐ No

Partner: *

Add partner »

Title ?

Main title: *

I

×₂

×²

Ω

<>

↔

↔

↔

↔

↔

↔

This is the title in the language of the thesis

Figure 8: Imported document (part 1)

Title ?

Main title: *

I

$\times_2$

$\times^2$

Ω

<>

Aa

This is the title in the language of the thesis

Subtitle:

I

$\times_2$

$\times^2$

Ω

<>

Aa

An subtitle in the language of the thesis

Language: *

English

Alternative title ?

Main title:

I

$\times_2$

$\times^2$

Ω

<>

Aa

Detta är den svenska översättningen av titeln

Subtitle:

I

$\times_2$

$\times^2$

Ω

<>

Aa

Detta är den svenska översättningen av undertiteln

Language:

Swedish

Degree ?

Level: *

Independent thesis Basic level (degree of Bachelor)

University credits: *

15 HE credits

Educational program:

Bachelor of Science - Information and Communication Techn

Subject / course:

Information and Communication Technology

Another degree =>

Content category ?

☐ Artistic work

Other information ?

Year: *

2021

Number of pages:

xxxiii,35

Series ?

Title of series/ISSN:

*

Figure 9: Imported document (part 2) – Note that both the English and Swedish titles are shown as well as the information about which level, the number of university credits, the educational program, and the subject.- Additionally, the year and number of pages are shown.

Title of series: TRITA-ICT-EX ISSN: EISSN: No. in series:
Identifiers ? URI: urn:nbn:se:kth:diva-213041 DIVA-ID: diva2-import:28212 ISRN: DOI: ☐ Free full-text URL: URL label: ☐ Free full-text Another URL »
National subject category * ? Suggestion for national subject category Choose national subject category » Datavetenskap (datalogi); Computer Science (10201) x Dator teknik; Computer Engineering (10206) x
Part of project ? If this publication is part of a project registered in DIVA, create a link by searching for the project below. Search and add project »
Part of other project ? Another other project »
Keywords 1 ? 5-6 Palabras claves Language: Spanish
Keywords 2 ? 5-6 Märksönad Language: Estonian
Keywords 3 ? 5-6 trefwoorden Language: Dutch; Flemish

Figure 10: Imported document (part 3) – Note that the number in the series is not being shown. The nation subject catergies specified in the JSON file are shown. Additionally, a number of the keyoards in different languages are shown.

Keywords 4 ?

5-6 mots-clés

Language:

French

Keywords 5 ?

Canvas Lärplattform, Dockerbehållare, Prestandajustering Nyckelord som beskriver innehåll

Language:

Swedish

Keywords 6 ?

5-6 Søgeord

Language:

Danish

Keywords 7 ?

Canvas Learning Management System, Docker containers, Performance tuning

Language:

English

Keywords 8 ?

5-6 parole chiave

Language:

Italian

Keywords 9 ?

5-6 Schlüsselwörter

Language:

German

Keywords 10 ?

5-6 nøkkelord

Language:

Norwegian

Keywords in another language »

Abstract 1 ?

B

I

$\times_2$

$\times^2$

☰

☷

Avsnitt

☰

☷

⋮

All theses at KTH are required to have an abstract in both English and Swedish.

Exchange students many want to include one or more abstracts in the language(s) used in their home institutions to avoid the need to write another thesis when returning to their home institution.

Keep in mind that most of your potential readers are only going to read your title and abstract. This is why it is important that the abstract give them enough information that they can decide is this document relevant to them or not. Otherwise the likely default choice is to ignore the rest of your document.

A abstract should stand on its own, i.e., no citations, cross references to the body of the document, acronyms must be spelled out, ...

Write this early and revise as necessary. This will help keep you focused on what you are trying to do.

Write an abstract with the following components:

- What is the topic area? (optional) Introduces the subject area for the project.
- Short problem statement

Figure 11: Imported document (part 4) – more keywords are shown as well as the abstract in English

Keywords in another language »

Abstract 1 ?

B I $\times_2$ $\times^2$ $\vdots$ $\vdots$ Avsnitt   ...

- Short problem statement
- Why was this problem worth a Bachelor's/Masters thesis project? (i.e., why is the problem both significant and of a suitable degree of difficulty for a Bachelor's/Masters thesis project? Why has no one else solved it yet?)
- How did you solve the problem? What was your method/insight?
- Results/Conclusions/Consequences/Impact: What are your key results/conclusions? What will others do based upon your results? What can be done now that you have finished - that could not be done before your thesis project was completed?

announcement of the oral presentation and for entering data into DiVA

$\pi \cdot r$ or $\int \sin(a)(b) x^2 dx$

A^{*}, A[®], and A[™].

first chemical formula can be handled, while the second will require hand editing

P 299 ORD

Language: English

Abstract 2 ?

B I $\times_2$ $\times^2$ $\vdots$ $\vdots$ Avsnitt   ...

Alla avhandlingar vid KTH måste ha ett abstrakt på både engelska och svenska.

Om du skriver din avhandling på svenska ska detta göras först (och placera det som det första abstraktet) - och du bör revidera det vid behov.

If you are writing your thesis in English, you can leave this until the draft version that goes to your opponent for the written opposition. In this way you can provide the English and Swedish abstract/summary information that can be used in the announcement for your oral presentation.

If you are writing your thesis in English, then this section can be a summary targeted at a more general reader. However, if you are writing your thesis in Swedish, then the reverse is true – your abstract should be for your target audience, while an English summary can be written targeted at a more general audience.

This means that the English abstract and Swedish sammanfattning or Swedish abstract and English summary need not be literal translations of each other.

The abstract in the language used for the thesis should be the first abstract, while the Summary/Sammanfattning in the other language can follow.

P 190 ORD

Language: Swedish

Abstract 3 ?

B I $\times_2$ $\times^2$ $\vdots$ $\vdots$ Avsnitt   ...

Résumé en français.

P 3 ORD

Language: French

Abstract 4 ?

B I $\times_2$ $\times^2$ $\vdots$ $\vdots$ Avsnitt   ...

Résumé en espagnol.

P 3 ORD

Language: Spanish

Figure 12: [Imported document (part 5)]Imported document (part 5) – the bottom of the English abstract is shown - showing some of the formatted information that can be shown. Additionally, the Swedish and other abstracts are shown.

The image shows a web interface with six abstracts listed vertically. Each abstract has a title, a language dropdown, and a word count. The abstracts are:

- Abstract 5 ?**: Title: Sommario in italiano. Language: Italian. Word count: 3 ORD.
- Abstract 6 ?**: Title: Sammenhang på norsk. Language: Norwegian. Word count: 3 ORD.
- Abstract 7 ?**: Title: (empty). Language: German. Word count: 0 ORD.
- Abstract 8 ?**: Title: Abstrakt på dansk. Language: Danish. Word count: 3 ORD.
- Abstract 9 ?**: Title: Zusammenfassung in Deutsch. Samenvatting in het Nederlands. Eesti keeles kokkuvõte. Language: Dutch; Flemish. Word count: 10 ORD.
- Abstract 10 ?**: Title: (empty). Language: Estonian. Word count: 0 ORD.

Figure 13: Imported document (part 6) – yet more abstracts

Figure 14 shows the three supervisors. The first two have KTH affiliations and

the third is from an external organization.

Supervisor 1 ?

[Connect authority record »](#) [Create authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name: Supervisor Academic title:

First name: A. Busy Local User Id: u100003

ORCID:

Department, unit or programme: Choose organisation »

E-mail: sa@kth.se

Other organisation: School of Electrical Engineering and Computer Science, Computer Science

Supervisor 2 ?

[Connect authority record »](#) [Create authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name: Supervisor Academic title:

First name: Another Busy Local User Id: u100003

ORCID:

Department, unit or programme: Choose organisation »

E-mail: sb@kth.se

Other organisation: School of Architecture and the Built Environment, Public Buildings

Supervisor 3 ?

[Connect authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name: Supervisor Academic title:

First name: Third Busy Local User Id:

ORCID:

Department, unit or programme: Choose organisation »

E-mail: sc@tu.va

Other organisation: Timbuktu University, Department of Pseudoscience

[Another supervisor »](#)

Examiner ?

[Connect authority record »](#) [Create authority record »](#)

Figure 14: Imported document (part 7) – the supervisors are shown

Examiner ?

Connect authority record »

Create authority record »

Get saved personal data »

Save personal data »

Last name:

Maguire Jr.

Academic title:

First name:

Gerald Q.

Local User Id:

u1d13i2c

ORCID:

Department, unit or programme:

Choose organisation »

E-mail:

maguire@kth.se

Other organisation:

KTH, School of Electrical Engineering and Computer Science, Cor

Another examiner »

Presentation ?

Date:

2021-03-15 13:00

Language:

English



Room:

via Zoom <https://kth-se.zoom.us/j/ddddddddddd>

Address:

Isafjordsgatan 22 (Kistagången 16)

City:

Stockholm

Note ?

B *I* $\times_2$ $\times^2$ Ω 

Figure 15: Imported document (part 8) – the examiner is shown along with the information about the oral presentation

8.3 Limitations

I was initially unable to get the KTH affiliations correctly entered. I did not understand the description of how to do this at: <https://wiki.epc.uu.se/pages/viewpage.action?pageId=27466001> where it says:

För att göra en sådan koppling skall dels "affiliation" finnas i personelementet dels ett name-element med ID i xlink:href samt ett namePart-element med samma namn som affiliation. Om man har bägge nedanstående element i exemplet kommer Urban Ericsson automatiskt att kopplas till Universitetsbiblioteket i Uppsala vid importen. Om enbart personelementet finns men inte det andra kommer Urban Ericsson att få Uppsala universitet, Universitetsbiblioteket som "Annat lärosäte" vid imported.

I did not find any examples of how to do what they describe. The result is that all of my affiliation information ended up in the "Other university" field.

I was also unable to get the issue number to appear, even though it seems to be set, since if you click on the box after the MODS import, it knows the value. Note that for testing, I forced the TRITA series to be "TRITA-ICT-EX" since this is the actual series that is in the earlier (test) version of DiVA.

Finally, it seems that DiVA does not import the cooperation data at all: according to <https://wiki.epc.uu.se/display/divainfo/Externt+samarbete>.

The first two limitations were overcome as described in Section 8.4.

The current version of the program does not support the UN's Sustainable Development Goals (SDGs) metadata (as this metadata field was introduced after the program was written).

8.4 Limitations overcome

This section describes how two of the limitations were overcome.

8.4.1 Entering number in series

The first limitation to be overcome was entering the particular number in the series for a TRITA number. Figure 16 shows the value being entered. The

error was in the structure of the object, the working version is shown in Listing 19. My earlier error was due to having the `<identifier type="issue number">2021:00</identifier>` *within* the `<titleInfo>`.

Other series ?

Title of series:

ISSN:

EISSN:

No. in series:

Figure 16: Number in series correctly entered

Listing 19: MODS data for series

```
<relatedItem type="series">
  <titleInfo><title>TRITA-EECS-EX</title>
  <identifier type="local">16855</identifier>
</titleInfo>
  <identifier type="issue_number">2021:00</identifier>
</relatedItem >
```

8.4.2 Including KTH affiliations

Earlier, those with KTH affiliations were appearing in the “Other organization” field. The error in my understanding was how the controlled fields for KTH authors was being used*. We will start by looking at the two KTH-affiliated supervisors (see Figure 17), the third supervisor (see Figure 18), and the examiner (see Figure 19). The MODS data for these four people is shown in Listing 20 and the “magic” **corporate name** MODS data is shown in Listing 21. The trick

*These so-called “controlled” fields have specific values that are permitted and these have numeric IDs assigned by the DiVA Consortium.

is that there needs to be corporate name MODS data for each of the possible affiliations – as this is the way that one passes the DiVA numeric code via the `xlink` information. Note that in the case of the examiner, two different affiliations were specified and there was a corporate name entry for each of them. Moreover, the key is just to have test extra corporate name entries, only one is needed for the publishers, as shown in Listing 21 – with the results in Figure 20 and shown Figure 21.

NB: The ICT affiliations (and DiVA codes) were used and the pictures are all from the test DiVA environment: <https://kth.test.diva-portal.org/dream/import/importList.jsf>.

Supervisor 1 ?

Connect authority record »

Create authority record »

Get saved personal data »

Save personal data »

Last name:

Supervisor

Academic title:

First name:

A. Busy

Local User Id:

u100003

ORCID:

Department, unit or programme:

Choose organisation »

Skolan för informations- och kommunikationsteknik (ICT), School of Information and Communication Technology (ICT)

x

E-mail:

sa@kth.se

Other organisation:

Supervisor 2 ?

Connect authority record »

Create authority record »

Get saved personal data »

Save personal data »

Last name:

Supervisor

Academic title:

First name:

Another Busy

Local User Id:

u100003

ORCID:

Department, unit or programme:

Choose organisation »

Skolan för arkitektur och samhällsbyggnad (ABE), School of Architecture and the Built Environment (ABE)

x

E-mail:

sb@kth.se

Other organisation:

Figure 17: Two KTH supervisors in different schools

Supervisor 3 ?

[Connect authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name: Academic title:

First name: Local User Id:

ORCID:

Department, unit or programme:
[Choose organisation »](#)

E-mail:

Other organisation:

[Another supervisor »](#)

Figure 18: Third supervisor

Examiner ?

[Connect authority record »](#) [Create authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name:

Academic title:

First name:

Local User Id:

ORCID:

Department, unit or programme:
[Choose organisation »](#)

x

x

E-mail:

Other organisation:

[Another examiner »](#)

Figure 19: Examiner

Listing 20: MODS data for the supervisors and examiners

```

<name type="personal" authority="kth"
  ↪ xlink:href="u1d13i2c">
  <namePart type="family">Maguire_Jr.</namePart>
  <namePart type="given">Gerald_Q.</namePart>
  <description>email=maguire@kth.se</description>
  <affiliation>KTH, Skolan för informations- och
    ↪ kommunikationsteknik (ICT), Kommunikationssystem,
    ↪ CoS</affiliation>
  <affiliation>KTH, Skolan för informations- och
    ↪ kommunikationsteknik (ICT)</affiliation>
  <role><roleTerm type="code"
    ↪ authority="marcrelator">mon</roleTerm></role>
</name>

<name type="personal" authority="kth" xlink:href="u100003">
  <namePart type="family">Supervisor</namePart>
  <namePart type="given">A._Busy</namePart>
  <affiliation>KTH, Skolan för informations- och
    ↪ kommunikationsteknik (ICT)</affiliation>
  <description>email=sa@kth.se</description>
  <role><roleTerm type="code"
    ↪ authority="marcrelator">ths</roleTerm></role>
</name>

<name type="personal" authority="kth" xlink:href="u100003">
  <namePart type="family">Supervisor</namePart>
  <namePart type="given">Another_Busy</namePart>
  <affiliation>KTH, Skolan för arkitektur och
    ↪ samhällsbyggnad (ABE)</affiliation>
  <description>email=sb@kth.se</description>
  <role><roleTerm type="code"
    ↪ authority="marcrelator">ths</roleTerm></role>
</name>

<name type="personal">
  <namePart type="family">Supervisor</namePart>
  <namePart type="given">Third_Busy</namePart>
  <affiliation>Timbuktu University, Department of
    ↪ Pseudoscience</affiliation>

```

```

<description>email=sc@tu.va</description>
<role><roleTerm type="code"
  ↪ authority="marcrelator">ths</roleTerm></role>
</name>

```


Listing 21: Corporate name data in MODS data for the affiliations

```

<name type="corporate" authority="kth" xlink:href="5994">
  <namePart>KTH</namePart>
  <namePart>Skolan för informations- och
    ↳ kommunikationsteknik (ICT)</namePart>
  <role><roleTerm type="code"
    ↳ authority="marcrelator">pbl</roleTerm></role>
</name>

<name type="corporate" authority="kth" xlink:href="5998">
  <namePart>KTH</namePart>
  <namePart>Skolan för informations- och
    ↳ kommunikationsteknik (ICT)</namePart>
  <namePart>Kommunikationssystem, CoS</namePart>
  <role><roleTerm type="code"
    ↳ authority="marcrelator">pbl</roleTerm></role>
</name>

<name type="corporate" authority="kth" xlink:href="5850">
  <namePart>KTH</namePart>
  <namePart>Skolan för arkitektur och samhällsbyggnad
    ↳ (ABE)</namePart>
  <role><roleTerm type="code"
    ↳ authority="marcrelator">pbl</roleTerm></role>
</name>

```

Listing 22: Reduced MODS for corporate names more

```

<name type="corporate" authority="kth" xlink:href="5994">
  <namePart>KTH</namePart>
  <namePart>Skolan för informations- och
    ↳ kommunikationsteknik (ICT)</namePart>
  <role><roleTerm type="code"
    ↳ authority="marcrelator">pbl</roleTerm></role>
</name>

<name type="corporate" authority="kth" xlink:href="5998">
  <namePart>KTH</namePart>
  <namePart>Skolan för informations- och
    ↳ kommunikationsteknik (ICT)</namePart>
  <namePart>Kommunikationssystem, CoS</namePart>
</name>

```

```
<name type="corporate" authority="kth" xlink:href="5850">  
  <namePart>KTH</namePart>  
  <namePart>Skolan för arkitektur och samhällsbyggnad  
    ↪ (ABE) </namePart>  
</name>
```

Last name:	Academic title:
Supervisor	
First name:	Local User Id:
A. Busy	u100003
ORCID:	
Department, unit or programme:	
Choose organisation »	
Skolan för informations- och kommunikationsteknik (ICT), School of Information and Communication Technology (ICT) x	
E-mail:	
sa@kth.se	
Other organisation:	

Supervisor 2 ?	
Connect authority record »	Create authority record »
Get saved personal data »	Save personal data »
Last name:	Academic title:
Supervisor	
First name:	Local User Id:
Another Busy	u100003
ORCID:	
Department, unit or programme:	
Choose organisation »	
Skolan för arkitektur och samhällsbyggnad (ABE), School of Architecture and the Built Environment (ABE) x	
E-mail:	
sb@kth.se	
Other organisation:	

Figure 20: The two supervisors with the reduced MODS for corporate names

Examiner ?

[Connect authority record »](#) [Create authority record »](#)

[Get saved personal data »](#) [Save personal data »](#)

Last name: Academic title:

First name: Local User Id:

ORCID:

Department, unit or programme:

[Choose organisation »](#)

x

x

E-mail:

Other organisation:

[Another examiner »](#)

Figure 21: The examiner with the reduced MODS for corporate names

9 Accessibility

The discussion of accessibility has been divided into accessibility of the calendar entries (see Section 9.1), the cover and PDF files (see Section 9.2), and the template itself (see Section 9.3). There is also a subsection regarding improving accessibility (see Section 9.4).

9.1 Calendar entries

The calendar entries that are generated in the Canvas course room are as accessible as all content in Canvas (as Instructure tries to follow the W3C Web Content Accessibility Guidelines (WCAG)). The European standard EN 301 549 V2.1.2 (based upon WACG 2.1) are the accessibility guidelines for web content that are recommended in Sweden by Myndigheten för digital förvaltning (DIGG), based upon the presentation by Tommy Olsson of DIGG to the SUNET SALSA group* on 2021-06-03. Moreover, the contents are HTML language-tagged, so that a text-to-speech program that has access within Canvas (such as ReaderSpaker (when run as an Learning Tools Interoperability (LTI) app)) could read the content with the correct pronunciation for each of the two languages. Note that KTH's current screen reader solution does **not** access the HTML of the page and hence it does not use the language tags, thus the user must manually choose the language for output.

The calendar entries in the KTH Calendar are in the same format as the current calendar entries. The structure of these entries has been developed in consultation with XXXXXX (who works with the KTH Web) and the KTH Calendar API developer XXXXX, together with the additional help of XXXXXX. Note that the last of these was the original developer of the KTH cover generator.

9.2 Cover and PDF file

The PDF metadata (author(s), title(s), keywords, *etc.*) is accessible to any programs that use the PDF metadata (this is a standard feature of PDF files). Unfortunately, there are no provisions for language markup for this data.

No investigation has been made of the accessibility of the KTH cover nor of the contents of the PDF file (*i.e.*, the thesis itself). The PDF output by Overleaf appears to be PDF version 1.5 (*i.e.*, thus it is accessible via Acrobat version 6 or later). The `examplethesis.tex` file has code to change this version.

*<https://wiki.sunet.se/pages/viewpage.action?pageId=30441654>

The template utilizes color in conjunction with the `hyperref` colors and `todonotes`. The `hyperref` colors are defined as shown in Listing 24.

Listing 23: `\hypersetup`

```
\hypersetup{
  colorlinks = true,
  breaklinks = true,
  linkcolor = \linkscolor,
  urlcolor = \urlscolor,
  citecolor = \refscolor,
  anchorcolor = black
}
```

Where the colors are defined as (note that the `ForestGreen` lacks sufficient contrast for readability – so another color should be used to replace it) as shown in Listing 24.

Listing 24: Some colors for hyper references

```
\definecolor{ForestGreen} {RGB}{34, 139, 34}
\definecolor{HeraldRed2} {rgb}{0.81, 0.12, 0.15}

\newcommand{\refscolor} {blue}
\newcommand{\linkscolor}{HeraldRed2}
\newcommand{\urlscolor} {ForestGreen}
```

Note that the colors do **not** encode any essential meaning (*i.e.*, they could all be turned to black), since the citations are recognizable by their format, the URL (and URI) by linking to an external document, and the links (by linking within the document).

Some additional colors are defined and used for `todonotes` – they should, of course, be removed before the thesis is finalized (as of course all of these issues should have been resolved!). The default background color for `todonotes` is orange (which is predefined as `#FF7F00`). The other colors are shown in Listing 25. The KTH colors are defined in `kthcolors.tex`.

Listing 25: Defines for todo notes

```
% Conventions for todo notes:
% Informational
%% \generalExpl{Comments/directions/... in English}
\newcommand*{\generalExpl}[1]{\todo[inline]{#1}}
```

```
% Language-specific information (currently in English or
↪ Swedish)
\newcommand*{\engExpl}[1]{\todo[inline, backgroundColor=kth
↪ -lightgreen40]{#1}} %% \engExpl{English descriptions
↪ about formatting}
\newcommand*{\sweExpl}[1]{\todo[inline, backgroundColor=kth
↪ -lightblue40]{\foreignlanguage{swedish}{#1}}} %% % \
↪ sweExpl{Text på svenska}

% warnings
\newcommand*{\warningExpl}[1]{\todo[inline, backgroundColor
↪ =kth-lightred40]{#1}} %% \warningExpl{warnings}
```

9.3 Template itself

The template itself is written in $\text{\LaTeX}$ using UTF-8 characters. The template uses packages from *TeX Live version 2024*. It must be compiled with $\text{\Lua\LaTeX}$ (which natively supports UTF-8 input).

The template is designed with a set of options to generate a thesis in English or Swedish* and to use either $\text{\BibTeX}$ or $\text{\Bib\LaTeX}$, as explained at the top of the `examplethesis.tex` file (see Listing 26).

Listing 26: `\documentclass`

```
% --- Configure the documentclass ---
%% The template is designed to handle a thesis in English
↪ or Swedish
%%
\documentclass[
English,           % the language of the body of the
↪ document: English or Swedish
biblatex,         % the bibliographic processor to use:
↪ biblatex or bibtex
%includepublications, % uncomment - if you want to include
↪ publications for a compilation thesis - also creates
↪ divider pages
%includeExtraAbstracts, % uncomment to allow abstracts in
↪ languages beyond English & Swedish
```

*The option names for both languages are must be given in title case, *i.e.*, with initial capital letters.

```
%nomenclature,      % add the option nomenclature
%digitaloutput,      % optimize for digital output (this
    ↪ changes the color palette)
%final,              % turn off all of the todo notes
]{kththesis}
```

The template is available from Overleaf (via a share URL).

The text in the template itself (*i.e.*, the `examplethesis.tex` file) is in Swedish and English. The notes regarding the $\text{\LaTeX}$ class file, the various `lib` files, and the various `README` files are currently only in English.

The default bibliographic style when using $\text{\BibTeX}$ is my own adaptation of the IEEE Transactions format (*i.e.*, numbered citations, numbered references, references in order of use) with the extension of adding DOIs, URLs, and ISBNs (when relevant).

9.4 Improving accessibility

One method for improving accessibility would be to include the accessibility package, *i.e.*, `\usepackage{axessibility}` as this would include a comment in the PDF file for each equation with the $\text{\LaTeX}$ that generated the equation. However, this package is no longer maintained and is incompatible with many other packages. For an introduction to what the $\text{\LaTeX}$ -project has been working on, see “ $\text{\LaTeX}$ Tagged PDF – A blueprint for a large Project”[2]. Based upon this article and the status of the `tagpdf` package, I conclude that it is too early to worry about properly tagging PDF files; thus, doing so will have to await the release of a package designed for production use.

10 The structure of the template and the report

The report in itself (*i.e.*, the thesis in `examplethesis.tex`) has a classical Introduction, Methods, Results, and Discussion (IMRAD) structure. In some subject areas, such as mathematics there is a tradition for another structure. The files and folders in Overleaf have the form shown in Figure 22. Some students add folders per chapter and reduce the main document to a skeleton that includes the other parts of the document.

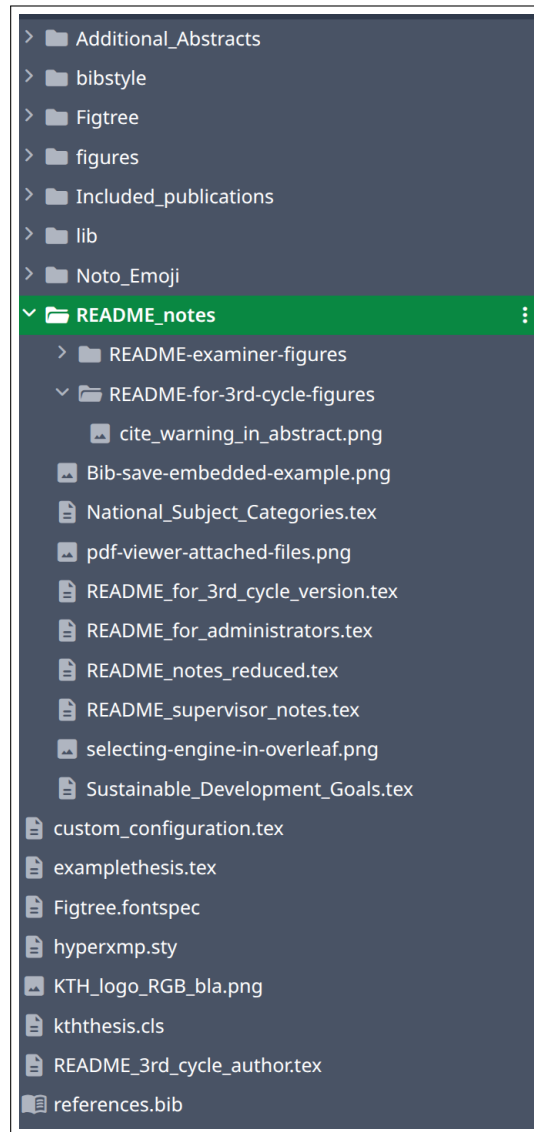


Figure 22: Files and folders in Overleaf template

There are currently instructions concerning the template in the `README_notes/README_notes_reduced` file*. Ideally, these instructions should be available in both English and Swedish; however, a Swedish version remains as future work.

*You will need to compile it separately.

Most students will only need to add acronyms to the `acronyms.tex` file and perhaps additional packages to `includes.tex` (in some cases, these packages may need to be added to the `includes-after-hyperref.tex` file if there is a conflict with `hyperref`).

The `school_subjects_3rdcycle.ins` contents were generated from the data in LADOK. Students should not have to change this file.

11 Generating LaTeX commands for in the student's thesis source files

The thesis makes use of a set of $\text{\LaTeX}$ commands to collect information about the student and the supervisor(s) in the `custom_configuration.tex` file. However, how can a student know what their KTHID and other necessary information is?

There are two alternatives:

- The main supervisor or someone can use the method described in section “Make it simple from the start” in the file `README_supervisor_notes`.
- One can use the program: `whoami_for_latex.py` - see the command shown in Listing 27*; as Canvas does not currently have any source for the student's program information.

Listing 27: Getting information regarding supervisor

```
./whoami_for_latex.py email course_id [supervisor_index]
```

The `course_id` has to be the `canvas_id` of a course where the supervisor is enrolled (as student, teacher, ...). If `supervisor_index` is not specified, it defaults to "A".

While it might seem desirable to incorporate this directly in the $\text{\LaTeX}$ project, there are two reasons to avoid this: (i) I am not sure that student's would want to or should put their Canvas access token into a project in Overleaf – as this seems to be a security issue (since others will potentially access this project.) and (ii) this only needs to be done at the start of the project and then does not need to be done again.

*As this version of the program only uses Canvas and does not use LADOK, it excludes program information

12 Copyright or Creative Commons License

It is possible to have several variants of the bookinfo page:

`copyright` If you want to have a bookinfo page, include the line saying `\bookinfo` page.

`Creative Commons (CC)` If you want to have a bookinfo page but want to have a Creative Commons license, then include `\bookinfo` page and use and configure the `doclicense` package as shown in Listings 28, 29, and 30.

`none` If you do **not** want to have a bookinfo page, comment the line saying `\bookinfo` page and add a `\cleardoublepage`.

For background about Creative Commons licenses, see: <https://www.kb.se/samverkan-och-utveckling/oppen-tillgang-och-bibsamkonsortiet/open-access-and-bibsam-consortium/open-access/creative-commons-faq-for-researchers.html> and <https://kib.ki.se/en/publish-analyse/publish-your-article-open-access/open-licence-your-publication-cc>.

Note that the lowercase version of the license has to be used in the modifier, *i.e.*, one of: `by`, `by-nc`, `by-nd`, `by-nc-nd`, `by-sa`, `by-nc-sa`, or `zero`. For the list of supported licenses, see the documentation for the `doclicense` package.

Listing 28: Example configuration to have a CC BY-NC-ND license

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  hyphenation={RaggedRight},
]{doclicense}
```

Note that the option “`hyphenation=RaggedRight`” can be used with the configuration of the package to set the license information with a ragged right margin rather than as a filled and justified paragraph.

Listing 29: Example configuration to have a CC BY-NC-ND license with a Euro symbol rather than a Dollar sign

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  imagemodifier={-eu-88x31}, % to get Euro symbol rather
    ↪ than Dollar sign
  hyphenation={RaggedRight},
]{doclicense}
```

Listing 30: Example configuration to have a CC0 license

```
\usepackage[
  type={CC},
  modifier={zero},
  version={1.0},
]{doclicense}
```

For someone who is maintaining or modifying the template, it is important to know that the `\bookinfoPageCC` command has a case statement in it that uses an internal variable (`\doclicense@modifier`) set by `doclicense` as shown in Listing 31. This case statement will need to be modified if there is a change in the set of Creative Commons licenses. While it might be possible to replace this case statement with `\doclicenseModifier` to directly produce the capitalized version of the license names or perhaps even `\doclicenseName` or `\doclicenseNameRef` to include the capitalized license name as well as the version number (the latter also including a URL to the license). However, I think that these alternatives are unnecessary given the `\doclicenseThis` that generates the display of the license with logo(s). More importantly, using these other commands rather than the unprocessed `\doclicense@modifier` leads to problems when writing out the `fordiva.json` file.

Note that if the `doclicense` package is used it automatically redefines `\bookinfoPage` to be `\bookinfoPageCC`.

Listing 31: Case statement in `\bookinfoPageCC`

```
\IfEqCase{\doclicense@modifier}{%
  {by}{CC BY}%
  {by-nc}{CC BY-NC}%
  {by-nd}{CC BY-ND}%
}
```

```

{by-nc-nd}{CC BY-NC-ND}%
{by-sa}{CC BY-SA}%
{by-nc-sa}{CC BY-NC-SA}%
{zero}{CC0}%
}{\typeout{Creative Commons code \doclicense@modifier not
↪ found}\copyright]

```

There is a new `\thesiscopyrightleft` command added to the `\bookinfo` and `\bookinfoCC` pages to store the value of the copyright or copyleft information that the user chose (if any). This information is later output as information for DiVA in the form shown in Listings 32 to 34

Listing 32: If the user chose to have a copyright on the bookinfo page

```
"Copyrightleft": "copyright"
```

Listing 33: If the user chose to have a CC BY-NC license on the bookinfo page

```
"Copyrightleft": "by-nc"
```

Listing 34: If the user chose **not** to have a bookinfo page

```
"Copyrightleft": "None"
```

As a result of adding support for the PDF metadata for the copyright, the template now uses the `hyperxmp` package. Details of the code to set the value of the `pdfcopyright` metadata can be seen in Listing 35. The use of `hyperxmp` caused a change in how the author's name is processed - for details about this see Section 16 starting on page 95.

Listing 35: Processing copyright PDF metadata information in `\bookinfoCC`

```

% Command to print out the standardized document
↪ information page
\newcommand{\bookinfo}{
\ifgfivepaper
  \newgeometry{top=140mm,bottom=30mm,left=12.25mm,right=35
↪ mm}
\else
  \newgeometry{top=250mm,bottom=30mm,left=74pt,right=35mm}
\fi
\thispagestyle{empty}
\begin{flushleft}
\sffamily

```

```

\copyright\enspace\the\year\quad\@authorsFirstname\space\
    ↪ @authorsLastname
\ifx\@secondAuthorsLastname\@empty\relax
\else
\ifinswedish
\enspace och\enspace\@secondAuthorsFirstname\space\
    ↪ @secondAuthorsLastname \\\
\else
\enspace and\enspace\@secondAuthorsFirstname\space\
    ↪ @secondAuthorsLastname \\\
\fi
\fi
\end{flushleft}
\thesiscopyrightleft{copyright}
\newcommand{\mycopyright}{%
{Copyright (c)}\ \the\year\ \@authorsFirstname\ \
    ↪ @authorsLastname %
\ifx\@secondAuthorsLastname\@empty\relax%
\else%
\ifinswedish%
\ och\ \@secondAuthorsFirstname\ \
    ↪ @secondAuthorsLastname%
\else%
\ and\ \@secondAuthorsFirstname\ \
    ↪ @secondAuthorsLastname%
\fi %
\fi %
}
\hypersetup{
pdfcopyright={\mycopyright}
}
\restoregeometry
\clearpage
}

% bookinfo page with CC license
% CC BY (Attribution)
% CC BY-NC (Attribution + Non-commercial)
% CC BY-ND (Attribution + No derivatives)
% CC BY-NC-ND (Attribution + Non-commercial + No

```

```

↪ derivatives)
% CC BY-SA (Attribution + ShareAlike)
% CC BY-NC-SA (Attribution + Non-commercial + ShareAlike)
% CC0 (Zero)
\newcommand{\bookinfoCC}{
\ifgfivepaper
  \newgeometry{top=130mm,bottom=30mm,left=12.25mm,right=35
    ↪ mm}
\else
  \newgeometry{top=220mm,bottom=30mm,left=74pt,right=35mm}
\fi
\thispagestyle{empty}
\begin{flushleft}
  \sffamily
  \IfEqCase{\doclicense@modifier}{%
    {by}{CC BY}%
    {by-nc}{CC BY-NC}%
    {by-nd}{CC BY-ND}%
    {by-nc-nd}{CC BY-NC-ND}%
    {by-sa}{CC BY-SA}%
    {by-nc-sa}{CC BY-NC-SA}%
    {zero}{CC0}%
  }[\typeout{Creative Commons code \doclicense@modifier not
    ↪ found}\copyright]
  \enspace\the\year\quad\@authorsFirstname\space\
    ↪ @authorsLastname
  \ifx\@secondAuthorsLastname\@empty\relax
  \else
    \ifinswedish
      \enspace och\enspace\@secondAuthorsFirstname\space\
        ↪ @secondAuthorsLastname \\\
    \else
      \enspace and\enspace\@secondAuthorsFirstname\space\
        ↪ @secondAuthorsLastname \\\
    \fi
  \fi
  \doclicenseThis
\end{flushleft}
\thesiscopyrightleft{\doclicense@modifier}

```

```

\let\@myCC\@empty
\ifdefstring{\doclicense@modifier}{by}{\def\@myCC{CC BY}}
\ifdefstring{\doclicense@modifier}{by-nc}{\def\@myCC{CC BY-NC
↪ }}
\ifdefstring{\doclicense@modifier}{by-nd}{\def\@myCC{CC BY-ND
↪ }}
\ifdefstring{\doclicense@modifier}{by-nc-nd}{\def\@myCC{CC BY
↪ -NC-ND}}
\ifdefstring{\doclicense@modifier}{by-sa}{\def\@myCC{CC BY-SA
↪ }}
\ifdefstring{\doclicense@modifier}{by-nc-sa}{\def\@myCC{CC BY
↪ -NC-SA}}
\ifdefstring{\doclicense@modifier}{zero}{\def\@myCC{CC0}}

\newcommand{\mycopyright}{%
\@myCC\ \the\year\ \@authorsFirstname\ \@authorsLastname%
\ifx\@secondAuthorsLastname\@empty\relax%
\else%
\ifinswedish%
\ och\ \@secondAuthorsFirstname\ \
↪ \@secondAuthorsLastname%
\else%
\ and\ \@secondAuthorsFirstname\ \
↪ \@secondAuthorsLastname %
\fi%
\fi%
}

\hypersetup{
pdfcopyright={\mycopyright}
}
\restoregeometry
\clearpage
}

% If the doclicense package is installed, then use the \
↪ bookinfoCC when encountering a \bookinfo command
\AtBeginDocument{\@ifpackageloaded{doclicense}
{%
\renewcommand{\bookinfo}{\bookinfoCC}

```



```

}
{\relax}
}

```

13 Inserting information into LADOK

As part of the effort to minimize the effort of cutting and pasting. I have made a program `JSON_to_ladok.py` that takes the extracted JSON information and uses the information about the author(s) and the title and alternative title to try to insert this information into LADOK for the module (*i.e.*, “moment” in Swedish) that requires a project title, *i.e.*, ‘KravPaProjekttitel’ is True. Listing 36 shows an example of using this program to try to put the title and alternative title into LADOK. Basically, the program logic should work, but I do **not** have the required permission to make entries of this sort of data for a degree project.

Note that this program uses the `ladok3` python library but extends it with some features that are not (yet) in the library. It should be regarded as very much a work in progress.

Listing 36: Using the extracted JSON to produce a LADOK entry

```

./JSON_to_ladok.py -c 11 --json xxx.json --code DA213X
testing=False
d={'Author1': {'Last_name': 'xxx', 'First_name': 'xxx', '
↪ Local_User_Id': 'yyyyy', 'E-mail': 'xxx@kth.se', '
↪ organisation': {'L1': 'School_of_Electrical_
↪ Engineering_and_Computer_Science_'}}, 'Degree': {'
↪ Educational_program': 'Masters_Programme,_Computer_
↪ Science,_120_credits', 'programcode': 'TCSCM', 'Level'
↪ : '2', 'Course_code': 'DA231X', 'Credits': '30.0', '
↪ Exam': 'Masters_Programme,_Computer_Science,_120_
↪ credits', 'subjectArea': 'Computer_Science'}, 'Title':
↪ {'Main_title': 'xxx', 'Subtitle': 'yyyy', 'Language': '
↪ eng'}, 'Alternative title': {'Main title': 'zzzz', '
↪ Subtitle': 'www', 'Language': 'swe'}, 'Supervisor1':
↪ {'Last_name': 'x', 'First_name': 'y', 'Local_User_Id':
↪ 'xx', 'E-mail': 'xxxxx@kth.se', 'organisation': {'L1'
↪ : 'School_of_Electrical_Engineering_and_Computer_Science
↪ ', 'L2': 'Computer_Science'}}, 'Supervisor2': {'Last_name
↪ : 'yyy', 'First_name': 'xxx', 'E-mail': 'xxx@zzzz.com'

```

```

↪ , 'Other_organisation': 'zzzz'}, 'Examiner1': {'Last name
↪ ': 'Maguire Jr.', 'First name': 'Gerald Q.', 'Local User
↪ Id': 'uld13i2c', 'E-mail': 'maguire@kth.se', 'organisation
↪ ': {'L1': 'School of Electrical Engineering and Computer
↪ Science ', 'L2': 'Computer Science'}}, 'Cooperation':
↪ {'Partner_name': 'xxx'}, 'National Subject Categories':
↪ '10201, 10206', 'Other information': {'Year': '2021',
↪ 'Number of pages': 'xix, 99'}, 'Opponents': {'Name': 'A.
↪ B. Normal & A. X. E. Normalè'}, 'Presentation': {'Date
↪ ': '2021-03-15 13:00', 'Language': 'eng', 'Room': 'via
↪ Zoom', 'Address': 'Isafjordsgatan 22 (Kistagången 16)',
↪ 'City': 'Stockholm'}, 'Number of lang instances': '2',
↪ 'abstracts': {'eng': 'xxxxx', 'swe': 'yyyy'}, 'keywords'
↪ : {'eng': 'x, y z', 'swe': 'x, y, z'}}
course_code=DA231X
author={'Last_name': 'xxx', 'First_name': 'yyy', 'Local_
↪ User_Id': 'ulxxxx', 'E-mail': 'xxxx@kth.se', '
↪ organisation': {'L1': 'School_of_Electrical_
↪ Engineering_and_Computer_Science_'}}
sortable_name=X,Y
Canvas user_id=dddd
integration_id=ggggggg-gggg-gggg-gggg-gggggggg
ladoK_course_info={'id': '6683207e-5a5d-11eb-9b32-
↪ eeb44fb14647', 'round_id': '8e15ae14-1d86-11ea-a622
↪ -3565135944de', 'education_id': '374ea085-73d8-11e8-
↪ afa7-8e408e694e54', 'instance_id': '8eee8da9-dd0a-11
↪ e8-bb7a-19f8cd1a470e', 'swe_name': 'Examensarbete_i_
↪ datalogi_och_datateknik, _avancerad_nivå', 'eng_name':
↪ 'Degree_Project_in_Computer_Science_and_Engineering, _
↪ Second_Cycle'}
moment code=PRO1, requires title=False
moment code=PRO2, requires title=False
moment code=PRO3, requires title=True
trying to store a passing grade for moment=PRO3
Traceback (most recent call last):
  File "./JSON_to_ladok.py", line 533, in <module>
    sys.exit(main(sys.argv[1:]))
  File "./JSON_to_ladok.py", line 519, in main
    status=save_result_degree_project3(ladok, integration_id
↪ , course_code, mom['Utbildningskod'], '2021-07-14'

```

```

    ↪ , 'P', "PF", main_title, alternative_main_title)
File ".\JSON_to_ladok.py", line 374, in
    ↪ save_result_degree_project3
    raise Exception("Couldn't register" + course_moment + "
    ↪ =" + grade_raw + "_" + result_date_raw + ":" + r.
    ↪ json()["Meddelande"])
Exception: Couldn't register PRO3=P 2021-07-14: Hinder mot
    ↪ skapa resultat påträffat: Rapporteringsrättighet saknas

```

14 Extending the set of languages supported for abstracts

Note for maintainers: The code that implements the `abstract` command, shown in Listing 37, may need to be extended if an additional language that requires a special font is added for abstracts. This is necessary for languages that use characters not available in Figtree. The basic approach is to look at the value of `\localename` (this is set when there is a call to `\selectlanguage{x}`), then use the sans serif font that `babel` has set up for this language; otherwise, set the heading using Figtree.

Listing 37: Modified abstract command

```

% This environment is to overcome the problem that babel
    ↪ changes the fonts and chapter heading formatting in
    ↪ language specific ways
% The method used here is to simply set the heading without
    ↪ using a heading command.
\renewenvironment{abstract}{%
    %\typeout{kththesisChip \f@encoding\ \f@family\ \
    ↪ f@series\ \f@shape}%
    \vspace*{178pt}% Space before the title - the ^is needed
    ↪ because \vspace is removed at page breaks
    \noindent
    \IfEqCase{\localename}{% \language name
        {hindi} { {\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{n}\
        ↪ selectfont \abstractname} }%
        {russian} { {\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{n}
        ↪ }\selectfont \abstractname} }%

```

```

{ukrainian} {{\sffamily \fontsize{24}{26}\fontseries{m}\fontshape
    ↪ {n}\selectfont \abstractname}}%
{greek} {{\sffamily \fontsize{24}{26}\fontseries{m}\selectfont \
    ↪ abstractname}}%
{chinese-simplified} {{\sffamily \fontsize{24}{26}\fontseries{m}
    ↪ }\fontshape{n}\selectfont \abstractname}}%
{arabic} {{\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{n}
    ↪ }\selectfont \abstractname}}%
{centralkurdish} {{\sffamily \fontsize{24}{26}\fontseries{m}\
    ↪ fontshape{n}\selectfont \abstractname}}%
{hebrew} {{\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{n}
    ↪ }\selectfont \abstractname}}%
%{hebrew} {{\hebrewfontsf \fontsize{24}{26}\fontseries{m}
    ↪ }\fontshape{n}\selectfont \abstractname}}%
{japanese} {{\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{
    ↪ n}\selectfont \abstractname}}%
% We have to use \NotoSansFont as Figtree is missing ă"
    ↪ and ô" from "Tóm átt öni dung"
{vietnamese} {{\sffamily \fontsize{24}{26}\fontseries{m}\
    ↪ fontshape{n}\selectfont \abstractname}}%
{yiddish} {{\sffamily \fontsize{24}{26}\fontseries{m}\fontshape{n}
    ↪ }\selectfont \abstractname}}%
% Add other specific languages here if they don't use
    ↪ FigtreeFont
][% Default case: If none of the above special languages
    ↪ match, use FigtreeFont.
% This covers: english, swedish, french, spanish,
    ↪ norwegian, german, italian, danish, dutch,
    ↪ estonian, finnish, icelandic, romanian, turkish,
    ↪ latin, latvian, hungarian
{\FigtreeFont \fontsize{24}{26}\fontseries{m}\selectfont \
    ↪ abstractname}%
]%

\vspace{20pt}% Space after the title, before the abstract
    ↪ content begins
\normalsize\normalfont % Reset font size for the content of
    ↪ the abstract
}{%

```

```
% No specific code needed after the abstract content ends,
↪ unless you have closing commands.
% The 'abstract' environment typically handles its own
↪ vertical spacing at the end.
}
```

15 Language codes

The US Library of Congress has a table of the language codes at https://www.loc.gov/standards/iso639-2/php/code_list.php.

Based upon the US Library of Congress “Codes for the Representation of Names of Languages: Codes arranged alphabetically by alpha-3/ISO 639-2 Code”, I have eliminated all of the rows that did not have both a three-letter ISO 639-2 Code **and** a two-letter ISO 639-1 Code. For the three-letter ISO 639-2 Codes, I used only the “B”, *i.e.*, bibliographic codes. I split the table into two parts: Languages already implemented (Table 5) and those remaining to be implemented (Table 7). The term “implemented” means that I have considered it in the current template, at least to some extent (but do not necessarily have a skeleton abstract in the language*). For example, babel supports northern sami and yiddish as languages you can select, and I have created skeleton abstracts for both languages. In the case of Yiddish, I added it to the language switch in the code described in Section 14. For Norwegian Bokmål, one can use the “nob” code for an abstract and specify the babel language Norwegian. In contrast, for Norwegian Nynorsk one uses the “nno” code for an abstract and the babel language nynorsk. However, I have only implemented a skeleton abstract for Norwegian.

I have a skeleton abstract in Kurdish Sorani (babel language centralkurdish) with language code “cbk”, I do not have any support for Kurmanji. However, as Kurmanji uses a Latin-based alphabet, it should not be a problem to add an abstract with the “kur” code.

According to SCB (<https://www.scb.se/contentassets/221c69e399f0424b852b21236952e300/sprakkoder.xlsx>), there are also ISO 639-2 codes for Yiddish with regions (see Table 2). For Romanian, the ISO 639-2 B code is “rum” and the ISO 639-1 code is “ro”. However, SCB gives additional ISO 639-2 codes (see Table 3). For Northern Sami (Samiska, (norra)),

*For example, I do not have a skeleton abstract in Belarusian - but babel supports belarusian and the template supports the use of the Cyrillic alphabet. However, I have not done an investigation of whether the font supports all of the glyphs used in Belarusian.

the ISO 639-2 is "sme" and the ISO 639-1 code is "se", but is there a region code that should be used? (See Table 4) For Tornedalen Finnish (Meänkieli), the ISO 639-2 code is "fi", but I am unable to determine what ISO 639-1 code should be used (see Table 6). Is it "fi" with some region code? If so, what region code?

I have added a skeleton abstract for Tornedalen Finnish (Meänkieli). It is probably incorrect, but it provides a starting point. My assumption is that the spacing, hyphenation, *etc.*, follow that of Finnish, but that there is a difference in words.

Table 2: SCB table - Yiddish with regions

ISO 639-2 Code	Swedish name of Language
YIH	Jiddisch (Väst)
YDD	Jiddisch (Öst)

Table 3: SCB table - Romainian with regions

ISO 639-2 Code	Swedish name of Language
ROM	Romani chib
RMO	Romani, Abbruzzesi, Serbisk romani, Slovensk-kroatisk romani
RMN	Romani, Arli, Dzambasi, Gurbeti
RMC	Romani, Bashaldo, Ungersk-slovakisk romani, Romungro
RMF	Romani, Kale, Kalo
RMY	Romani, Lovara, Kalderash
RML	Romani, Polsk romani, Estnisk, Lettisk, Nordrysk, Vitrysk
RMU	Romani, resande romani, Svensk romani, Romani tavringer
RMW	Romani, Walesisk

Table 4: SCB table - Sami with regions

ISO 639-2 Code	Swedish name of Language
SMI	Samiska
SIA	Samiska, Akkalamiska
SMN	Samiska, Enaresamiska
SJK	Samiska, Kemisamiska
SJD	Samiska, Kildinsamiska
SMJ	Samiska, Lulesamiska(Luleå)
SJE	Samiska, Pitesamiska
SMS	Samiska, Skoltsamiska
SMA	Samiska, sydsamiska (södra)
SJT	Samiska, Tersamiska
SJU	Umesamiska

Table 5: Languages already implemented

ISO 639-2 Code	ISO 639-1 Code	English name of Language
alb	sq	Albanian
ara	ar	Arabic
bel	be	Belarusian
bul	bg	Bulgarian
cat	ca	Catalan; Valencian
chi	zh	Chinese
cze	cs	Czech
dan	da	Danish
dut	nl	Dutch; Flemish
eng	en	English
est	et	Estonian
fin	fi	Finnish
fre	fr	French
ger	de	German
gre	el	Greek, Modern (1453-)
heb	he	Hebrew
hin	hi	Hindi
hun	hu	Hungarian
ice	is	Icelandic
ita	it	Italian
jpn	ja	Japanese
lat	la	Latin
lav	lv	Latvian
lit	lt	Lithuanian
nno	nn	Norwegian Nynorsk; Nynorsk, Norwegian
nob	nb	Bokmål, Norwegian; Norwegian Bokmål
nor	no	Norwegian
per	fa	Persian
pol	pl	Polish
por	pt	Portuguese
rum	ro	Romanian; Moldavian; Moldovan
rus	ru	Russian
sme	se	Northern Sami
srp	sr	Serbian
slo	sk	Slovak
spa	es	Spanish; Castilian
swe	sv	Swedish
swa	sw	Swahili
tur	tr	Turkish
ukr	uk	Ukrainian
vie	vi	Vietnamese
yid	yi	Yiddish

Table 6: Missing ISO 639-1 Code from the Library of Congress list

ISO 639-2 Code	ISO 639-1 Code	English name of Language
fit	???	Meänkieli

As to the language tag to use for Meänkieli, one can simply skip the ISO 639-1 Code and use "fit" - as done in <https://www.msb.se/sv/om-msb/information-pa-andra-sprak/meankieli/om-msb/>. However, it seems strange that there is no ISO 639-1 Code. Interestingly, they use 'smi' for Lule Sami (lulesamiska) in <https://www.msb.se/sv/om-msb/information-pa-andra-sprak/lulesamiska/om-msb/>.

I have added a skeleton abstract for Meänkieli. I had help from a former professor at Stockholm's university and sprakcentrummeankieli@isof.se for the abstract and keywords in Meänkieli (although there is a small difference between their spellings of two words).

Table 7: Remaining languages to implement

ISO 639-2 Code	ISO 639-1 Code	English name of Language
aar	aa	Afar
abk	ab	Abkhazian
afr	af	Afrikaans
aka	ak	Akan
amh	am	Amharic
arg	an	Aragonese
arm	hy	Armenian
asm	as	Assamese
ava	av	Avaric
ave	ae	Avestan
aym	ay	Aymara
aze	az	Azerbaijani
bak	ba	Bashkir
bam	bm	Bambara
baq	eu	Basque
ben	bn	Bengali
bih	bh	Bihari languages
bis	bi	Bislama

Continued on next page

Table 7 – continued from previous page

ISO 639-2 Code	ISO 639-1 Code	English name of Language
bos	bs	Bosnian
bre	br	Breton
bur	my	Burmese
cha	ch	Chamorro
che	ce	Chechen
chu	cu	Church Slavonic; Old Slavonic; Church Slavonic; Old Bulgarian; Old Church Slavonic
chv	cv	Chuvash
cor	kw	Cornish
cos	co	Corsican
cre	cr	Cree
div	dv	Divehi; Dhivehi; Maldivian
dzo	dz	Dzongkha
epo	eo	Esperanto
ewe	ee	Ewe
fao	fo	Faroese
fij	fj	Fijian
fry	fy	Western Frisian
ful	ff	Fulah
geo	ka	Georgian
gla	gd	Gaelic; Scottish
gle	ga	Irish
glg	gl	Galician
glv	gv	Manx
grn	gn	Guarani
guj	gu	Gujarati
hat	ht	Haitian; Haitian
hau	ha	Hausa
her	hz	Herero
hmo	ho	Hiri Motu
hrv	hr	Croatian
ibo	ig	Igbo
ido	io	Ido
Continued on next page		

Table 7 – continued from previous page

ISO 639-2 Code	ISO 639-1 Code	English name of Language
iii	ii	Sichuan Yi; Nuosu
iku	iu	Inuktitut
ile	ie	Interlingue; Occidental
ina	ia	Interlingua (International Auxiliary Language Association)
ind	id	Indonesian
ipk	ik	Inupiaq
jav	jv	Javanese
kal	kl	Kalaallisut; Greenlandic
kan	kn	Kannada
kas	ks	Kashmiri
kau	kr	Kanuri
kaz	kk	Kazakh
khm	km	Central Khmer
kik	ki	Kikuyu; Gikuyu
kin	rw	Kinyarwanda
kir	ky	Kirghiz; Kyrgyz
kom	kv	Komi
kon	kg	Kongo
kor	ko	Korean
kua	kj	Kuanyama; Kwanyama
kur	ku	Kurdish
lao	lo	Lao
lim	li	Limburgan; Limburger; Limburgish
lin	ln	Lingala
ltz	lb	Luxembourgish; Letzeburgesch
lub	lu	Luba-Katanga
lug	lg	Ganda
mac	mk	Macedonian
mah	mh	Marshallese
mal	ml	Malayalam
mao	mi	Maori
Continued on next page		

Table 7 – continued from previous page

ISO 639-2 Code	ISO 639-1 Code	English name of Language
mar	mr	Marathi
may	ms	Malay
mlg	mg	Malagasy
mlt	mt	Maltese
mon	mn	Mongolian
nau	na	Nauru
nav	nv	Navajo; Navaho
nbl	nr	Ndebele, South; South Ndebele
nde	nd	Ndebele, North; North Ndebele
ndo	ng	Ndonga
nep	ne	Nepali
nya	ny	Chichewa; Chewa; Nyanja
oci	oc	Occitan (post 1500)
oji	oj	Ojibwa
ori	or	Oriya
orm	om	Oromo
oss	os	Ossetian; Ossetic
pan	pa	Panjabi; Punjabi
pli	pi	Pali
pus	ps	Pushto; Pashto
que	qu	Quechua
roh	rm	Romansh
run	rn	Rundi
sag	sg	Sango
san	sa	Sanskrit
sin	si	Sinhala; Sinhalese
slv	sl	Slovenian
smo	sm	Samoan
sna	sn	Shona
snd	sd	Sindhi
som	so	Somali
sot	st	Sotho, Southern
srd	sc	Sardinian
Continued on next page		

Table 7 – continued from previous page

ISO 639-2 Code	ISO 639-1 Code	English name of Language
ssw	ss	Swati
sun	su	Sundanese
tah	ty	Tahitian
tam	ta	Tamil
tat	tt	Tatar
tel	te	Telugu
tgk	tg	Tajik
tgl	tl	Tagalog
tha	th	Thai
tib	bo	Tibetan
tir	ti	Tigrinya
ton	to	Tonga (Tonga Islands)
tsn	tn	Tswana
tso	ts	Tsonga
tuk	tk	Turkmen
twi	tw	Twi
uig	ug	Uighur; Uyghur
urd	ur	Urdu
uzb	uz	Uzbek
ven	ve	Venda
vol	vo	Volapük
wel	cy	Welsh
wln	wa	Walloon
wol	wo	Wolof
xho	xh	Xhosa
yor	yo	Yoruba
zha	za	Zhuang; Chuang
zul	zu	Zulu

16 Adding keywords and other data to PDF meta-data

In an effort to add the PDF metadata via the `hyperref` package, I decided to add the English and Swedish keywords part of the PDF metadata. However, in order to

do this, I had to know the keywords *before* the `\begin{document}` command in the $\text{\LaTeX}$ file. To do so, I added three new commands to the `kththesis.cls` file, as shown in Listing 38. The commands are used in the `examplethesis.tex` file to set up the keywords in both English and Swedish as well as include a new set of $\text{\LaTeX}$ commands to store the PDF metadata (as shown in Listing 39) using a file called `lib/pdf_related_includes.tex` (shown in Listing 40). Later, the keywords that have been stored are inserted into the $\text{\LaTeX}$ after their respective language abstracts. The title page of the thesis and the PDF metadata are shown in Figure 23. Note that both the English and Swedish versions of the keywords are included in the metadata. Finally, the keywords appear (as expected) in the “For DiVA” data at the end of the PDF file.

Note that `\makeatletter` and `\makeatother` are used to include the character “@” as a letter and then return “@” to being a punctuation code. This use of “@” protects the internal names from being accessed outside of these two commands. More explicitly, `\EnglishKeywords` is a function that takes one argument, the text of the English keywords, and then stores them into “@EnglishKeywords”. Later, the text can be retrieved with the command `\InsertKeywords{english}` or `\InsertKeywords{swedish}`.

Listing 38: New commands in `kththesis.cls`

```
% Keywords
\let\@EnglishKeywords\@empty
\newcommand{\EnglishKeywords}[1]{\def\@EnglishKeywords{#1}}

\let\@SwedishKeywords\@empty
\newcommand{\SwedishKeywords}[1]{\def\@SwedishKeywords{#1}}

\makeatletter
\newcommand{\InsertKeywords}[1]{
  \IfEqCase{#1}{%
    {english}{\@EnglishKeywords}
    {swedish}{\@SwedishKeywords}
  }[\typeout{argument must be english or swedish}]
}
```

Listing 39 shows the storing of the keywords using the above commands and the include of the library to set up the PDF metadata.

Listing 39: shows the storing of the keywords using the above commands and the include of the library to set up the PDF metadata

```
% Enter the English and Swedish keywords here for use in
  ↪ the PDF metadata _and_ for later use
% following the respective abstract.
% Try to put the words in the same order in both to
  ↪ facilitate matching.
\EnglishKeywords{Canvas Learning Management System, Docker
  ↪ containers, performance tuning}
\SwedishKeywords{Canvas Lärplattform, Dockerbehållare,
  ↪ prestandajustering}

% Put the title, author, and keyword information into the
  ↪ PDF meta information
\include{lib/pdf_related_includes}
```

The `lib/pdf_related_includes.tex` file contains the $\LaTeX$ - to add information to the PDF file (specifically, author(s), title(s), and keywords). It uses the `hyperref` package and should be included before the `\begin{document}`. I want to acknowledge the inspiration of Karl Voit's template for TU Graz which inspired me to add the PDF document information. For more information about his template, see <https://github.com/novoid/LaTeX-KOMA-template> Note that my template does not use anything from his template other than the names of the information for the PDF meta fields, *i.e.*, `mytitle`, `myauthor`, and `mykeywords`, together with the idea of defining the corresponding `newcommand` to set the relevant `hyperref` parameters. A result of both these decisions is that these command names are visible to the rest of the $\LaTeX$ file. See Listing 40 for the code.

In keeping with the Swedish standard (Svenska skrivregler [Språkrådet 1.6.3 och 12.11.5]) a “mellanslag tankstreck” with a space before and after, *i.e.*, “ – ” is used to separate the title and subtitle when in Swedish, this leads to a condition based on whether the document is in Swedish or not in the computation of `mytitle` and `myalttitle`.

As a result of adding support for the PDF metadata for the copyright notice, the template now uses the `hyperxmp` package. Because of this, if the author's name includes a suffix such as “, Jr.” or “Jr.”, *i.e.*, the suffix can be separated with

a comma or not as the author prefers to write their name; thus, there was a need to change the processing of `pdfauthor` information, as `hyperxmp` treats comma-separated strings in the author or keywords fields as being a list and converts these into a sequence of data in the metadata. This change is shown in Listing 40 where the `\hyxmp@comma` macro is used to insert something that is later replaced by a comma but is not treated as a comma when processing the `pdfauthor` information. The command `\xmpquote` is used to wrap the whole name.

Listing 40: lib/pdf_related_includes.tex (edited for readability and to avoid problems when rendering)

```
% This file contains the LaTeX to add information to the PDF
% ↪ file (specifically, author(s), title(s), and keywords
% It uses the hyperref package and should be included
% ↪ before the \begin{document}
%
% I want to acknowledge the inspiration of Karl Voit's
% ↪ template for TU Graz that inspired me to add the PDF
% ↪ document information
% For more information about his template see https://
% ↪ github.com/novoid/LaTeX-KOMA-template
% Note that this template does not use anything from his
% ↪ template other than the names of the information for
% ↪ the PDF meta fields, i.e., mytitle, myauthor, and
% ↪ mykeywords together with the idea of defining the
% ↪ corresponding newcommand to set the relevant hyperref
% ↪ parameters.

\makeatletter
\ifx\@subtitle\@empty
  \newcommand{\mytitle}{\@title}
\else
  \ifinswedish
    \newcommand{\mytitle}{\@title\-\xspace\xspace\
      ↪ @subtitle}
  \else
    \newcommand{\mytitle}{\@title: \@subtitle}
  \fi
\fi
\makeatother

% Put the alternative title (and subtitle) into the PDF
% ↪ Subject meta
\makeatletter
\ifx\@altsubtitle\@empty\relax
  \newcommand{\myalttitle}{\@alttitle}
\else
  \ifinswedish
    \newcommand{\myalttitle}{\@alttitle: \@altsubtitle}
```

```

\else
\newcommand{\myalttitle}{\@alttitle\-\xspace\xspace\
  ↪ @altsubtitle}
\fi

\fi
\makeatother
\hypersetup{
  pdfsubject={\myalttitle} % Subject field
}

\ifinswedish
\XMPLangAlt{en}{pdfsubject={\myalttitle}}
\else
\XMPLangAlt{sv}{pdfsubject={\myalttitle}}
\fi

\ifinswedish
\hypersetup{%
  pdflang={sv},
  pdfmetalang={sv},
  pdftitle={\mytitle} % Title field
}
\XMPLangAlt{en}{pdftitle={\myalttitle}}
\else
\hypersetup{%
  pdflang={en},
  pdfmetalang={en},
  pdftitle={\mytitle} % Title field
}
\XMPLangAlt{sv}{pdftitle={\myalttitle}}
\fi

\makeatletter
\ltx@ifpackageloaded{hyperxmp}{
\ifx\@secondAuthorsLastname\@empty
% Note that \hyxmp@comma is used explicitly rather than \
  ↪ xmpcomma
% As the later will simply turn into a comma in this

```

```

↪ context.
\StrSubstitute{\@authorsLastname}{,}{\hyxmp@comma}[\
↪ @authorsLastnameXMP]
\newcommand{\myauthor}{\xmpquote{\@authorsFirstname\
↪ space\@authorsLastnameXMP}}
\else
% Note that \hyxmp@comma is used explicitly rather than \
↪ xmpcomma
% As the later will simply turn into a comma in this
↪ context.
\StrSubstitute{\@authorsLastname}{,}{\hyxmp@comma}[\
↪ @authorsLastnameXMP]
\StrSubstitute{\@secondAuthorsLastname}{,}{\hyxmp@comma}[\
↪ @secondAuthorsLastnameXMP]
\newcommand{\myauthor}{\xmpquote{\@authorsFirstname\
↪ space\@authorsLastnameXMP},
\xmpquote{\@secondAuthorsFirstname\space\
↪ @secondAuthorsLastnameXMP}}
\fi
}{
\ifx\@secondAuthorsLastname\@empty
\newcommand{\myauthor}{\@authorsFirstname\space\
↪ @authorsLastname}
\else
\newcommand{\myauthor}{\@authorsFirstname\space\
↪ @authorsLastname,
\space\@secondAuthorsFirstname\space\@secondAuthorsLastname}
\fi
}% end of ifpackage conditional
\makeatother

\hypersetup{
pdfauthor={\myauthor} % Author field
}

\makeatletter
\ifx\@EnglishKeywords\@empty
\ifx\@SwedishKeywords\@empty
\newcommand{\mykeywords}{}

```

```

\else
\newcommand{\mykeywords}{\@SwedishKeywords}
\fi
\else
\ifx\@SwedishKeywords\@empty
\newcommand{\mykeywords}{\@EnglishKeywords}
\else
\ifswedish
\newcommand{\mykeywords}{\@SwedishKeywords, \
↪ \@EnglishKeywords}
\else
\newcommand{\mykeywords}{\@EnglishKeywords, \
↪ \@SwedishKeywords}
\fi
\fi
\fi
\makeatother

\hypersetup{
pdfkeywords={\mykeywords} % Keywords field
}
% I have _not_ set the following fields:
% pdfcreator % Creator field
% pdfproducer % Producer field

%% Note that the copyright information is added to the PDF
↪ file inside bookinfo{}
%% as until then, the copyright information is unknown.

% Put the alternative title (and subtitle) into the PDF
↪ Subject meta
\makeatletter
\ifx\@secondkthid\@empty\relax
\newcommand{\mykthids}{author: \@kthid}
\else
\newcommand{\mykthids}{author: \@kthid, \xspace
secondauthor: \@secondkthid}
\fi
\makeatother

```

```

\hypersetup{
  pdfcontactemail={\mykthids} % Subject field
}

% Add the TRITA number to the metadata
% Get and store information about the series and the number
  ↪ within this series, i.e, TRITA numbers
%"Series": \{
% "Title of series": "TRITA-ICT-EX"
% "No. in series": "2019:00"
\makeatletter
\ifswedish
\hypersetup{
  pdfvolumenum={\@thesisSeries}, % put the series in
    ↪ the volume field
  pdfissuenum={\@thesisSeriesNumber},
  pdfpublisher={Kungliga Tekniska högskolan (KTH)},
  pdfpubtype={report}
}
\else
\hypersetup{
  pdfvolumenum={\@thesisSeries}, % put the series in
    ↪ the volume field
  pdfissuenum={\@thesisSeriesNumber},
  pdfpublisher={KTH Royal Institute of Technology},
  pdfpubtype={report}
}
\fi
\makeatother

```

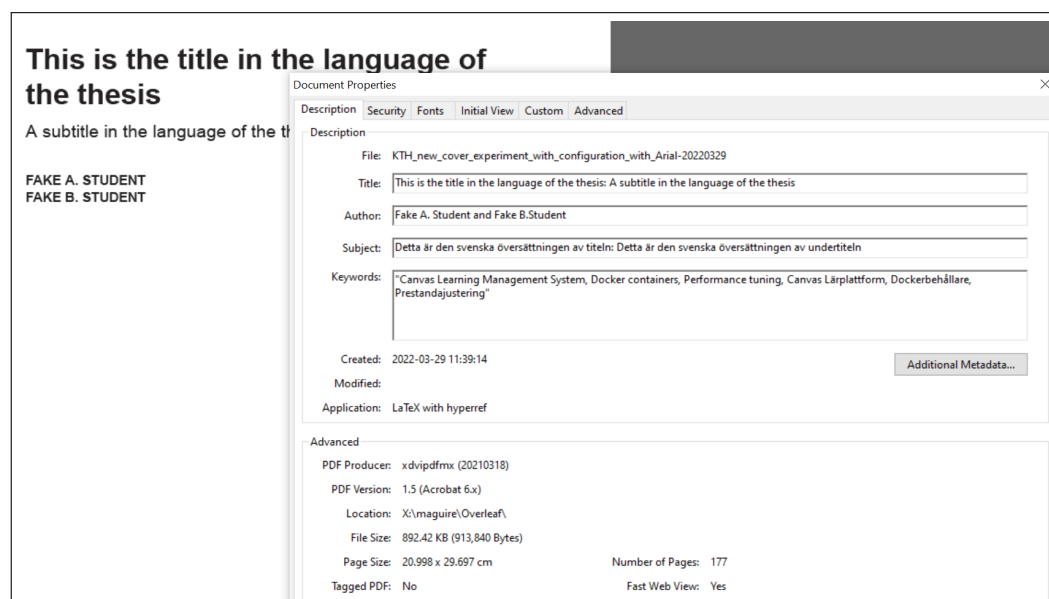


Figure 23: The title page of the thesis and the PDF metadata. This title page is from a first-cycle thesis. What is relevant here is the metadata shown as part of the PDF document description.

16.1 Adding additional metadata to the PDF file

Increasingly metadata is being embedded into digital objects to provide information about the creator (author, photographer, *etc.*), date of creation, copyright notices, *etc.* In a typical PDF file produced by $\text{\LaTeX}$ there is both PDF-specific metadata and additional metadata. This additional metadata is stored using Adobe's Extensible Metadata Platform (XMP). Extensible Metadata Platform (XMP) is a means of embedding metadata in XMP format into PDF documents so that tools can extract it mechanically.

An early paper about automatically extracting the title, authors(s), and abstract from theses was written by Mao Ni of the University of North Carolina at Chapel Hill, School of Library and Information Science in 2004 [3]. The idea was to collect the metadata from the thesis using an Adobe Acrobat plugin and then write this data to XMP (which could be embedded into the PDF file). The schema used for the metadata[4] * was that of Networked Digital Library of Theses and

*<https://ndltd.org/metadata/>

Dissertations*.

More recently, Elsevier's Digital Commons[†] uses an extension of the Dublin Core[‡] shown in Table 8 to collect metadata. We will see the Dublin Core elements and other metadata schema used in XMP in the remainder of this section.

Bepress has written about Elsevier's Digital Commons in a webpage "Digital Commons and OAI-PMH: Outbound Harvesting of Repository Records" https://bepress.com/reference_guide_dc/digital-commons-oai-harvesting/. The Open Archives Initiative Protocol for Metadata Harvesting (OAI-PMH) is to foster interoperability between repositories[§] to enable harvesters to collect metadata from a repository. Over the last 20 years, there has been an evolution from simply having metadata in a database that is associated with a publication to an entire ecosystem of metadata and applications – see for example, the twentieth International Conference on Dublin Core and Metadata Applications (DCMI 2022) - <https://www.dublincore.org/conferences/2022/programme/>.

*<https://ndltd.org/>

†<https://www.elsevier.com/solutions/digital-commons>

‡<https://www.dublincore.org/>

§<https://www.openarchives.org/pmh/>

Table 8: Dublin Core Elements in Digital Commons[5] - items in bold are from the so-called “simple Dublic Core”

Element	Element
dc.contributor	dc.rights
dc.contributor.role	dc.rights.accessRights
dc.coverage	dc.rights.license
dc.coverage.spatial	dc.source
dc.coverage.spatial.lat	dc.subject
dc.coverage.spatial.long	dc.title
dc.coverage.temporal	dc.title.alternative
dc.creator	dc.type
dc.date	dcam.memberOf
dc.date.available	dcterms.accrualMethod
dc.date.created	dcterms.accrualPeriodicity
dc.date.dateAccepted	dcterms.accrualPolicy
dc.date.dateCopyrighted	dcterms.audience
dc.date.dateSubmitted	dcterms.audience.educationLevel
dc.date.issued	dcterms.audience.mediator
dc.date.modified	dcterms.instructionalMethod
dc.date.valid	dcterms.provenance
dc.description	dcterms.rightsHolder
dc.description.abstract	geo.alt
dc.description.note	thesis.degree
dc.description.release	thesis.degree.discipline
dc.description.tableOfContents	thesis.degree.grantor
dc.format	thesis.degree.level
dc.format.extent	thesis.degree.name
dc.format.medium	
dc.identifier	
dc.identifier.bibliographicCitation	
dc.language	
dc.publisher	
dc.relation	
dc.relation.conformsTo	
dc.relation.hasFormat	
dc.relation.hasPart	
dc.relation.hasVersion	
dc.relation.isFormatOf	
dc.relation.isPartOf	
dc.relation.isReferencedBy	
dc.relation.isReplacedBy	
dc.relation.isRequiredBy	
dc.relation.isVersionOf	
dc.relation.references	
dc.relation.replaces	
dc.relation.requires	

Figure 24 shows the properties of a PDF file when using an old version of Adobe Acrobat with the command: File→Properties. This figure shows the Title, Author, Subject, and Keyword fields (among other properties of this PDF file). Because this file was generated in PDF 2.0 format, Acrobat says that you cannot edit the information, and it indicates this by displaying the information in gray.

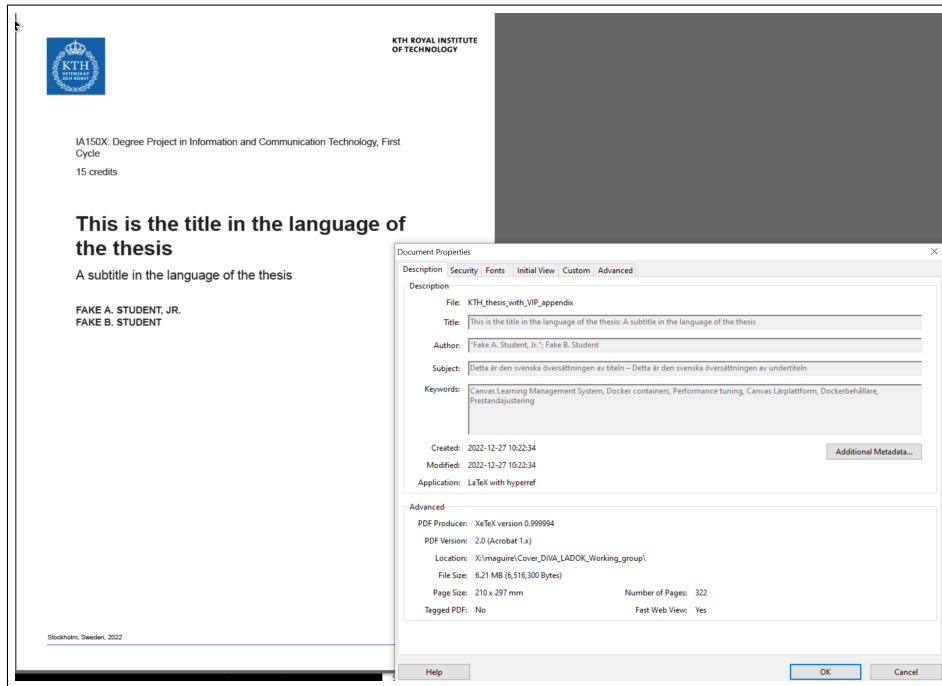


Figure 24: PDF file properties

Figure 25 shows similar data for another PDF file generated with an earlier PDF version. Because this PDF file uses PDF-1.5, Acrobat displays the information in dark letters, and it is possible to edit the metadata using Acrobat.

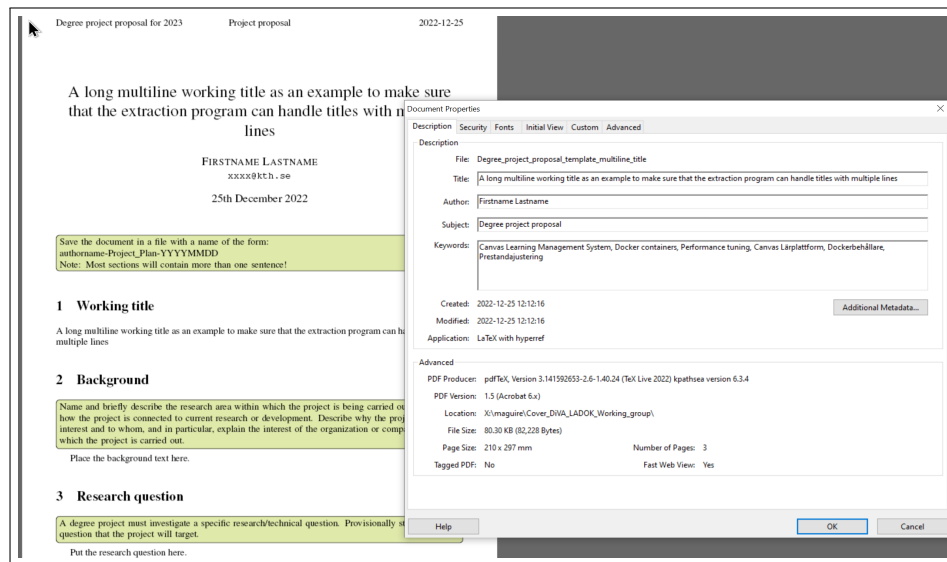


Figure 25: PDF file properties of another PDF file, in this case, the proposal template

The XMP data is in a stream indicated in the PDF as Metadata of the subtype XML, as shown in Listing 41. In the listing, the actual length in bytes has been replaced by dddd.

Listing 41: PDF stream where the XMP is found

```
<< /Subtype /XML /Type /Metadata /Length dddd >>
stream
```

In the PDF file, the XML encoded metadata is between `<?xpacket begin="yyyy" id="xxxx" ?>` and `<?xpacket end="w" ?>` (where xxxx is some unique ID). It is important to note that the value for yyyy in the previous is a so-called byte order marker (BOM) with the value 0xFEFF. While this character is a valid UTF-8 character, the `lstlisting` environment cannot handle this value; therefore, it has been replaced by the string “BOM 0xFEFF”. An example of this is shown in Listing 42.

Listing 42: The first part of the XML metadata embedded in a PDF file (some reformatting has been done to fit the text in the borders)

```
<?xpacket begin="BOM 0xFEFF"
  ↪ id="W5M0MpCehiHzreSzNTczkc9d"?>
<x:xmpmeta xmlns:x="adobe:ns:meta/">
  ☐☐<rdf:RDF
    ↪ xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#">
  ☐☐☐☐<rdf:Description rdf:about=""
    xmlns:pdf="http://ns.adobe.com/pdf/1.3/"
    xmlns:xmpRights="http://ns.adobe.com/xap/1.0/rights/"
    xmlns:dc="http://purl.org/dc/elements/1.1/"
    xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/"
    xmlns:xmp="http://ns.adobe.com/xap/1.0/"
    xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/"
    xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#"
    xmlns:pdfaid="http://www.aiim.org/pdfa/ns/id/"
    xmlns:pdfuaid="http://www.aiim.org/pdfua/ns/id/"
    xmlns:pdfx="http://ns.adobe.com/pdfx/1.3/"
    xmlns:pdfxid="http://www.npes.org/pdfx/ns/id/"
    xmlns:prism="http://prismstandard.org/namespaces/basic/3.0/"
    xmlns:jav="http://www.niso.org/schemas/jav/1.0/"
    xmlns:xmpTPg="http://ns.adobe.com/xap/1.0/t/pg/"
    xmlns:stFnt="http://ns.adobe.com/xap/1.0/sType/Font#"
    xmlns:Iptc4xmpCore="http://iptc.org/std/Iptc4xmpCore/1.0/xmlns/"
    xmlns:pdfaExtension="http://www.aiim.org/pdfa/ns/extension/"
    xmlns:pdfaSchema="http://www.aiim.org/pdfa/ns/schema\#"
    xmlns:pdfaProperty="http://www.aiim.org/pdfa/ns/property#"
    xmlns:pdfaType="http://www.aiim.org/pdfa/ns/type#"
    xmlns:pdfaField="http://www.aiim.org/pdfa/ns/field#">
```

Some of these XML elements include "rdf" elements referring to the Resource Description Framework (RDF)*.

The XMP data utilizes a number of different schema and namespaces. These namespaces are shown as "xmlns:xxxx" in Listing 42. One namespace is xmlns:dc and this refers to the scheme is 'Dublin Core'†. The information about this namespace can be found at <http://purl.org/dc/elements>

*<https://www.w3.org/RDF/>

†see Dublin Core Metadata Initiative <https://www.dublincore.org/>

/1.1/. Each of the schema and namespace are described in the next part of the XML, see Listing 43.

Listing 43: Second part of the XML metadata embedded in a PDF file (some reformatting has been done to fit the text in the borders)

```

... continuation of the above listing
  <pdfaExtension:schemas>
    <rdf:Bag>
      <rdf:li rdf:parseType="Resource">
        <pdfaSchema:schema>Adobe_PDF_
          ↪ Schema</pdfaSchema:schema>
        <pdfaSchema:prefix>pdf</pdfaSchema:prefix>
        <pdfaSchema:namespaceURI>http://ns.adobe.com/pdf/1.3/
          ↪ </pdfaSchema:namespaceURI>
        <pdfaSchema:property>
          <rdf:Seq>
            <rdf:li rdf:parseType="Resource">
              <pdfaProperty:name>Trapped</pdfaProperty:name>
              <pdfaProperty:valueType>Text</pdfaProperty:valueType>
              <pdfaProperty:category>internal</pdfaProperty:category>
              <pdfaProperty:description>Indication_if_
                ↪ the_document_has_been_modified_to_include_trapping_
                ↪ information</pdfaProperty:description>
            </rdf:li>
          </rdf:Seq>
        </pdfaSchema:property>
      </rdf:li>
    </rdf:Seq>
  </pdfaSchema:schema>
  <pdfaSchema:prefix>xmpMM</pdfaSchema:prefix>
  <pdfaSchema:namespaceURI>http://ns.adobe.com/xap/1.0/mm/
    ↪ </pdfaSchema:namespaceURI>
  <pdfaSchema:property>
    <rdf:Seq>

```

```

<rdf:li rdf:parseType="Resource">
  <pdfaProperty:name>DocumentID</pdfaProperty:name>
  <pdfaProperty:valueType>URI</pdfaProperty:valueType>
  <pdfaProperty:category>internal</pdfaProperty:category>
  <pdfaProperty:description>UUID-based
  identifier for all versions and renditions of a
  document</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
  <pdfaProperty:name>InstanceID</pdfaProperty:name>
  <pdfaProperty:valueType>URI</pdfaProperty:valueType>
  <pdfaProperty:category>internal</pdfaProperty:category>
  <pdfaProperty:description>UUID-based
  identifier for specific incarnation of a
  document</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
  <pdfaProperty:name>VersionID</pdfaProperty:name>
  <pdfaProperty:valueType>Text</pdfaProperty:valueType>
  <pdfaProperty:category>internal</pdfaProperty:category>
  <pdfaProperty:description>Document
  version identifier</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
  <pdfaProperty:name>RenditionClass</pdfaProperty:name>
  <pdfaProperty:valueType>RenditionClass</pdfaProperty:valueType>
  <pdfaProperty:category>internal</pdfaProperty:category>
  <pdfaProperty:description>The manner in

```

```

    ↪ which_a_document_is_
    ↪ rendered</pdfaProperty:description>
    </rdf:li>
  </rdf:Seq>
</pdfaSchema:property>
</rdf:li>
<rdf:li rdf:parseType="Resource">
  <pdfaSchema:schema>IPTC_Core_
    ↪ Schema</pdfaSchema:schema>

  <pdfaSchema:prefix>Iptc4xmpCore</pdfaSchema:prefix>

  <pdfaSchema:namespaceURI>http://iptc.org/std/Iptc4xmpCore/1.0/xmlns/
  ↪</pdfaSchema:namespaceURI>
</pdfaSchema:property>
<rdf:Seq>
  <rdf:li rdf:parseType="Resource">
    ↪ <pdfaProperty:name>CreatorContactInfo</pdfaProperty:name>
    ↪ <pdfaProperty:valueType>ContactInfo</pdfaProperty:valueType>
    ↪ <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Document_
      ↪ creator's_contact_
      ↪ information</pdfaProperty:description>
    </rdf:li>
  </rdf:Seq>
</pdfaSchema:property>
<pdfaSchema:valueType>
  <rdf:Seq>
    <rdf:li rdf:parseType="Resource">
      ↪ <pdfaType:type>ContactInfo</pdfaType:type>
      ↪ <pdfaType:namespaceURI>http://iptc.org/std/Iptc4xmpCore/1.0/xmlns/
      ↪</pdfaType:namespaceURI>
      ↪ <pdfaType:prefix>Iptc4xmpCore</pdfaType:prefix>
      <pdfaType:description>Basic_set_of_

```

```

    ↪ information_to_get_in_contact_with_a
    ↪ person</pdfaType:description>
    #####<pdfaType:field>
    #####<rdf:Seq>
    #####<rdf:li rdf:parseType="Resource">
    #####
    ↪ <pdfaField:name>CiAdrCity</pdfaField:name>
    #####
    ↪ <pdfaField:valueType>Text</pdfaField:valueType>
    #####<pdfaField:description>Contact_
    ↪ information_city</pdfaField:description>
    #####</rdf:li>
    #####<rdf:li rdf:parseType="Resource">
    #####
    ↪ <pdfaField:name>CiAdrCtry</pdfaField:name>
    #####
    ↪ <pdfaField:valueType>Text</pdfaField:valueType>
    #####<pdfaField:description>Contact_
    ↪ information_country</pdfaField:description>
    #####</rdf:li>
    #####<rdf:li rdf:parseType="Resource">
    #####
    ↪ <pdfaField:name>CiAdrExtadr</pdfaField:name>
    #####
    ↪ <pdfaField:valueType>Text</pdfaField:valueType>
    #####<pdfaField:description>Contact_
    ↪ information_address</pdfaField:description>
    #####</rdf:li>
    #####<rdf:li rdf:parseType="Resource">
    #####
    ↪ <pdfaField:name>CiAdrPcode</pdfaField:name>
    #####
    ↪ <pdfaField:valueType>Text</pdfaField:valueType>
    #####<pdfaField:description>Contact_
    ↪ information_local_postal_code</pdfaField:description>
    #####</rdf:li>
    #####<rdf:li rdf:parseType="Resource">
    #####
    ↪ <pdfaField:name>CiAdrRegion</pdfaField:name>
    #####

```

```

    ↪ <pdfaField:valueType>Text</pdfaField:valueType>
    <pdfaField:description>Contact_
    ↪ information_regional_information_such_as_state_or_
    ↪ province</pdfaField:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaField:name>CiEmailWork</pdfaField:name>
    <pdfaField:valueType>Text</pdfaField:valueType>
    <pdfaField:description>Contact_
    ↪ information_email_address(es)</pdfaField:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaField:name>CiTelWork</pdfaField:name>
    <pdfaField:valueType>Text</pdfaField:valueType>
    <pdfaField:description>Contact_
    ↪ information_telephone_
    ↪ number(s)</pdfaField:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaField:name>CiUrlWork</pdfaField:name>
    <pdfaField:valueType>Text</pdfaField:valueType>
    <pdfaField:description>Contact_
    ↪ information_Web_URL(s)</pdfaField:description>
    </rdf:li>
    </rdf:Seq>
  </pdfaType:field>
</rdf:li>
</rdf:Seq>
</pdfaSchema:valueType>
</rdf:li>
<rdf:li rdf:parseType="Resource">
  <pdfaSchema:schema>PRISM_Basic_
  ↪ Metadata</pdfaSchema:schema>
  <pdfaSchema:prefix>prism</pdfaSchema:prefix>

```



```

#####
  ↪ <pdfaSchema:namespaceURI>http://prismstandard.org/namespaces/basic/3.0/
  ↪ </pdfaSchema:namespaceURI>
#####<pdfaSchema:property>
#####<rdf:Seq>
#####<rdf:li rdf:parseType="Resource">
#####
  ↪ <pdfaProperty:name>complianceProfile</pdfaProperty:name>
#####
  ↪ <pdfaProperty:valueType>Text</pdfaProperty:valueType>
#####
  ↪ <pdfaProperty:category>internal</pdfaProperty:category>
#####<pdfaProperty:description>PRISM_
  ↪ specification_compliance_profile_to_which_this_
  ↪ document_adheres</pdfaProperty:description>
#####</rdf:li>
#####<rdf:li rdf:parseType="Resource">
#####
  ↪ <pdfaProperty:name>publicationName</pdfaProperty:name>
#####
  ↪ <pdfaProperty:valueType>Text</pdfaProperty:valueType>
#####
  ↪ <pdfaProperty:category>external</pdfaProperty:category>
#####<pdfaProperty:description>Publication_
  ↪ name</pdfaProperty:description>
#####</rdf:li>
#####<rdf:li rdf:parseType="Resource">
#####
  ↪ <pdfaProperty:name>aggregationType</pdfaProperty:name>
#####
  ↪ <pdfaProperty:valueType>Text</pdfaProperty:valueType>
#####
  ↪ <pdfaProperty:category>external</pdfaProperty:category>
#####<pdfaProperty:description>Publication_
  ↪ type</pdfaProperty:description>
#####</rdf:li>
#####<rdf:li rdf:parseType="Resource">
#####
  ↪ <pdfaProperty:name>bookEdition</pdfaProperty:name>
#####

```

```

    <pdfaProperty:valueType>Text</pdfaProperty:valueType>
    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Edition of the
    book in which the document was
    published</pdfaProperty:description>
  </rdf:li>
  <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>volume</pdfaProperty:name>
    <pdfaProperty:valueType>Text</pdfaProperty:valueType>
    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Publication
    volume number</pdfaProperty:description>
  </rdf:li>
  <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>number</pdfaProperty:name>
    <pdfaProperty:valueType>Text</pdfaProperty:valueType>
    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Publication
    issue number within a
    volume</pdfaProperty:description>
  </rdf:li>
  <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>pageRange</pdfaProperty:name>
    <pdfaProperty:valueType>Text</pdfaProperty:valueType>
    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Page range for
    the document within the print version of its
    publication</pdfaProperty:description>
  </rdf:li>
  <rdf:li rdf:parseType="Resource">

```

```

<pdfaProperty:name>issn</pdfaProperty:name>
<pdfaProperty:valueType>Text</pdfaProperty:valueType>
<pdfaProperty:category>external</pdfaProperty:category>
<pdfaProperty:description>ISSN_for_the_
printed_publication_in_which_the_document_was_
published</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
<pdfaProperty:name>eIssn</pdfaProperty:name>
<pdfaProperty:valueType>Text</pdfaProperty:valueType>
<pdfaProperty:category>external</pdfaProperty:category>
<pdfaProperty:description>ISSN_for_the_
electronic_publication_in_which_the_document_was_
published</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
<pdfaProperty:name>isbn</pdfaProperty:name>
<pdfaProperty:valueType>Text</pdfaProperty:valueType>
<pdfaProperty:category>external</pdfaProperty:category>
<pdfaProperty:description>ISBN_for_the_
publication_in_which_the_document_was_
published</pdfaProperty:description>
</rdf:li>
<rdf:li rdf:parseType="Resource">
<pdfaProperty:name>doi</pdfaProperty:name>
<pdfaProperty:valueType>Text</pdfaProperty:valueType>
<pdfaProperty:category>external</pdfaProperty:category>
<pdfaProperty:description>Digital_Object_

```

```

    ↪ Identifier_for_the_
    ↪ document</pdfaProperty:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>url</pdfaProperty:name>
    <pdfaProperty:valueType>URL</pdfaProperty:valueType>
    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>URL_at_which_
    ↪ the_document_can_be_found</pdfaProperty:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>byteCount</pdfaProperty:name>
    <pdfaProperty:valueType>Integer</pdfaProperty:valueType>
    <pdfaProperty:category>internal</pdfaProperty:category>
    <pdfaProperty:description>Approximate_
    ↪ file_size_in_octets</pdfaProperty:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>pageCount</pdfaProperty:name>
    <pdfaProperty:valueType>Integer</pdfaProperty:valueType>
    <pdfaProperty:category>internal</pdfaProperty:category>
    <pdfaProperty:description>Number_of_
    ↪ pages_in_the_print_version_of_the_
    ↪ document</pdfaProperty:description>
    </rdf:li>
    <rdf:li rdf:parseType="Resource">
    <pdfaProperty:name>subtitle</pdfaProperty:name>
    <pdfaProperty:valueType>Text</pdfaProperty:valueType>

```

```

    <pdfaProperty:category>external</pdfaProperty:category>
    <pdfaProperty:description>Document 's_u
    subtitle</pdfaProperty:description>
  </rdf:li>
</rdf:Seq>
</pdfaSchema:property>
</rdf:li>
</rdf:Bag>
</pdfaExtension:schemas>

```

Listing 44 shows the third part of the XMP data. In this part, we can see information in the pdf namespace. Information about this namespace can be found at <https://developer.adobe.com/xmp/docs/XMPNamespaces/pdf/>. The Adobe PDF namespace has defined four names: pdf:Keywords, pdf:PDFVersion, pdf:Producer, and pdf:Trapped. In this listing, we can see two of these: pdf:Producer and pdf:Keywords. The latter of these is a comma-separated list of keywords.

Listing 44: The pdf-prefix metadata embedded in a PDF file (some reformatting has been done to fit the text in the borders)

```
... continuation of the above listing
      <pdf:Producer>XeTeX version 0.999994</pdf:Producer>
uuuuuu<pdf:Keywords>Canvas Learning Management System,
    ↪ Docker containers, Performance tuning, Canvas
    ↪ Lärplattform, Dockerbehållare,
    ↪ Prestanda justering</pdf:Keywords>
```

Listing 45 shows the next part of the XMP data. In this case, we see a new name prefix `xmpRights`, specifically the name `xmpRights:Marked`. This namespace is described at <https://developer.adobe.com/xmp/docs/XMPNamespaces/xmpRights/>. This namespace defines the following names: `xmpRights:Certificate`, `xmpRights:Marked`, `xmpRights:Owner`, `xmpRights:UsageTerms`, and `xmpRights:WebStatement`. In this case, `xmpRights:Marked` is marked as “True” which “indicates that this is a rights-managed resource. When false, indicates that this is a public-domain resource. Omit if the state is unknown.”. So in this case, it is a right-managed resource.

Listing 45: The `xmpRights:Marked` metadata embedded in a PDF file

```
... continuation of the above listing
  <xmpRights:Marked>True</xmpRights:Marked>
```

Listing 46 shows the next part of the XMP data. In this case, we can see an element from the `dc` namespace. The name `dc:format` indicates the format of the document. As per <https://www.dublincore.org/specifications/dublin-core/format-element/> in the case of an electronic document, the format is described using the Internet Media Types (*i.e.*, MIME values).

Listing 46: The `dc:format` metadata embedded in a PDF file

```
... continuation of the above listing
<dc:format>application/pdf</dc:format>
```

Listing 47 shows the next part of the XMP data. In this case, we can see another element from the `dc` namespace. More specifically, the element named `dc:title` and this contains the title of the document. This also illustrates another feature of XML – it is possible to have the title in several different languages. The RDF tag `rdf:Alt` indicates that there are a list of alternatives. In this case, the items in the list are tagged with a language tag. The language tag `xml:lang="x-default"` is the default language of the document. This is followed by items that contain the English (tagged “en”) and Swedish (tagged “sv”). The language tags are an ISO 639-1 two-letter language code with an optional ISO 3166-1 two-letter region code.

Listing 47: The `dc:title` metadata embedded in a PDF file (some reformatting has been done to fit the text in the borders)

```
... continuation of the above listing
<dc:title>
  <rdf:Alt>
    <rdf:li xml:lang="x-default">This is the title
      ↪ in the language of the thesis: A subtitle in the
      ↪ language of the thesis</rdf:li>
    <rdf:li xml:lang="en">This is the title in the
      ↪ language of the thesis: A subtitle in the language
      ↪ of the thesis</rdf:li>
    <rdf:li xml:lang="sv">Detta är den svenska ö
      ↪ versättningen av titeln - Detta är den svenska ö
      ↪ versättningen av undertiteln</rdf:li>
  </rdf:Alt>
</dc:title>
```

Listing 48 shows the `dc:rights`, `dc:publisher`, `dc:date`, and `dc:type` elements. The `\bookinfomacro` was extended to put the copyright information into the metadata. This metadata and the `xmpRights:Marked` value being **True** enables the display of the copyright data as shown in Figure 26.

Listing 48: The `dc:rights`, `dc:publisher`, `dc:date`, and `dc:type` metadata embedded in a PDF file (some reformatting has been done to fit the text in the borders)

```
... continuation of the above listing
    <dc:rights>
uuuuuuuuuu<rdf:Alt>
uuuuuuuuuu<rdf:li xml:lang="en">Copyright (c) 2022 Fake A.
    ↪ Student, Jr. and Fake B. Student</rdf:li>
uuuuuuuuuu</rdf:Alt>
uuuuuuuu</dc:rights>
uuuuuuuu<dc:publisher>
uuuuuuuuuu<rdf:Bag>
uuuuuuuuuuuu<rdf:li>KTH Royal Institute of
    ↪ Technology</rdf:li>
uuuuuuuuuu</rdf:Bag>
uuuuuuuu</dc:publisher>
uuuuuuuu<dc:date>
uuuuuuuuuu<rdf:Seq>
uuuuuuuuuuuu<rdf:li>2022-12-27T10:22:34Z</rdf:li>
uuuuuuuuuu</rdf:Seq>
uuuuuuuu</dc:date>
uuuuuuuu<dc:type>
uuuuuuuuuu<rdf:Bag>
uuuuuuuuuuuu<rdf:li>Text</rdf:li>
uuuuuuuuuu</rdf:Bag>
uuuuuuuu</dc:type>
```

Listing 49 shows the `dc:creator` metadata. In this case the comma separated list of authors has been rendered as an `rdf:Seq` which is an **ordered** list of items.

The screenshot shows a metadata editor window with the following fields and values:

- Description:** This is the title in the language of the thesis: A subtitl
- Author:** Fake A. Student, Jr.; Fake B. Student
- Author Title:**
- Description:** Detta är den svenska översättningen av titeln – Detta är den svenska översättningen av undertiteln
- Description Writer:**
- Keywords:** Canvas Learning Management System; Docker containers; Performance tuning; Canvas Lärplattform; Dockerbehållare; Prestandajustering
- Copyright Status:** Copyrighted
- Copyright Notice:** Copyright (c) 2022 Fake A. Student, Jr. and Fake B. Student
- Copyright Info URL:**

At the bottom, it shows: Created: 2022-12-27 13:27:34, Modified: 2022-12-27 13:27:34, Application: LaTeX with hyperref, Format: application/pdf. There are 'OK' and 'Cancel' buttons at the bottom right.

Figure 26: Example of the copyright information being displayed

Listing 49: The `dc:creator` metadata embedded in a PDF file

```
... continuation of the above listing
    <dc:creator>
      <rdf:Seq>
        <rdf:li>Fake A. Student, Jr.</rdf:li>
        <rdf:li>Fake B. Student</rdf:li>
      </rdf:Seq>
    </dc:creator>
```


Listing 50 shows the `dc:subject` metadata. In this case the comma separated list of keywords has been rendered as an `rdf:Bag` which is an **unordered** list of items.

Listing 50: The `dc:subject` metadata embedded in a PDF file

```
... continuation of the above listing
    <dc:subject>
UUUUUUUUUU<rdf:Bag>
UUUUUUUUUU<rdf:li>Canvas_Learning_Management_
    ↪ System</rdf:li>
UUUUUUUUUU<rdf:li>Docker_containers</rdf:li>
UUUUUUUUUU<rdf:li>Performance_tuning</rdf:li>
UUUUUUUUUU<rdf:li>Canvas_Lärplattform</rdf:li>
UUUUUUUUUU<rdf:li>Dockerbehållare</rdf:li>
UUUUUUUUUU<rdf:li>Prestandajustering</rdf:li>
UUUUUUUU</rdf:Bag>
UUUUUU</dc:subject>
```

Listing 51 shows the `dc:source` and `dc:language` metadata. The value of `dc:source` was not set explicitly, so this seems to have been derived in some way. Additionally, the `dc:language` `rdf:Bag` was automatically computed and includes all of the languages used via `BABEL` or `POLYGLOSSIA` in the thesis.

Listing 51: The `dc:source` and `dc:language` metadata embedded in a PDF file

```
... continuation of the above listing
      <dc:source>output.tex</dc:source>
uuuuuu<dc:language>
uuuuuuuu<rdf:Bag>
uuuuuuuuuu<rdf:li>fr-FR</rdf:li>
uuuuuuuuuu<rdf:li>es-ES</rdf:li>
uuuuuuuuuu<rdf:li>nb</rdf:li>
uuuuuuuuuu<rdf:li>de-DE</rdf:li>
uuuuuuuuuu<rdf:li>sv</rdf:li>
uuuuuuuuuu<rdf:li>it</rdf:li>
uuuuuuuuuu<rdf:li>da</rdf:li>
uuuuuuuuuu<rdf:li>nl</rdf:li>
uuuuuuuuuu<rdf:li>et</rdf:li>
uuuuuuuuuu<rdf:li>en-US</rdf:li>
uuuuuuuu</rdf:Bag>
uuuuuu</dc:language>
```

Listing 52 shows the xmp-prefix elements in the file. This namespace is supposed to be defined at <http://ns.adobe.com/xap/1.0/>; however, this URL is no longer valid but the Adobe XMP Basic namespace is described at <https://developer.adobe.com/xmp/docs/XMPNamespaces/xmp/>. In this specific case, the file provides the `CreateDate`, `xmp:ModifyDate`, `xmp:MetadataDate`, and `xmp:CreatorTool`. For the last of these the value is shown as “LaTeX with hyperref” even though the $\text{\LaTeX}$ engine had been used.

Listing 52: The final set of metadata embedded in a PDF file

```
... continuation of the above listing
    <xmp:CreateDate>2022-12-27T10:22:34</xmp:CreateDate>
UUUUUU<xmp:ModifyDate>2022-12-27T10:22:34Z</xmp:ModifyDate>
UUUUUU
    ↪ <xmp:MetadataDate>2022-12-27T10:22:34Z</xmp:MetadataDate>
UUUUUU<xmp:CreatorTool>LaTeX with
    ↪ hyperref</xmp:CreatorTool>
```

Listing 53 shows the xmpMM-prefix metadata. The information about this namespace should be at <http://ns.adobe.com/xap/1.0/mm/>, however, this URL is no longer valid but the description of the XMP Media Management namespace is available at <https://developer.adobe.com/xmp/docs/XMPNamespaces/xmpMM/>.

Listing 53: The xmpMM-prefix metadata embedded in a PDF file (two of the lines have been manually broken to fit the test in the margins)

```
... continuation of the above listing
    <xmpMM:DocumentID>uuid:d7a82bbb-620a-439b-ac90-5413339b620a
    </xmpMM:DocumentID>
UUUUUU
    ↪ <xmpMM:InstanceID>uuid:620005f7-f75f-4dce-a820-a8888d18205f
    </xmpMM:InstanceID>
UUUUUU<xmpMM:VersionID>1</xmpMM:VersionID>
UUUUUU<xmpMM:RenditionClass>default</xmpMM:RenditionClass>
```

Listing 54 shows the Iptc4xmpCore-prefix metadata. The IPTC Core Schema namespace used by the IPTC Photo Metadata standards is described briefly at <http://iptc.org/std/Iptc4xmpCore/1.0/xmlns/>. A full description is available at https://www.iptc.org/std/Iptc4xmpCore/1.0/specification/Iptc4xmpCore_1.0-spec-XMPSchema_8

.pdf. The `Iptc4xmpCore:CreatorContactInfo` is supposed to provide enough information to enable someone reading the file to be able to contact the creator of the content. For the purposes of the thesis I have identified the authors using the KTH_id, since one student has completed their studies at KTH their accounts are deactivated; hence, an e-mail address is no longer meaningful.

Listing 54: The `Iptc4xmpCore`-prefix metadata embedded in a PDF file

```
... continuation of the above listing
    <Iptc4xmpCore:CreatorContactInfo
      ↪ rdf:parseType="Resource">
uuuuuuuu<Iptc4xmpCore:CiEmailWork>author:␣
      ↪ u100001, secondauthor:␣
      ↪ u100002</Iptc4xmpCore:CiEmailWork>
uuuuuu</Iptc4xmpCore:CreatorContactInfo>
```

Listing 55 shows the `prism`-prefix metadata. The Publishing Requirements for Industry Standard Metadata (PRISM) namespace should be described at <http://prismstandard.org/namespaces/basic/3.0/>; however, this URL is no longer valid but the PRISM Basic Metadata Specification can be found at <https://www.w3.org/Submission/2020/SUBM-prism-20200910/prism-basic.html>.

Listing 55: The `prism`-prefix metadata embedded in a PDF file

```
... continuation of the above listing
    <prism:complianceProfile>three</prism:complianceProfile>
uuuuuu<prism:subtitle xml:lang="en">A␣subtitle␣in␣the␣
      ↪ language␣of␣the␣thesis</prism:subtitle>
uuuuuu<prism:aggregationType>report</prism:aggregationType>
uuuuuu<prism:volume>TRITA-EECS-EX</prism:volume>
uuuuuu<prism:number>2022:0000</prism:number>
uuuuuu<prism:pageCount>322</prism:pageCount>
uuuuuu<xmpTPg:NPages>322</xmpTPg:NPages>
```

Listing 56: The final part of the metadata embedded in a PDF file

```
... continuation of the above listing
    </rdf:Description>
uu</rdf:RDF>
</x:xmpmeta>
```

16.2 Extending the set of elements for theses

Even with all of the various schema included in XMP, there is a lot of metadata that is needed for DiVA that can not yet easily be included and embedded in XMP. Thus, for the present, the extra metadata needs to be collected and processed via some mechanism other than XMP.

For XMP, there are some specific areas that might be of interest in the future:

- Adding new dc elements (§ 16.2.1)
- Adding references to the metadata (§ 16.2.2), and
- Adding citation data to the metadata (§ 16.2.3)

16.2.1 Adding new dc elements

In Table 8 on 106 we saw five elements that are perhaps worth adding, specifically: `thesis.degree`, `thesis.degree.discipline`, `thesis.degree.grantor`, `thesis.degree.level`, and `thesis.degree.name`. It may be worth adding these to the XMP data for a thesis in the future. For some discussion of these elements, please see the discussion of EDT-MS metadata elements in the paper by Ivanović, Ivanović, and Surla entitled “A data model of theses and dissertations compatible with CERIF, Dublin Core and EDT-MS”[6]. EDT-MS is defined in “ETD-MS v1.1: an Interoperability Metadata Standard for Electronic Theses and Dissertations” [4] and available at <https://ndltd.org/wp-content/uploads/2021/04/etd-ms-v1.1.html>.

16.2.2 Adding references to the metadata

Increasingly, there is an interest in being able to easily know what references have been cited by a thesis. Part of this interest is to be able to compute citation maps (who cites whom) and to examine the types of sources that are cited.

There are several possibilities for storing information about references. One is the element `dc.relation.references`. Another possibility is to use the `dcterms:references` element refinement of `dc:relation` as proposed in 1998 by Ann Apps of the DCMi Citations Working Group (<https://www.dublincore.org/groups/citation/>).

More recently, a new schema was proposed by CrossRef, see <https://www.crossref.org/documentation/schema-library/markup-guide-metadata-segments/references/>. Note that the focus of this work is to facilitate depositing with CrossRef the list of references used in a publication. Their set of elements is shown in Table 9*. Note that while the focus seems to be on journal articles, they also have a “Metadata best practices” website with lots of examples of metadata for different types of documents, see <https://www.crossref.org/documentation/principles-practices/best-practices/>.

Table 9: CrossRef elements for citation tagging

Element
issn
journal_title
author
volume
issue
first_page
cYear
article_title
isbn
series_title
volume_title
edition_number
article_title
std_designator
standards_body_name
standards_body_acronym
component_number
unstructured_citation
doi

*<https://www.crossref.org/documentation/schema-library/markup-guide-metadata-segments/references/#00178>

16.2.3 Adding citation data to the metadata

There is also a desire by many to include bibliographic citation information into the XMP so that the PDF carries this data and one does *not* need to access some other source to find this information. The element `DCTERMS.bibliographicCitation` is proposed for this purpose in a document by Ann Apps, “Guidelines for Encoding Bibliographic Citation Information in Dublin Core™ Metadata” [7].

Note that to include the URN and DiVA identifiers assigned by DiVA, you first need to create a DiVA entry and then use the assigned values to update the XMP information in the PDF file *before* uploading the full text to DiVA.

16.2.4 Adding abstracts via the `dc.description`

The code shown in Listing 57 can be used to add the abstracts to the XMP data. While this code works, it has **not** been added as a feature in the current template, but is a candidate for a future addition.

Note that region codes have been used for several of the languages, this is to distinguish the abstract entries from alternate titles, which also seem to end up in the same part of the XMP.

Unfortunately, it is not possible to add language alternates for keywords as *hyoerxmp* does not support language alternatives for keywords. Instead, it expects all the keywords to be a single comma-separated list; see the documentation for the package for the reason behind this.

Listing 57: Using Lua it is possible to add the abstracts to the XMP data

```
% --- LUA BACKEND ---
\begin{luacode*}
-- This uses the global data table thesisData

-- This function will generate all the XMP metadata
function generate_xmp_data()
  local xmp_lang_map = {
    eng="en-US",
    swe="sv-SE",
    fre="fr-FR",
    spa="es-ES",
    ita="it",
    nno="nn",
```

```
nob="nb",
nor="no",
ger="de-DE",
dan="da",
dut="nl",
est="et",
ukr="uk",
lat="la",
fin="fi",
ice="is",
yid="yi",
lit="lt",
pol="pl",
por="pt",
per="fa",
rum="ro",
hin="hi",
hun="hu",
bul="bg",
alb="sq",
cat="ca",
chi="zh",
ara="ar",
bel="be",
rus="ru",
vie="vi",
scc="sr", -- deprecated B code
cze="cs",
gre="el",
ara="ar",
tur="tr",
lav="lv",
jpn="ja",
kur="ku",
ckb="ku",
heb="he",
slo="sk",

}
```

```

-- Iterate through the abstracts
if thesisData.abstracts then
  for lang_code, content in pairs(thesisData.abstracts) do
    local xmp_tag = xmp_lang_map[lang_code]
    if xmp_tag then
      -- Call the LaTeX helper with the corrected key
      tex.sprint("\StoreXMPAbstract{" .. xmp_tag .. "}{ "
        ↪ .. content .. "}")
    end
  end
end
\end{luacode*}
% --- LATEX FRONTEND ---

% Corrected helper commands using the right keys for
  ↪ hyperxmp
\newcommand{\StoreXMPAbstract}[2]{%
  % Use 'pdfsubject' for the abstract/description
  \XMPLangAlt{#1}{pdfsubject={#2}}%
}

% This command will trigger the process in the preamble
%\directlua{generate_xmp_data()}
% add this command before the end of the document.

```

17 Some enhancements

Enhancements were made to the template and supporting programs to avoid the user having to enter information on the command line and to make the information more available for a future DiVA entry. The enhancements described in Section 17.2 are to better support mathematical expressions in abstracts. Section 17.3 describes support for acronyms in abstracts. Section 17.4 describes the state of support for URLs in abstracts.

17.1 Enhancements to template and supporting programs

Add better support for the subject area (or areas in the case of a student doing both a Civ. Ing. Degree and a Master's degree).

Not relevant for the third-cycle template.

17.2 Better support for mathematical expressions in abstracts

$\text{\LaTeX}$ in the abstracts is passed through to the pseudo JSON in the “For DiVA” text at the end of the thesis. When preparing this text for Cortina calendar entries or for the announcement in the Canvas course room (and for the calendar event in the course room) some simple $\text{\LaTeX}$ commands are converted to HTML. As few abstracts (in the many abstracts that I looked at in DiVA) have equations I have only done these few transformations. However, if there were to be more use of equations, then there is probably a need to support them for the different platforms (Cortina (see Appendix 17.2.2), DiVA (see Appendix 17.2.3), and Canvas (see Appendix 17.2.1)).

17.2.1 Better support for mathematics in Canvas course announcement and course calendar

As of 2021-06-18, MathJAX and URLs are now supported in the Canvas course room announcement and calendar. Using the Overleaf project: <https://www.overleaf.com/read/qsyddnhhvkggr> to provide a test source document. The results can be seen in Figure 27. These results and the results shown in Appendix 17.2.2 were generated using the commands in Listing 58.



Bachelor's thesis presentation/Kandidate Examensarbete presentation on 2021-03-15 at 13:00

[Gerald Quentin Maguire Jr](#)

[All Sections](#)

Jun 20 at 4:47pm

Abstract

All theses at KTH are required to have an abstract in both *English* and *Swedish*.

Exchange students many want to include one or more abstracts in the language(s) used in their home institutions to avoid the need to write another thesis when returning to their home institution.

$\pi \cdot r$ or

$$\int_a^b x^2 dx$$

Some more text: A^{*}, A[®], and A[™].

Keywords: *Canvas Learning Management System, Docker containers, Performance tuning*

Figure 27: Examples of equations in an announcement

Listing 58: Commands to produce the JSON and make the calendar entries and announcement

```
./extract_pseudo_JSON-from_PDF.py --pdf
  ↪ abstracts_with_equations_in_them.pdf --json eqtest.
  ↪ json
./JSON_to_calendar.py -c 11 --config config-test.json --json
  ↪ eqtest.json
```

Note that block/display math are displayed in the Canvas summary for the announcements cause Canvas to stop summarizing the abstract. The cause for this is not yet known, but has been reported to e-learning ([#ID:KTH-INC-3677258#]) and I have blogged about it in the Canvas Community. An example of the equation being displayed in the summary of the announcement is shown in Figure 28. The announcement is shown in Figure 29 while Listing 59 shows the HTML for this announcement.

Bachelor's thesis presentation/Kandidat Examensarbete presentation on 2021-03-15 at 13:00

[All Sections](#)

Abstract All theses at KTH are required to have an abstract in both English and Swedish. Exchange stud...

$$\int_a^b x^2 dx$$

Some more text: A*, A®, and A™. Keywords: Canvas Learning Management System, Docker containers, ...

Figure 28: Announcement summary



Bachelor's thesis presentation/Kandidat Examensarbete presentation on 2021-03-15 at 13:00

[Gerald Quentin Maguire Jr](#)

[All Sections](#)

Jun 20 at 4:47pm

Abstract

All theses at KTH are required to have an abstract in both *English* and *Swedish*.

Exchange students many want to include one or more abstracts in the language(s) used in their home institutions to avoid the need to write another thesis when returning to their home institution.

$\pi \cdot r$ or

$$\int_a^b x^2 dx$$

Some more text: A*, A®, and A™.

Keywords: *Canvas Learning Management System, Docker containers, Performance tuning*

Figure 29: Announcement with equation

Listing 59: HTML for the announcement

```

<h2 lang="en">Abstract</h2>
<p>All theses at KTH are required to have an abstract in
  ↪ both <i>English</i> and <i>Swedish</i>.</p>
<p>Exchange students many want to include one or more
  ↪ abstracts in the language(s) used in their home
  ↪ institutions to avoid the need to write another
  ↪ thesis when returning to their home institution.</p>
<p><span class="math-tex">\(\pi \cdot r\)</span> or <span class
  ↪ ="math-tex">\[ \int_{a}^{b} x^2 \, dx \]</span></p>
<p>Some more text: A<sup>*</sup>, A<sup>&reg;</sup>, and A&
  ↪ trade;.</p>
<p><strong>Keywords:</strong> <em>Canvas Learning Management
  ↪ System, Docker containers, Performance tuning </em></
  ↪ p>

```

Via XXXXX at IT-support, I raised the question about why the block equations were appearing in the summary of the announcement.

The response from Instructure support regarding the block equations showing up in the summary of the announcements was*:

Thank you for reaching back out. I did some testing on this and I have a few things to point out and some follow up questions.

First off, the syntax of the block equation isn't the officially supported format. The inline equation is correct with the $(...)$ delimiters, but the block equation should be formatted with dollar signs, as $$$...$$$. I see the square bracket delimiters $[...]$ are working currently, but this is not the officially supported format and may not always work. Information on this can be found in these release notes (<https://community.canvaslms.com/t5/Canvas-Releases/Canvas-Release-Notes-2021-02-20/ta-p/434781#toc-hId-698876024>).

Regardless of that, I wanted to get clarification on the issue at hand. When I open the announcement, I see both the inline and block equations, but I see only the block equation when viewing it on the main Announcements page. Is this specifically what you're reporting? (sandbox screenshot for reference (<https://share.getcloud>

*Note that the URLs have been added in parenthesis.

app.com/jkuPOKRD)) If so, this is because the Announcements page only shows a snippet of the text, so an inline equation beyond the point where the text is truncated isn't treated any differently than the rest of the text not shown. It is interesting that Canvas decides to show the block equation in this view. Having that removed from the preview would best be submitted as a feature request (<https://community.canvaslms.com/t5/forums/postpage/board-id/ideas/search-before-post-mode/true>).

If the issues you're reporting are when the full announcement is open, I'm not seeing the same behavior on this end. If that's the case, can you provide a screenshot of what you're seeing? And does the behavior persist in a different browser?

Let me know if I've misinterpreted anything or if there are any other questions as well. We'll be happy to help. Thanks and have a great day!"

The response from someone at Instructure shows that Instructure got the same behavior from the block equations in the snippets of the announcements. A screenshot of this behavior is shown in Figure 30. However, their solution for this is to suggest that we submit a feature request.

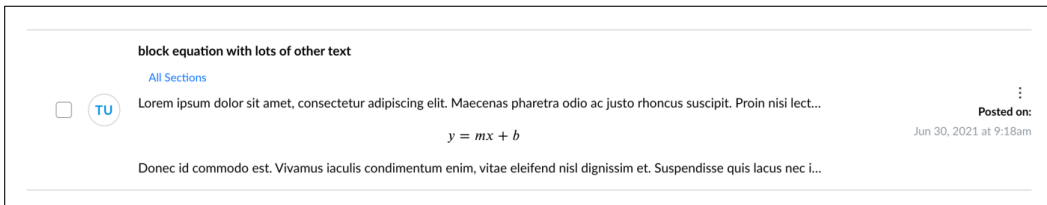


Figure 30: The screenshot mentioned above

17.2.2 Better support for mathematics in Cortina

YYYYYY* indicated that CORTINA supports mathematical expressions using a class name of “math-tex” and gave an example:

*An employee who wished to remain unnamed.

```
<span class="math-tex">\(x = -b \pm \sqrt{b^2-4ac} \over 2a\)</span>
```

He also noted that Cortina does not allow images in paragraphs, so the solutions for Cortina and DiVA have to be different.

An important note about the above example is that `\over` is deprecated and one should use `\frac` or one of its variance instead; hence, I used this version of the equation in my source $\text{\LaTeX}$ file.

Figure 31 shows the entry in the Cortina calendar, while Figure 32 and Figure 33 show zoomed in views of the equations.

However, an acceptable inline equation can use the back slash parenthesis syntax, such as

$$area = \int_a^b f(x)dx \text{ and } c = 2 \cdot \pi \cdot r$$

Some examples from <https://s3.amazonaws.com/tr-learncanvas/docs/CanvasEquationEditorAdvanced.pdf>

Intervals: (x, y) , $[x, y)$, $(-\infty, 4]$, $(\sqrt{5}, \frac{3}{4}]$

Block examples:

$$y = ax + b$$

$$c = 2 \cdot \pi \cdot r$$

$$\int_a^b x^2 dx$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For all $\mathbb{R}^n$ there is an α ...

Result: For all $\mathbb{R}^n$ there is an α ...

$$\sum_{i=0}^n i^2 = \frac{(n^2 + n)(2n + 1)}{6}$$

Piecewise function:

$$f(x) = \begin{cases} 1 + x & \text{if } x < 3 \\ 2 & \text{if } x = 3 \\ 1 - x & \text{if } x > 3 \end{cases}$$

Figure 32: Zoom in on part of the Cortina calendar entry

Piecewise function:

$$f(x) = \begin{cases} 1+x & \text{if } x < 3 \\ 2 & \text{if } x = 3 \\ 1-x & \text{if } x > 3 \end{cases}$$

Augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & \pi \\ 4 & 5 & 6 & \sqrt{5} \\ 7 & 8 & 9 & -3 \end{array} \right)$$

Two chemical formulas: H_2O or $(\text{C}_5\text{O}_2\text{H}_8)_n$

Some of the symbols and superscripts and subscripts that can be handled include: copyright symbol: © Maguire 2021, and some superscripts: $^{99\text{m}}\text{Tc}$, A^* , A° , and A^TM .

www.kth.se

[To the calendar](#)

Page responsible: kth-kalendern@kth.se

Belongs to: News & Events

Last changed: Jun 18, 2021

Figure 33: Zoom in on the lower part of the Cortina calendar entry

17.2.3 Support for mathematics in DiVA

In contrast to Canvas and Cortina, DiVA uses pictures (although these can have a $\text{\LaTeX}$ expression as an “alt” description of the image contents). The code does not (yet) support the addition of equations to the DiVA entry of the abstracts. The current DiVA user interface seems to strip out the math-tex classes, *i.e.*,

```
<span class="math-tex">\[ \int_{a}^{b} x^2 \, dx ]</span>
```

becomes simply:

```
\[ \int_{a}^{b} x^2 \, dx ]
```

However, I do not know what happens when this is later displayed by DiVA, *i.e.*, whether it will appear as an equation or not.

Fortunately, one can use the picture support as shown in the example below to generate $\widehat{\text{Err}}$ over Err:

```

```

This is an area where this likely to be a change for the new Cora-based DiVA.

17.3 Acronyms in abstracts

As students may use acronyms in abstracts, there is support for the commands: `\gls`, `\glspl`, `\Gls`, `\Glspl`, `\acrlong`, `\acrshort`, and `\acrfull`. There is also an optional argument to the extraction program to specify the name of the file with definitions of acronyms using the `\newacronyms{label}{acronym}{phrase}` form of definitions. This optional argument is shown in Listing 60. Note that expanding acronyms are handled independently for the abstracts, so that they are spelled out on first use in each abstract. Note also that there is no support for multiple languages for the phrase that is used, *i.e.*, the expansion simply uses the phrase defined in the `\newacronyms/*` definition.

Listing 60: Specifying acronyms file

```
./extract_pseudo_JSON-from_PDF.py --pdf xxxx.pdf --json
↪ xxxx.json --acronyms acronyms.tex
```

Additionally, the commands (from `defines.tex`) are supported in abstracts: `\ie`, `\eg`, `\etc`, `\etal`, `\first`, `\second`, `\third`, Now that all the acronyms are spelled out, there is no problem with them when making a calendar entry or MODS file.

17.4 URLs in abstracts

Currently, URLs in abstracts are supported in Canvas course room announcements and the Canvas calendar, but *not* in the Cortina calendar – where the URL is simply shown as text.

18 Future work and work in progress

There are several areas where the template, supporting programs, and documentation can be enhanced. The programs and notebooks can be found in <https://github.com/gqmaguirejr/Canvas-tools> or <https://github.com/gqmaguirejr/E-learning>.

18.1 Automating creation of `custom_configuration.tex` file

To help collect information for the `custom_configuration.tex` file, I have written some programs.

One of these programs is `extract_data_from_ISP.py`. It extracts some information from a PDF file of the student's Individual Study Plan (ISP) and generates a JSON file. This is in an early stage, but it shows what can be done to extract data from a PDF file of an ISP.

The program `ISP_JSON_to_configuration_info.py` takes the JSON produced by `extract_data_from_ISP.py` and produces much of the information for the `custom_configuration.tex` file (with respect to the information available at the time the writing of the thesis starts).

To get additional information for the author and supervisors, the program uses information from either a Canvas course room **or** requires Canvas administrative rights to be able to look up the user's KTHID using the Canvas API and the user's email address. In addition, the program utilizes the KTH Profile API to obtain further information about each person's organization, given an email address (if they are affiliated with KTH).

18.2 Automating creation of list of publications

The Jupyter notebook `Compute_publications_list_from_DiVA-Notebook-20250907.ipynb` uses data from `fordiva.json` and `references.bib` files to look up the student's publications in DiVA and generate the LaTeX that might be used in a list of publications in a third-cycle thesis.

This could be used for the "List of Publications" for a compilation thesis or for a list of papers and contributions in a monograph thesis. Note that the author will need to edit the resulting list to indicate which publications are included in a compilation thesis and which papers have not been included. The author may also need to move items from one category to another.

18.3 Helping students find the DiVA IDs of previous publications

The Jupyter notebook `citations-to-DiVA-Notebook-20250811-all.ipynb` use data from the `citedtags.bib` and `fordiva.json` files look up DiVA IDs for these publications.

This information is valuable when the student is entering the metadata for their thesis in DiVA.

18.4 JSON_to_Cora_DiVA

Implemented JSON_to_Cora_DiVA remains as future work.

18.5 Better exploiting unicode

The current design store the $\LaTeX$ commands in abstract *without* expanding the $\LaTeX$ macros This makes it possible to post-process these commands and make content that is suited for making announcements of the oral presentation, making calendar entries for this presentation, and generating the abstracts for the MODS file for DiVA. However, if one wants to take the full leap to unicode, this is perhaps not a good design choice - as it requires that all of the processing that $\LaTeX$ could do of the unicode is not done and that the programs that extract the unexpanded contents will have to keep changing to keep up with want $\LaTeX$ macros are used.

There might seem to be two alternatives:

1. Keep modifying the extraction programs as needed or
2. Only allow the use of $\LaTeX$ macros that expand into unicode; thus, not allowing environments and commands such as `\textbf{ }`, `\textit{ }`, etc. that do not translate directly into characters that can be written in unicode.

In the second alternative, one might think about translating the string inside a `\textbf{ }` into unicode characters in the Bold, Latin, uppercase & lowercase range, *i.e.*, U+1D400 to U+1D433 (there is a similar transformation possible for Greek). The characters are defined in the `UNICODE-MATH` package. Logically, this would be similar to redefining `\textbf{ }` as `\mathtextbf{ }`*.

*`\mathtextbf{ }` cannot be used directly as it only works in math mode.

For `\textit{ }` one could utilize the Italic, Latin, uppercase and lowercase range, *i.e.*, U+1D434 to U+1D467 (there is a similar transformation possible for Greek). Similar transformations are possible for `\texttt{ }` using Typewriter, Latin, uppercase & lowercase and for `\textsc{ }`

Additionally, there is a Bold italic, Latin, uppercase & lowercase range, *i.e.*, U+1D468 to U+1D49B.

However, the above two alternatives are not actual alternatives – as the $\LaTeX$ commands do not get expanded at all. One possibility is to utilize the expanded output in the DiVA metadata for abstracts. Also note that these character ranges are for mathematics (hence, they use mathematical glyphs, *i.e.*, the block Mathematical Alphanumeric Symbols).

This suggests a third alternative: have all of the contents (for titles, subtitles, keywords, and abstracts) be directly in unicode (or as characters specified as described in section “Configure the title, subtitle, alternate title, and alternate subtitle” in the file `README_3rd_cycle_author.tex`). See also Section 4.5. Perhaps some limited use of some macros can be allowed, as long as they can easily be processed into HTML with unicode contents.

This third alternative does **not** address the issue of equations. As noted earlier, this will depend on whether the new Cora-based DiVA will support MathJax or similar (see Section 18.6). Although perhaps just leaving the equations in place is the correct solution. However, limiting abstracts to unicode text with possible $\LaTeX$ equations seems to be too limited. Therefore, some subset of $\LaTeX$ has to be allowed, basically limited to what can be easily transformed to HTML, *i.e.*, the *current* solution!

One can split the text into math and text blocks - as done in my LaTeXtoHTML notebook.

18.6 Better support for mathematical expressions in DiVA

With the future introduction of the new Cora-based DiVA, there may be better support for mathematical expressions via MathJax. This should enable the $\LaTeX$ expressions to just be passed through to DiVA as MathJax runs in the user’s browser.

This does raise a question of what happens to these expressions in titles and subtitles when they have to be printed on the student’s diploma!

18.7 Further automation to generate additional parts of the `custom_configuration.tex` file

There is an open question of the extent to which the National Subject Category information, the UN's Sustainable Development Goals (SDGs) list, and English and Swedish keywords can be automatically extracted or guessed (perhaps with the help of a large language model).

This needs further research and prototyping. Note that my earlier experiments with detecting national subject category information for first- and second-cycle theses using the University of Linköping's tool yielded good correlation with the manually assigned codes and the top 5 codes generated by the tool. The tool performed badly on theses written in Swedish; I suspect that this is because it had limited training data for such theses.

Acronyms

API	Application Programming Interface
BOM	byte order marker
DIGG	Myndigheten för digital förvaltning
DiVA	Digitala Vetenskapliga Arkivet
DOCX	Microsoft Word Open XML Document
HTML	Hypertext Mark-up Language
IMRAD	Introduction, Methods, Results, and Discussion
ISBN	International Standard Book Number
ISP	Individual Study Plan
JSON	JavaScript Object Notation
KTH	KTH Royal Institute of Technology
LADOK	Lokalt adb-baserat dokumentationssystem
LTI	Learning Tools Interoperability
MODS	Metadata Object Description Schema
OAI-PMH	Open Archives Initiative Protocol for Metadata Harvesting
PDF	Portable Document Format
RDF	Resource Description Framework
WCAG	Web Content Accessibility Guidelines
XML	Extensible Markup Language
XMP	Extensible Metadata Platform

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